FITTER-EU

DELIVERABLE 4.1

Report on the hazards' probability rates and negative impacts' severity rate

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Abstract

This deliverable presents a structured, empirically grounded methodology for estimating perceived probability of hazards and severity of their negative impacts arising from the twin (green and digital) transition. Focusing on three priority sectors — Energy (photovoltaic — PV — installations), Housing (retrofitting of houses), and Transport (acquisition of electric vehicles — EVs) — the report details the design, implementation, and analysis of sector-specific questionnaires administered across six use-case countries (Ireland, Italy, Spain, Germany, Hungary, and Portugal). The questionnaires, informed by WP2 case studies and aligned with EU-SILC variables, comprise three parts: access to policy interventions, sociodemographic profiling, and 5-point Likert-scale assessments of 26 hazards (7 in Energy, 13 in Housing, 6 in Transport) — the full questionnaires are reported in Annex I, and a complete description of variables and coding is provided in Annex II (Survey Variable Dictionary). Stratified-quota sampling via the Prolific platform was planned for n=456 valid responses per sector (designed at α=0.05 / 95% confidence and $1-\beta$=0.80 / 80% power, detailed sampling design and stratification procedures are documented in Section 2.2.3, Annex III, Annex V and Annex VII); after the three-step quality-control protocol (Section 2.3.2), the actual analytical dataset comprises n=408 (Energy), n=264 (Housing) and n=359 (Transport) — a total of 1,031 unique verified respondents. Because the achieved samples fall below the n=456 design value, the actually achieved statistical power is lower than originally planned: at α=0.05 and the planned effect-size detection criterion (margin of error $e \approx 0.16$ on the 1–5 Likert scale, $\sigma \approx 1.2$), the achieved post-hoc power is approximately 75% (Energy), 57% (Housing), and 70% (Transport); equivalently, holding power at 80%, the detectable margin of error widens to ±0.17 (Energy), ±0.21 (Housing) and ±0.18 (Transport) — see Section 2.2.3 and the transparency reflection in Section 2.3.2. Results comprise sectoral mean probability and severity scores (with standard deviations and variance), visualisations per hazard, socio-demographic breakdowns, and ANOVA tests comparing Disadvantaged Groups (DGs) against the remaining population. The quality-controlled and validated probability and severity indices, together with direct sociodemographic variables, are structured and formatted for direct ingestion into the FITTER Digital Platform (with full statistical outputs reported in Annex VI). This work supports anticipatory governance by enabling policymakers to identify at-risk groups and inform mitigation measures for a fair and inclusive twin transition.

Executive Summary

This deliverable, D4.1, presents the methodological foundations, implementation process, and analytical synthesis produced under Task 4.1 of the FITTER-EU project. It develops and applies a structured, consortium-wide methodology for estimating hazards' perceived probability and negative impacts' severity associated with the twin transition in three core sectors: Energy, Housing, and Transport. The document refines and clarifies methodological decisions, expands details on sampling and scoring, and strengthens the connection to subsequent tasks in WP4. As part of WP4 ("Inequalities Risk Assessment and Intersectional Mitigation Plans"), it details the design and implementation of quantitative surveys targeted at Adverse Effects on Disadvantaged Groups (AE DGs), utilizing 5-point Likert scales for perceptual data on hazard likelihood and impact severity across these sectors in six use cases (Ireland, Italy, Spain, Germany, Hungary, and Portugal). Sampling methodologies are outlined, including a sample size of 456 respondents per sector calculated using a null-hypothesis testing approach (mathematically equivalent to the Cochran formula, justified in Section 2.2.3), stratified selection on Prolific aligned with Eurostat sociodemographic proportions, and rigorous recruitment workflows with quality checks to ensure data integrity. The methodology draws on EU-SILC guidelines for variable selection and stratification. The deliverable provides quality-controlled and structured results ready for integration into the FITTER Digital Platform and for further analytical use within WP4.

Table of Contents

Abstract 4
Executive Summary 4
Table of Contents 5
List of Figures 8
List of Tables 10
Abbreviations and Acronyms 11
Copyright and disclaimer 12
Disclaimer on the use of Generative AI tools 12
1 Introduction 13
1.1 Overview of FITTER-EU 13
1.2 Purpose and scope of the Deliverable 13
1.3 Structure of the Deliverable 14
1.4 Connections with other WPs 14
2 Methodology 16
2.1. T4.1 survey design 16
2.1.1. Initial methodological decisions and inputs from other WPs for T4.1 survey design 16
2.1.2. Operational steps to design T4.1 questionnaire 17
2.1.3. Summary of questionnaire's content 18
2.1.4. Detailed process for the development of questions 19
2.2. Sampling Strategy 20
2.2.1. Target population and selection of sampling frame 21
2.2.2. Sampling techniques 22
2.2.3. Sample size calculation 22
2.2.4. Stratification of the sample 26
2.3. Data collection process 29
2.3.1. Identification of indirect indicators for the predictive framework 29
2.3.2. Monitoring Response Rates 31
Sample composition limitations: low eligibility of sociodemographic targets on Prolific. 31
2.3.3. Removal of Low-Quality Responses 39
2.3.4. Metadata Storage and Documentation 41
2.3.5. Expected Attrition and Replacement Procedures 41
2.3.6. Measurement and scoring (probability & severity index construction) 42
2.4. Data Analysis Methodology (overview) 43

2.4.1. Methodology of the Analysis of Variance 44
2.4.2. Sample Size for the Analysis of Variance 45
3 Results 47
3.1. Perception of Hazards by Sector 49
3.1.1. Energy Sector – PV Installation (n=408; DE, ES, HU, IE, IT, PT) 49
3.1.2. Housing Sector – Retrofitting (n=264; DE, IE, PT) 50
3.1.3. Transport Sector – Electric Vehicles (n=359; ES, HU, IT) 52
3.1.4. Cross-Sector Comparison of Overall Risk Perception 53
3.2. Perception of Hazards by Socio-Demographic Profiles 54
3.2.1. Energy Sector: Women vs. Men 55
3.2.2. Energy Sector: Youth vs. Elders 56
3.2.3. Housing Sector: Women vs. Men 56
3.2.4. Housing Sector: Low Income vs. Rest 57
3.2.5. Transport Sector: Rural vs. Urban 57
3.2.6. Transport Sector: Disability vs. No Disability 58
3.2.7. Tenure (Tenant vs Homeowner) — Energy and Housing 58
3.3. Perception of Hazards by Country 59
3.3.1. Energy Sector by Country 59
3.3.2. Housing Sector by Country 60
3.3.3. Transport Sector by Country 61
3.3.4. Country-Level Synthesis and Statistical Caveats 62
3.4 Results of the Analysis of Variance (ANOVA) 62
4. Discussion 64
5. Conclusions and further developments 65
Key Findings 66
Implications for WP4 and the FITTER Digital Platform 67
Further Developments 67
6. References 68
7. Annexes 69
I. Annex I – T4.1 Survey Questionnaires (Energy, Housing and Transport sectors) 69
Annex II – Survey Variable Dictionary 70
III. Annex III – Technical Derivation of the Sample-Size Formula 70
IV. Annex IV – Sociodemographic CrossCheck Verification Report 74
Annex V – EU-SILC Stratification Data Analysis 75

Annex VI – D4.1 Statistical Tables (Hazard, ANOVA and Subgroup Statistics) 76
Annex VII – Prolific Sub-Studies Quota Specification (Recruitment Plan) 77
Annex VIII – Prolific Sub-Studies Configuration Files 78
Annex IX – Prolific Recruitment Dashboard Evidence 78
Annex X – Raw Survey Response Data (LimeSurvey Exports) 78
Annex XI – DMP Annex: Joint Ownership and Restricted-Access Data Agreement 79

List of Figures

Figure 1. Interconnectivity of FITTER work packages, tasks and deliverables. 16
Figure 2. Sample process steps 21
Figure 3. Prolific dashboard — Project view (page 1). DG-targeted FITTER sub-studies, with completion-rate badges indicating partial fill rates for several sociodemographic targets. 35
Figure 4. Prolific dashboard — Project view (page 2). Additional FITTER sub-studies, several with completion-rate constraints due to small eligible pools. 35
Figure 5. Prolific study detail — Study 85 (DG sub-study). Only 82 of 333,736 platform participants are eligible after applying the joint sociodemographic and country filters; 20 of 44 submissions in progress. 36
Figure 6. Prolific study detail — Study 86. Eligible pool 26 of 333,736; 8 of 42 submissions; illustrative of intersectional DG profiles with very narrow eligibility. 36
Figure 7. Prolific study detail — Study 81. Eligible pool 63 of 333,736; 26 of 34 submissions, showing recovery from constrained pool through extended recruitment. 37
Figure 8. Prolific study detail — Study 82. Eligible pool 22 of 333,736; 3 of 38 submissions; evidence that some DG profiles are nearly impossible to fully populate on Prolific. 37
Figure 9. LimeSurvey administration dashboard — final completed responses per sector: Transport 539, Housing 413, Energy 553 (counts include responses before final cross-check exclusions; see Annex IV). . 38
Figure 10. F and F' probability distributions with Error type 1 and Error type 2 46
Figure 11. ANOVA sample size calculator 47
Figure 12. Power analysis of ANOVA test 47
Figure 13. Energy sector: mean probability and severity per hazard (n=408). Blue=Probability; Red=Severity. Error bars=±1 SD. 50
Figure 14. Housing sector: mean probability and severity per hazard (n=264). Blue=Probability; Red=Severity. Error bars=±1 SD. 52
Figure 15. Transport sector: mean probability and severity per hazard (n=359). Blue=Probability; Red=Severity. Error bars=±1 SD. 53
Figure 16. Cross-sector comparison of overall mean risk perception (probability+severity)/2 with ±1 SD. ANOVA across sectors: F=48.96, p<0.001. 54
Figure 17. Energy sector: mean probability and severity – women (n=195) vs. men (n=209). Blue=Women; Orange=Men. Error bars=±1 SD. [Production note: this figure spans two side-by-side panels (Probability | Severity) 56
Figure 18. Energy sector: mean probability and severity – youth 18–34 (n=199) vs. elders 65+ (n=13). Blue=Youth; Orange=Elders. Error bars=±1 SD. [Production note: this figure spans two side-by-side panels (Probability | Severity) 56
Figure 19. Housing sector: mean probability and severity – women (n=85) vs. men (n=175). Blue=Women; Orange=Men. Error bars=±1 SD. [Production note: this figure spans two side-by-side panels (Probability | Severity) 57
Figure 20. Housing sector: mean probability and severity – low income <€20k (n=43) vs. rest (n=221). Blue=Low income; Orange=Rest. Error bars=±1 SD. [Production note: this figure spans two side-by-side panels (Probability | Severity) 57
Figure 21. Transport sector: mean probability and severity – rural (n=55) vs. urban/suburban (n=304). Blue=Rural; Orange=Urban. Error bars=±1 SD. [Production note: this figure spans two side-by-side panels (Probability | Severity) 58
Figure 22. Transport sector: mean probability and severity – disability (n=33) vs. no disability (n=320). Blue=Disability; Orange=No disability. Error bars=±1 SD. [Production note: this figure spans two side-by-side panels (Probability | Severity) 58

Figure 23. Energy sector — mean hazard probability and severity by country (Hungary absent; n shown per country). Error bars = ±1 SD. 60
Figure 24. Housing sector — mean hazard probability and severity by country (Italy, Spain, Hungary not in the Housing plan). Error bars = ±1 SD. 61
Figure 25. Transport sector — mean hazard probability and severity by country (only Italy and Spain in the Transport plan). Error bars = ±1 SD. 61
Figure 26. Probability distribution of the variable "$\bar{X}$ = proportion of success in the sample" 72
Figure 27. Representation of H0 and H1 in the Null Hypothesis Test in a No Standard Normal distribution 72
Figure 28. Representation of H0 and H1 in the Null Hypothesis Test in a Standard Normal distribution... 73

List of Tables

Table 1. Summary of T4.1 questionnaires..... 18
Table 2. List of references about the sample size linked to the number of predictors. Source: Memon et al. (2020). Citation convention: in this table all source references are given at the end of the rule in parentheses, e.g. '(Hair et al., 2018)'. 24
Table 3. Other references about sample size not linked to the number of predictors. Source: Memon et al. (2020). 25
Table 4. EU-SILC stratification variables by sector, level and expansion factor..... 27
Table 5. Steps in the indirect indicator identification and selection. 30
Table 6. Planned versus achieved respondents per country and sector 33
Table 7. Observations, Totals and Means for two-group ANOVA..... 44
Table 8. Energy sector – mean probability and severity per hazard (n=408). 50
Table 9. Housing sector – mean probability and severity per hazard (n=264). 51
Table 10. Transport sector – mean probability and severity per hazard (n=359). 53
Table 11. Mapping of each analytical subgroup to its EU-SILC stratification status, sectors analysed, and ANOVA outcome. Primary = EU-SILC-derived stratification variable; secondary analytical lens = pre-registered exploratory variable not used for quota stratification. 55
Table 12. ANOVA results per potential DG and sector. ✓ YES = significant difference at $p<0.05$; NO = no significant difference; — = insufficient data. Power column: — = $n_i \geq 64$ (≥80% power); % = recalculated power for $n_i<64$. 62

Abbreviations and Acronyms

Abbreviation and Acronyms	Description
AE DGs	Adverse Effects on Disadvantaged Groups
AGs	Advantaged Groups
ANOVA	Analysis of Variance
API	Application Programming Interface
CG	Cluster Group (FITTER Cluster Group)
CCGS	Co-Creation and Governance Structures
CLT	Central Limit Theorem
DG	Disadvantaged Group(s)
EU-SILC	European Union Statistics on Income and Living Conditions
EV	Electric Vehicle(s)
GA	Grant Agreement
GDPR	General Data Protection Regulation
H0	Null Hypothesis
H1	Alternative Hypothesis
ISO 31000	International Standard for Risk Management
KMT	Krejcie and Morgan Table
LGBTIQA+	Lesbian, Gay, Bisexual, Transgender, Intersex, Queer, Asexual (+)
NAP-DG	National Action Plan for Disadvantaged Groups (referenced via D2.6)
PDSS	Predictive Decision Support System
PV	Photovoltaic
SD	Standard Deviation
SEM	Structural Equation Modeling
Zα	Z-score for confidence level (Type I error)
Zβ	Z-score for statistical power (Type II error)

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Disclaimer on the use of Generative AI tools

In the preparation of this deliverable, generative AI tools were used in a limited and supervised capacity to assist with: (i) language editing and proof-reading of author-written text; (ii) formatting and consistency checks of tables, figures and cross-references; and (iii) drafting support for the structuring of methodological descriptions subsequently reviewed, corrected and validated by the authors. All statistical analyses, data-processing decisions, interpretations, conclusions and scientific claims in this deliverable were defined, verified and approved by the human authors. The empirical results derive exclusively from the T4.1 survey data analysed with established statistical software (Stata and Python); generative AI was not used to generate, simulate or alter any primary data or statistical result. The authors take full responsibility for the content of this deliverable.

1 Introduction

1.1 Overview of FITTER-EU

The FITTER-EU project aims to contribute to existing research on the origins, dynamics and determinants of inequalities and enable anticipatory governance to support a fair and inclusive twin transition across Europe. The project innovates through the formulation and development of an ecosystem (i.e., the FITTER ecosystem) that pivotally includes a highly interactive and gamified Digital Platform powered by an Intelligent Predictive Decision Support System. By applying a co-creative methodological approach, the ecosystem aims to enable policymakers to predict which social groups perceive being at risk of being adversely affected by twin transition policies under different scenarios, with the option to simulate the implementation of these policies for the assessment of inequalities and social exclusion risks. The Digital Platform within the ecosystem aims to provide proposed mitigation measures and better practice guides that can be used to curb potential negative effects of policies on identified at-risk groups. In addition, the FITTER ecosystem will be tested and validated with the FITTER Cluster Group (CG), which helps to support the needs of the engaging stakeholders within the community of the FITTER Cluster Group (CG). This testing and validation process ensures that the platform's features are rigorously evaluated in real-world scenarios involving policymakers, civil society organizations, and other key members, thereby enhancing its overall effectiveness and alignment with user requirements across the project's objectives. The group includes key national and pan-European members such as policymakers and civil society organizations.

1.2 Purpose and scope of the Deliverable

D4.1 reports on Task 4.1 of the FITTER-EU project, whose purpose is to estimate the perceived probability of hazards and severity of negative impacts arising from three priority twin-transition policy interventions: photovoltaic (PV) system installations (Energy sector), retrofitting of houses (Housing sector), and electrification of vehicles (EVs, Transport sector). The three final questionnaires — as uploaded in LimeSurvey — are provided in full in Annex I. The surveys were administered across six FITTER use-case countries: Ireland, Italy, Spain, Germany, Hungary, and Portugal.
The deliverable covers: (i) the design and implementation of three sector-specific questionnaires (one per sector), informed by WP2 case studies and aligned with EU-SILC sociodemographic variables; (ii) a stratified-quota sampling strategy using the Prolific platform; and (iii) analysis of hazard probability and severity scores, including breakdowns by sociodemographic subgroups associated with potential disadvantage in the twin transition.
Alignment with the Grant Agreement (GA) on the identification of Disadvantaged-Group / Hazard pairs. The GA assigns the formal identification of Disadvantaged-Group / Hazard (DG–Hazard) pairs to Task 3.2 (twin-transition scenarios) and Task 3.3 (mapping of DG–Hazard pairs per scenario): "Taking as inputs the scenarios developed in T3.2 and the identified Disadvantaged Group–Hazard pairs per scenario in T3.3". Accordingly, T4.1 does NOT identify DG–Hazard pairs in this deliverable. Instead, T4.1 takes as inputs (a) the hazards drawn from the WP2 case studies — operationalised here in Section C of each questionnaire — and (b) a set of sociodemographic variables aligned with the EU-SILC framework, which are used both as quota-stratification variables during recruitment (Section 2.2.4) and as analytical subgroups in the post-hoc comparisons reported in Section 3.2. The final and authoritative DG–Hazard pairs will be established in D3.3 (WP3) and operationalised in T4.3 (predictive modelling, D4.3). Throughout this deliverable, the term "DG" therefore refers, more precisely, to analytical sociodemographic subgroups defined by the EU-SILC-aligned variables collected in Section B of the survey, not to confirmed vulnerability profiles in the GA/WP3. sense.
The survey data serve two purposes: (1) statistical estimation of perceived probability and severity of

hazards per sector and sociodemographic subgroup (reported in this deliverable); and (2) structured input for the predictive modelling work planned in Task 4.3 (D4.3).

1.3 Structure of the Deliverable

The Deliverable D4.1 is structured as follows: Section 1 introduces the project overview (1.1), purpose and scope of the deliverable (1.2), structure (1.3), and connections with other WPs (1.4); Section 2 details the methodology, including survey design and development process (Section 2.1, with subsections 2.1.1 to 2.1.4), sampling strategies (Section 2.2: target population, sampling techniques, sample-size calculation, and stratification), and data collection and measurement approaches (Section 2.3); Section 3 presents the results, including hazard perceptions by sector (3.1) and by sociodemographic profiles (3.2); Section 4 provides analytical insights through ANOVA tests and discussion; Section 5 summarises conclusions and implications for further developments; followed by references and the following eleven annexes: Annex I (T4.1 Survey Questionnaires), Annex II (Survey Variable Dictionary), Annex III (Technical Derivation of the Sample-Size Formula), Annex IV (Sociodemographic CrossCheck Verification Report), Annex V (EU-SILC Stratification Data Analysis), Annex VI (D4.1 Statistical Tables), Annex VII (Prolific Sub-Studies Quota Specification — Recruitment Plan), Annex VIII (Prolific Sub-Studies Configuration Files), Annex IX (Prolific Recruitment Dashboard Evidence), Annex X (Raw Survey Response Data), and Annex XI (DMP Annex: Joint Ownership and Restricted-Access Data Agreement). Supporting elements like tables, figures, and statistical outputs are integrated throughout to enhance clarity and substantiate findings. All eleven annexes are explicitly cross-referenced from the relevant sections of the main text (Annex III from Section 2.2.3; Annex IV from Section 2.3.2; Annexes V and VII from Section 2.2.4; Annex VI from Sections 3 and 4; Annex VIII from Sections 2.2.2 and 2.3.1; Annex IX from Section 2.3.1 and Figures 3–9; Annex X from Section 2.3.2; Annex XI from Section 2.3.4) so that the reader can navigate directly from each methodological or analytical claim to the underlying evidence.

1.4 Connections with other WPs

This deliverable (D4.1) connects explicitly with other Work Packages across the FITTER project. The main links are:

- **WP1 (Project management & quality assurance):** D4.1 feeds into the project quality and risk monitoring process by flagging methodological risks (e.g., sampling limitations, representativeness issues) that WP1 should track; outputs and lessons learned from D4.1 will be referenced in updates to the project Quality, Risk & Management Assessment Plan (D1.2/D1.3).
- WP2 (Methodology, stakeholder engagement & Co-Creation and Governance Structures — CCGS): Sampling choices, questionnaire design, in D4.1 are informed by WP2 state of the art and theoretical framework on just transitions, Feedback loops between WP2 and WP4 ensure survey instruments are aligned with engagement strategies and that identified DGs are those prioritized by stakeholders. The list of potential critical and disadvantaged groups has been studied and identified in D2.1 "Embedding social justice in twin transition research: Concepts, definitions and methodological approach" and D2.6 "Methodological guidelines for the identification and involvement of DGs and action plans" in WP2. These groups have been listed per each sector of transport, housing and energy. The calculation of the perceived hazard for each of these disadvantaged groups is further studied in section 3 of this deliverable.
- WP3 (Vulnerability mapping & scenarios): D4.1 draws on the background context provided by WP3 outputs (D3.1, D3.2, D3.3), including DG lists and hazard definitions, as contextual reference. However, the survey design primarily drew from WP2 case study findings, and the probability and severity rates estimated in D4.1 are not directly or quantitatively linked to WP3 scenario outputs.

The WP3 deliverables operate at a broader sectoral level, while the T4.1 survey focuses on specific policy interventions as defined in the WP2 case study approach. WP3 outputs served as useful background to inform the general understanding of vulnerabilities, without being integrated directly into the survey instrument or its results.

- WP5 (FITTER Digital Platform development): The quality-controlled and validated probability and severity indices from D4.1 will be formatted for ingestion into the Digital Platform (D5.1–D5.3). Interactions with WP5 are needed to define API requirements, data schema, and the representation of risk indices in the Predictive Decision Support System (PDSS) and user interface.
- WP6 (Validation and demonstration): The probability and severity indices produced in D4.1 will be integrated into the FITTER Digital Platform and the Predictive Decision Support System (PDSS). It is in that capacity—as part of the platform—that the data will ultimately be subject to validation. The platform itself (beta version) will be validated in T6.2 and T6.3 (D6.3). Direct validation of raw survey data collected in T4.1 is not envisaged in WP6 workshops (T6.1), which focus on co-creation and stakeholder engagement rather than data quality assessment.
  Note on terminology — In this deliverable, tasks are referred to as Tn.n and deliverables as Dn.n, where the first digit indicates the Work Package number and the second indicates the sequential task or deliverable within that WP (e.g., T4.1 = Task 1 of WP4; D4.1 = Deliverable 1 of WP4). The most-referenced items in this deliverable are: T2.1 (case-study methodology, WP2); T3.2 / T3.3 (scenarios and hazard-DG pair identification, WP3); T4.1 (this task: survey design and data collection); T4.2 (mitigation plans); T4.3 (predictive modelling); T5.1–T5.3 (digital platform development); T6.1–T6.3 (validation and demonstration); together with their corresponding deliverables (D2.1, D3.1–D3.3, D4.1–D4.3, D5.1–D5.3, D6.1–D6.3). The full list is in the Grant Agreement (GA).
- **WP7 (Dissemination & policy uptake):** Key findings from WP4 (including D4.1, e.g., top hazards per DG, sectoral differences) will inform policy briefs and dissemination activities (D7.8–D7.9) and will be presented to the CCGS and other stakeholders through WP7 channels. These findings encompass aggregated insights on hazard perceptions, probability rates, and severity impacts, contributing to broader WP4 results that can be shared in workshops, reports, and online platforms to foster policy uptake and stakeholder engagement across national and European levels.

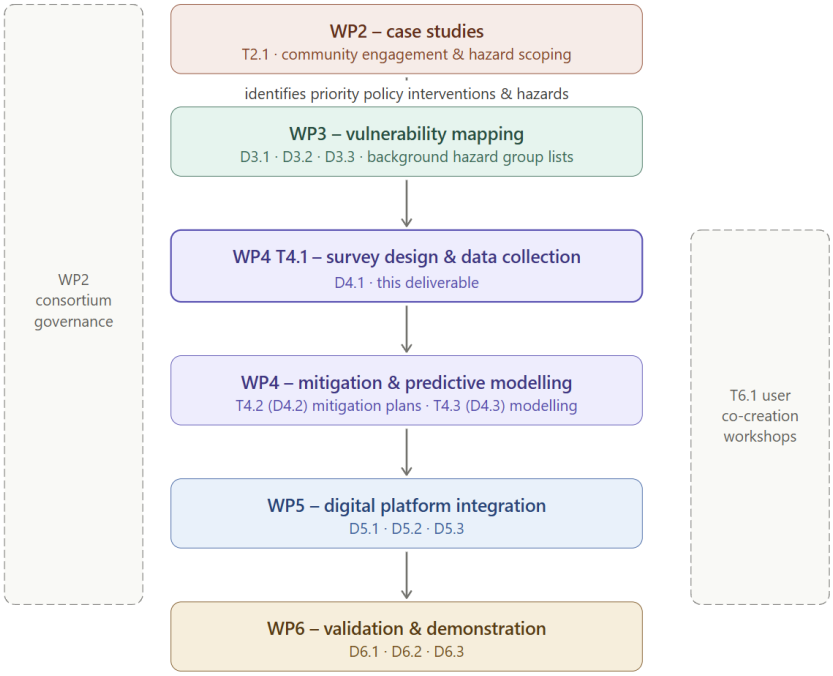


Figure 1. Interconnectivity of FITTER work packages, tasks and deliverables.

2 Methodology

This section describes the approaches used to develop surveys and implement sampling, ensuring robust data collection for calculating probability and severity rates. The undertaking was developed into five stages, detailed in the following sub-sections: (1) survey design and content development (Section 2.1); (2) questionnaire translation and national adaptation (Section 2.1.2); (3) sampling strategy, including target population, sampling techniques, sample-size calculation and stratification (Section 2.2); (4) data collection via LimeSurvey and Prolific, with response monitoring (Section 2.3.1); and (5) data quality control and analytical-dataset construction (Section 2.3.2).

2.1. T4.1 survey design

2.1.1. Initial methodological decisions and inputs from other WPs for T4.1 survey design

Design of the T4.1 survey had as a main objective the development of a survey to collect perceptions of DGs regarding the negative effects of the implementation of twin transition policies. In particular, as detailed in the Grant Agreement (GA): "Taking as inputs the scenarios developed in T3.2 and the identified "Disadvantaged Group-Hazard" pairs per scenario in T3.3, T4.1 calculate hazards' probability rates and impacts' severity rates, which are performed with quantitative data on inequality indicators. This quantitative data is collected from DGs by surveys developed in each target sector." FITTER - energy sector case study template for initial steps march 2025.docx

In January 2025, the scientific coordinators (TU Dublin) and the consortium collectively agreed on applying an integrated and revisited methodological approach to all WPs, based on ongoing WP2 findings and a more defined theoretical framework emerging from D2.1. This led to the development of a step-by-step methodological FITTER-EU approach was developed in March and validated by the whole FITTER-EU consortium including a decision: this to implement the project methodology through specific case studies per sector, focused on policy interventions related to the prioritised policy objectives in T3.1. This decision

implied narrowing down of the methodology towards a more focused and concrete approach analysing the concrete effects of policy interventions. first template was provided to partners with the aim of analysing specific policy mechanisms and actions to pursue the transition objectives.

Within this approach, policy interventions were defined as "a combination of instruments designed to achieve a particular specific policy objective (e.g. increase the number of electric vehicles by X% by 2030)". The prioritised policy interventions selected for the T4.1 survey were:

- Installation of photovoltaic (PV) systems for the energy sector.
- Retrofitting of houses for the housing sector.
- Electrification of vehicles for the transport sector.

This implied aligning the hazards and negative impacts to be included in the T4.1 survey with the case study approach and the prioritised policy interventions. In particular, perceptions of the probability of hazards and the severity of mitigation measures to be identified in T4.1 would assess the negative effects of implementing the selected policy interventions. Therefore, unlike the GA proposal of aligning the survey design with WP3 results, it was decided to align the T4.1 survey design with the research findings of the WP2 case study approach.

In addition, the timing of WP3 outputs — specifically D3.2 and D3.3, which were intended to feed into the identification of hazards and DGs for the survey — had been updated in a revised WP3 Gantt Chart in January, with delivery planned for 31 August 2025. This revised schedule reinforced the need for WP4 to establish an earlier process for identifying hazards and DGs so that the survey design could be finalised before the summer break of 2025 and data collection could begin as planned.

The methodological steps continued with TU Dublin providing inputs for identifying the list of hazards and DGs, leading to a first proposed l — Template for the identification of vulnerable groups and the associated risk assessment ENERGY-IRE test AB 010525.docx— and a final template provided to case study leaders for the identification of DGs: FITTER Identifying Structural Advantage Disadvantage v2 copy.docx.

The first version of the questionnaire was then crafted based on the completed template for the Irish case of the energy sector by TU Dublin: FITTER Identifying Structural Advantage Disadvantage - Energy IRE example TU Dublin .docx.

2.1.2. Operational steps to design T4.1 questionnaire

- The design of the T4.1 questionnaire was led by SV as task leaders, in close coordination with AITEC and TU Dublin, taking the Irish case template for the energy sector (see above) as a starting point and producing a first version of the survey tailored to the Irish case which was subject to iterative feedback loops and reviews. The Irish energy questionnaire—developed by SV—served as the example for partners on how to adapt the WP2 template into survey questions.
- A set of guidelines (Guidelines for partners T4.1 questionnaire v2.docx) was thus designed and shared with partners (June 2025) to develop adapted national questionnaires for the energy sector. Annexing the revised Irish energy questionnaire served as an example
- Guidelines for developing questionnaires for the housing and transport sectors were provided (Guidelines for partners T4.1 questionnaire other questionnaires.docx).
- National questionnaires submitted by partners were revised and a final revision and harmonization was performed by SV. In all sectors, hazards and negative impacts were deemed relevant across countries prioritising the sector, leading to the development of three questionnaires—one per sector—for T4.1 data collection.

- In August 2025 partners translated questionnaires into their national languages to ensure readiness for the data collection process.
- In parallel to the questionnaire design, methodological discussions were held on:
    - Sampling strategy, sample representativeness and required data analysis in EU-SILC (detailed in Section 2.2 Sampling Strategy). This includes calculations for sample sizes, stratification methods based on sociodemographic variables, and alignment with EU-SILC data to ensure balanced representation across key groups.
    - Implementation of the questionnaire and data collection given the high number of respondents aimed for, decision was made to rely on an external online participant recruitment platform for academic and scientific research "Prolific" integrated with a Survey platform "Limesurvey" (detailed in Section 2.3 Data collection process). This integration facilitates efficient handling of large-scale surveys, with built-in features for quota management, real-time monitoring, and data export to maintain integrity throughout the collection process.

2.1.3. Summary of questionnaire's content

Each questionnaire was structured into four subsections:

- **Introduction to the questionnaire:** including contextual information about the project, the focus of the specific case study, the purpose of the questionnaire, target population, structure, and confidentiality statement - a specific informed consent form was prepared by SV for data collection. Information on the policy intervention considered in each sector was included in the introduction: installation of PV systems, house retrofitting, and the acquisition of electric vehicles for the energy, housing and transport sectors respectively. Specifications referring to national policy interventions were included in the adapted sectoral versions per country case studies.
- **Section A:** questions to identify access to the prioritised policy intervention.
- **Section B:** sociodemographic questions.
- **Section C:** dedicated questions to measure perceived probability of hazards and the severity of negative impacts of twin transition hazards.

Questions in Sections A and B allow the identification of DGs in each sector.

Three questionnaires —one per sector— were developed, aiming to assess the same hazards across all countries studying the relevant sectors. A validation process with national case study leaders was carried out to ensure the relevance of hazards across countries. This analysis was part of the adaptation of questionnaires to national context conducted by partners following the provided guidelines by SV. Further details of this process are provided in the next subsection. National policy specificities were addressed in the introduction.

The table below summarises the content and number of questions in each sectoral questionnaire.

Table 1. Summary of T4.1 questionnaires

Description of questionnaire	Energy	Housing	Transport
Implemented in countries	DE, ES, HU, IE, IT, PT	DE, IE, PT	ES, HU, IT

Transition policies in scope	Installation of PV system	Retrofitting of houses	Acquisition of Evs
Number of questions, of which:			
• Criteria to access the policy intervention (Section A)	28	28	30
• Sociodemographic questions (Section B)	25	25	25
• Hazards assessed (Section C; each rated for probability and severity)	7	13	6

Target population of the questionnaire are both advantaged groups (AGs) and disadvantaged groups (DGs), defined respectively as those who are able or unable to access the prioritised policy interventions. In other words, AGs are benefited from the access to the policy intervention, while DGs are disadvantaged because they are not able to access the policy intervention. In addition, DGs are also determined as population groups negatively affected by the spillover effects of implementing the policy intervention under analysis.

Respondents are not excluded from any part of the survey based on this categorisation—all respondents are expected to complete all relevant sections of the questionnaire. Specific questions in Parts A and B of the questionnaire are used to determine whether a respondent belongs to an AG or a DG. It should be noted that, in practice, the identification of DGs from survey data is carried out at the sectoral level, aligned with the target populations defined in the policy factsheets developed in T4.2. The formal linkage of specific hazards with specific DG profiles at the policy-intervention level was not completed in this task and is planned for development in T4.3. For the analyses presented in Section 3, the key sociodemographic variables used to identify DG subgroups from the survey include: income level (HY020 — equivalised disposable income), ability to keep the home adequately warm (HH050), gender (RB090), disability status (PH030), and other EU-SILC-aligned variables from Section B. As part of the guidelines, partners were asked to identify which variables could represent vulnerability conditions in their national contexts and to evidence this in their questionnaire adaptations.

The final version of the questionnaire which was uploads on Limesurvey is available at the following link: Final T4.1 integrated all questionnaires 09-10.xlsx. A dictionary of variables is included in Annex II. This dictionary provides detailed definitions, codes, and mappings for all variables used across the questionnaires, including sociodemographic items from Part B, access determinants from Part A, and Likert-scale perceptions from Part C, to facilitate accurate data interpretation and replication in subsequent analyses or platform integrations.

2.1.4. Detailed process for the development of questions

Following, details are provided describing the development of questions in each section of the questionnaire. Each component was elaborated from the content of the WP2 template (FITTER Identifying Structural Advantage Disadvantage - Energy IRE example TU Dublin .docx). The Irish energy questionnaire —developed by SV— served as the example for partners on how to adapt the WP2 template into survey questions.

Section A of the questionnaire - Determinants to access the policy intervention

This section includes questions to determine respondents' capacity to access the policy intervention. In the energy and housing sectors, the focus was on housing conditions, while in the transport sector, questions focused on transport patterns and access to private and public transport. When adapting the Irish energy questionnaire as a template, partners were also asked to identify possible interlinkages between questions and to propose a logical ordering.

After proposing an initial set of questions, a harmonisation with EU-SILC was undertaken by AITEC and SV, focusing on the ad hoc module "Energy and the Environment" included in EU-SILC 2025, which contains questions on transport patterns — such as the use of public transport, car ownership, frequency of private transport use — and housing conditions — including environmental or weather-related damage, energy efficiency renovations, and obstacles to improving energy efficiency.

Section B of the questionnaire – Sociodemographic questions

Section B includes sociodemographic questions aimed at collecting relevant variables that may determine whether respondents belong to a DG. Some questions capture intersectional characteristics, such as gender, belonging to the LGBTIQ+ community, or belonging to an ethnic minority. Questions were harmonised with EU-SILC, particularly with the personal register module focused on individual household members. A common set of sociodemographic questions was included in all three sectoral questionnaires.

Section C of the questionnaire – Assessment of the probability and severity rates

Section C is divided into two sets of questions on possible negative effects respondents may experience from not accessing the policy intervention — i.e. being unable to improve the energy efficiency of their dwelling (due to not installing a PV system or not being able to retrofit) or being unable to acquire an EV. This section uses Likert scales to appraise perceptions:

- Question C1: Assesses how realistic the respondent considers a set of negative impacts derived from the policy intervention.
- Question C2: Measures how harmful these negative effects could be for the respondent or their household.

For the energy questionnaire, SV proposed the initial list of hazards based on the WP2 case study template for Ireland. For the housing and transport sectors, the TU Dublin team proposed the potential hazards. These lists were validated by partners during questionnaire adaptation for their national case studies. In all sectors, hazards and negative impacts were deemed relevant across countries prioritising the sector, leading to the development of three questionnaires — one per sector — for T4.1 data collection.

2.2. Sampling Strategy

The sampling strategy for T4.1 follows the four-step framework proposed by Khalid (2024) and illustrated in Figure 2 below. Before describing each step, it is useful to clarify the key concepts involved: **Target population:** The full set of individuals about whom we wish to make inferences—in this case, all residents aged 18 and over in the six FITTER use-case countries (Ireland, Italy, Spain, Germany, Hungary, and Portugal) who are potentially affected by the twin transition policy interventions under study. **Sampling frame:** The list or mechanism through which members of the target population can be accessed for recruitment. For FITTER, this is the Prolific platform, which provides access to a large, pre-registered panel of participants who can be filtered by country and sociodemographic characteristics. **Sampling technique:** The method by which individuals are selected from the sampling frame. As described

in Section 2.2.2, FITTER uses non-probabilistic quota sampling via Prolific. **Sample size:** The number of respondents needed to achieve the desired statistical precision and power, as calculated in Section 2.2.3.

Fig. 2 below illustrates the sampling strategy (steps 1 to 4) addressed in this sub-section 2.2, and the data collection process (step 5) addressed in sub-section 2.3:

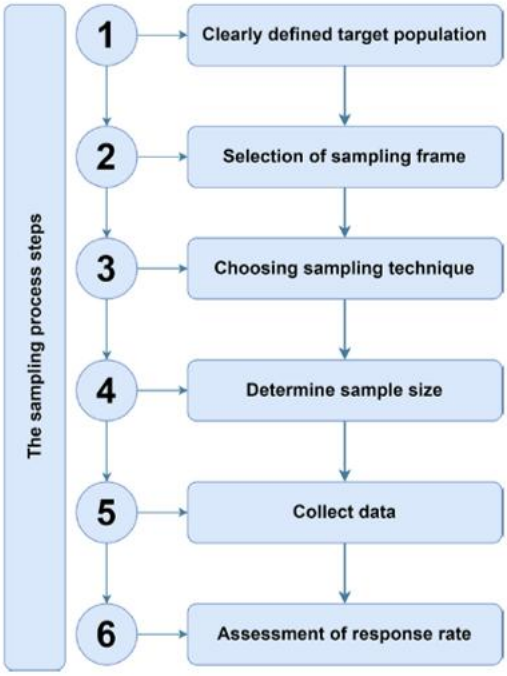


Figure 2. Sample process steps

2.2.1. Target population and selection of sampling frame

Definition of the Target Population: All citizens above 18 years old in the 6 FITTER use cases

Selection of Sampling Frame: FITTER will use Prolific (https://www.prolific.com/) to recruit participants, which is a pre-registration and compensation-based platform. The use of this platform has the next limitations regarding Bias:

- Bias due to recruit participants by using a digital platform based on an economical compensation: Recruiting participants through Prolific, an online platform that offers monetary compensation, introduces a form of **self-selection bias**. Individuals who regularly participate in paid online studies tend to be more **digitally literate**, younger on average, and more **extrinsically motivated by financial rewards** than the general population. They may also differ in personality traits (e.g., higher need for cognition and lower altruism). This can result in an over-representation of digitally skilled and economically motivated respondents, and a potential under-representation of certain structurally disadvantaged groups identified in D2.6 (e.g., older adults or people with lower digital access/skills).

FITTER accepts this limitation as a pragmatic methodological trade-off. The survey methodology literature consistently shows that **unpaid or voluntary recruitment methods** (e.g., open web surveys, social media dissemination, or snowball sampling) typically produce **even stronger**

selection biases, often severely under-representing lower-income groups, older adults, ethnic minorities, and less digitally literate populations while over-representing highly educated or politically engaged individuals. When combined with strict quota controls and EU-SILC-aligned stratification, paid recruitment via Prolific improves response rates, sample diversity, and feasibility while keeping overall bias comparatively lower — especially for the large target sample of 456 respondents per sector[1][2][3].

- Bias due to a lack of random in the process of recruiting participants: In national-statistical surveys, such as EU-SILC, full probability sampling (where every member of the target population has a known, non-zero probability of selection) is achievable through population-frame access and probability-based recruitment. This level of methodological rigour is, however, not viable within a limited-scope research project such as FITTER, where the consortium does not have direct access to national-frame data nor the operational capacity for probability-based recruitment across six countries. Although people registered in Prolific did their registration freely, without coercion, and there is no impediment to platform registration, not all of the target population has access to or knowledge of the platform. The use of non-probability quota sampling via Prolific is therefore adopted as a pragmatic and defensible compromise, with the limitations explicitly documented (see Section 2.2.2).

To avoid other kind of bias, FITTER is recruiting participants in the same socio-demographic proportions than the total target population according to EU-SILC.

2.2.2. Sampling techniques

Prolific is, by definition, a non-probabilistic sampling method. The platform operates as an online access panel where participants self-register and opt into studies; neither the probability of platform registration nor the probability of being selected for a given study is known or controlled by the researcher. As such, Prolific cannot be considered probabilistic sampling in the strict statistical sense.
For FITTER, Prolific is used as a non-probability quota sampling approach combined with an online access panel. Researchers apply sociodemographic filters (country, gender, age, income level, etc.) and enforce quota limits so that the final sample mirrors, as closely as possible, the Eurostat population distributions for each country. This approach is standard practice in academic and policy research with limited resources, and has been validated in the sampling literature (Peer et al., 2017; Hsieh & Kocielnik, 2016). However, the inherent limitations of non-probabilistic sampling—including self-selection bias and unknown inclusion probabilities—must be acknowledged as a constraint on the generalisability of findings, as noted in Section 2.2.1. The operational implementation of the Prolific-based non-probability quota sampling for FITTER — including the exact quota targets per country and sociodemographic combination and the 76 sub-studies actually launched — is documented in Annex VII (Prolific Sub-Studies Quota Specification — Recruitment Plan) and Annex VIII (Prolific Sub-Studies Configuration Files).

2.2.3. Sample size calculation

FITTER will follow a 3 steps methodology for the sampling size calculation:

[1] Hsieh, G., & Kocielnik, R. (2016). You get who you pay for: The impact of incentives on participation bias. In Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing (pp. 823–835). https://doi.org/10.1145/2818048.2819278

[2] Peer, E., Brandimarte, L., Samat, S., & Acquisti, A. (2017). Beyond the Turk: Alternative platforms for crowdsourcing behavioral research. Journal of Experimental Social Psychology, 70, 153–163. https://doi.org/10.1016/j.jesp.2017.01.006

[3] Singer, E., & Ye, C. (2013). The use and effects of incentives in surveys. The ANNALS of the American Academy of Political and Social Science, 645(1), 112–141. https://doi.org/10.1177/0002716212458082

Step 1 – Sample size for statistical estimation of perceived probability and severity.

The first purpose of the T4.1 data collection is to estimate, with sufficient statistical precision, the mean perceived probability and severity of each hazard across the target population and key subgroups. This is a classical estimation-of-population-parameters problem.
Choice of formula. We adopt the Cochran (1977) sample-size formula in its null-hypothesis-testing form (Eq. 2 below, derived in Annex III). This formula is widely used and well-validated in survey research and is appropriate when (i) the parameter of interest is a population mean (or proportion) estimable on a bounded measurement scale (here, the 1–5 Likert scale used in Section C) and (ii) the researcher wishes to control both the Type-I error (α) and the Type-II error (β) of the inference. Alternative formulas considered and not adopted included: (a) Cochran's simpler proportion formula $n = Z^2 \cdot p \cdot (1-p) / e^2$, which controls only α and is therefore less transparent about statistical power; (b) Krejcie & Morgan tables, which fix α at 0.05 and provide tabulated minima but do not explicitly accommodate power and SD; and (c) effect-size-based formulas (e.g., Cohen's d-based n), which are appropriate for two-group comparisons rather than population-mean estimation.
Limitations of the chosen formula. The Cochran formula assumes simple random sampling from a homogeneous population. FITTER uses a non-probability quota sampling approach via Prolific (Section 2.2.2), which does not strictly satisfy this assumption. We accept this departure as a pragmatic constraint: the formula provides a principled lower bound on the required sample size for each sector, and the stratified quota approach (Section 2.2.4) and the post-hoc quality-control protocol (Section 2.3.2) further mitigate sampling bias. The achieved statistical power after the quality-control protocol is reported transparently in Sections 2.2.3 and 2.3.2.

> *Null Hypothesis (H_0)*: Given the use of a five-point Likert scale in the assessment of the perception of each hazard in the "*Section C"* of the questionnaire, each point of this assessment has the same probability to be selected by the respondents of the sample with a p = 0.20 (=1/5) *(see annex III for a more technical explanation).*

- ***Type 1 Error*** means that we reject this H_0 when the hypothesis is true.
- ***Type 2 Error*** means that we accept this H_0 when it is false.
- The ***size effect*** means that we accept that the probability of selecting 1 to 5 for a specific Hazard can vary a 5% and the H_0 of assuming a P=0.2 continue being considered as valid, even when p=0,2 (-5%) and p=0,2 (+5%).

This *Null Hypothesis test* provides us with the next formula for the estimation of the simple size (= number of respondents). See **annex III** for a more technical explanation about how this formula is got (Eq.2):

$$n = (Z_\alpha + Z_\beta)^2 \cdot \varrho^2 / e^2 \quad [4] \qquad \text{(Eq.2)}$$

Where:

Note on Eq. 2: the squared term $(Z\alpha/2 + Z\beta)^2$ is the standard form for null-hypothesis-testing sample-size calculations because both the Type-I and Type-II error allowances enter the formula as standardised normal deviates that must be summed (not multiplied) and the whole sum is then squared because the sample-size equation is derived from squaring the standardised difference between the null and alternative distributions. The '2' in the exponent is therefore not arbitrary: it follows directly from the variance expression in the central limit theorem (σ^2/n) and from algebraically isolating n. A single-power formula

[4] *Probability and Statistics for Engineers and Scientists* by Mc Millan Publishing company. ISBN: 0-02-424210-1. Pag. 332

(Zα/2 only, no Zβ, no exponent 2) would underestimate n because it would ignore the power requirement (β) entirely.

- Z_α is the Z-score in a *Standard normal distribution* corresponding to the chosen confidence level.
- Z_β is the Z-score in a *Standard normal distribution* corresponding to the desired statistical power.
- φ^2 is the Variance of the probabilistic distribution associated to each test based on selecting a score from 1 to 5 per each Hazard in the survey by a respondent, which is P.(1-P) in a *Bernoulli process* with 2 possible results: "Success" with a probability "P" or "Fail" with a probability "(1-P)".
- e is the size effect or accepted margin of error.

A **95% confidence level** and **85% statistical power** were selected, corresponding to $\mathbf{Z_\alpha = 1.645}$ and $\mathbf{Z_\beta = 1.035}$ (in a one-tailed normal distribution). Being P = 0.20 (=1/5), this yields **P. (1 – P) = 0.16** as the **Variance**. The **size effect** was set at **e = 0.05**, consistent with standard practice in perception and social survey research.

- With all these considerations, we will get a sample size **n=460 per sector**.

Step 2 – Sample size adequacy for predictive modelling:

The second use of T4.1 data is as input for the predictive models planned in Task 4.3. Predictive models (such as regression or Bayesian networks) require a minimum number of observations relative to the number of predictor variables (also referred to as independent variables, IVs) included. The concept of 'predictors' here refers to the sociodemographic and contextual variables collected in Sections A and B of the T4.1 survey that will be used to predict individual hazard perception scores in Section C.

The references below (Table 3) provide established rules of thumb for the minimum sample size relative to the number of predictors. These are included here to demonstrate that the sample of 456 respondents per sector is sufficient not only for the statistical estimation in Step 1, but also for the predictive modelling purposes of T4.3. Note that the specific predictive modelling approach and the number of predictors will be determined in D4.3; this section documents the compatibility of the sample size with foreseeable modelling requirements. The 'Applicable model types' of column was added to indicate which estimation methods each rule is most relevant to, as requirements differ across OLS regression, PLS-SEM, and multilevel models.

Table 2. List of references about the sample size linked to the number of predictors. Source: Memon et al. (2020). Citation convention: in this table all source references are given at the end of the rule in parentheses, e.g. '(Hair et al., 2018)'.

#	Sample size rule	Applicable model types
1	15 to 20 observations per independent variable are strongly recommended *(Hair et al., 2018)*.	OLS regression Multiple regression PLS-SEM
2	Green (1991) proposed N ≥ 50+8m (where m refers to the number of predictors in the model).	OLS regression Multiple regression
3	The 10-times rule recommends that the minimum "sample size should be equal to the larger of (1) 10 times the largest number of formative indicators used to measure one construct or (2) 10 times	PLS-SEM Formative measurement models

#	Sample size rule	Applicable model types
	the largest number of structural paths directed at a particular latent construct in the structural model" *(Hair et al., 2017, p. 24)*.	
4	Harris (1975) recommended a minimum sample size of N ≥ 50+m (where m is the number of predictors).	OLS regression Multiple regression
5	Nunnally suggested that if one wants to select the best variables from as many as 10 possible ones, there should be between 400 and 500 respondents.	Variable selection Exploratory factor analysis Scale development

Step 3 – Cross-validation against additional benchmarks:

To confirm that the sample size of 456 (from Step 1) is well-justified, it is compared against additional criteria from the literature (Table 3, Memon et al. 2020). The criteria in Table 3 are not linked to the number of predictors; rather, they provide general minimum thresholds for survey-based research. Each criterion is met by the design sample of 456 respondents per sector: 456 > 200 (minimum for correlation analysis), > 384 (KMT for populations over 1 million), within the 30–500 range recommended by Roscoe for behavioural studies, > 160 (inverse square root method), and > 100 (Kline's 'large' SEM sample). The 456 respondents per sector, divided across 6 countries (~76 per country), also respect the Roscoe 30-respondent minimum per subgroup for comparative analysis (e.g., ~87 women in the Energy sector).

Table 3. Other references about sample size not linked to the number of predictors. Source: Memon et al. (2020).

1	Regression analysis needs at least 50 samples and generally 100 samples for most research situations (Hair et al., 2018). An absolute minimum of 200 samples is required for Pearson Correlation analysis (Guilford, 1954).
2	The Krejcie and Morgan table (KMT, Krejcie & Morgan, 1970) suggests that a sample of 384 is sufficient for a population of 1,000,000 or more.
3	Roscoe suggested that a sample size greater than 30 and less than 500 is suitable for most behavioral studies, while a sample size larger than 500 may lead to a Type II error (Sekaran & Bougie, 2016). Roscoe also posited that for comparative analysis, if the data set needs to be broken into several subgroups (e.g., male/female, rural/urban, local/international, etc.), 30 respondents should be considered the minimum for each group. The logic behind the rule of 30 is based on the Central Limit Theorem (CLT). The CLT assumes that the distribution of sample means approaches (or tends to approach) a normal distribution as the sample size increases.
4	Hadaya (2018) proposed that the minimum sample size required would be 160 based on the inverse square root. However, if researchers use the gamma exponential method, the sample size would be 146.
5	Kline (2005) offered sample size guidelines for analyzing structural equation models, suggesting that a sample of 100 is considered small, a sample of 100 to 200 is medium, and a sample over 200 is considered large.
6	Kreft (1996) recommended the 30/30 rule for multilevel models, which dictates that 30 groups with 30 individuals per group should be the minimum sample size for a multilevel study. Later, Hox (2010) modified Kreft's 30/30 rule into a more conservative 50/20 rule, such that 50 groups with 20 individuals per group should be the minimum sample size for cross-level interactions

2.2.4. Stratification of the sample

Sample sizes per granularity were determined by stratifying the overall 456 respondents per sector to ensure proportionate responses per key subgroup (e.g., gender, age groups, income), providing adequate power for comparative analyses without over-representing any single group. Note that due to distribution across 6 countries, subgroup sizes may be smaller in practice (e.g., ~76 per country in Energy), which is acknowledged as a limitation for subgroup-level inferences.

Sequential stratification approach grounded in EU-SILC microdata.

The stratification targets for each country were calculated by SV using EU-SILC microdata, analysed in Stata, following a sequential population → household → individual stratification approach. This design ensures that each variable is applied at its appropriate conceptual level and that the corresponding EU-SILC expansion factor is consistently aligned with the universe it represents. The selection of stratification variables is grounded in the analytical work conducted under WP2, which identified a set of socio-economic and demographic characteristics as particularly relevant for capturing vulnerability across the three sectors. The sampling strategy operationalises these findings by translating the WP2 vulnerability dimensions into a structured, multi-level stratification scheme.

Step 1 – Inter-country allocation (population level). The total sample size is first allocated across the six countries (Germany, Hungary, Ireland, Italy, Spain and Portugal) proportionally to the relative size of each country's population. This allocation is estimated using the EU-SILC personal cross-sectional weight **RB050**. Because the most recent EU-SILC data available for Hungary refer to 2018, population distributions for this inter-country allocation are calculated using **2018 data for all six countries**, ensuring temporal consistency at this initial stage.

Step 2 – Intra-country stratification (household level). Within each country, stratification is refined using variables defined at the household level, principally total disposable household income (**HY020**) and, for the Housing and Energy sectors, housing tenure status (**HH021**). These variables are calculated exclusively within the household data file and weighted with the EU-SILC **household cross-sectional weight (DB090)**. This choice avoids distortions that would arise from household size effects if household-level characteristics were recalculated at the individual level. For this intra-country stratification, the most recent EU-SILC data available are used: **2024** for Germany, Ireland, Italy, Spain and Portugal, and **2018** for Hungary (the most recent available for that country).

Step 3 – Individual-level stratification. In the final step, stratification is completed at the individual level, incorporating personal characteristics such as gender (**PB150**) and, for the Transport sector, a proxy for disability based on activity limitations (**PH030**). These variables are analysed among individuals aged 16 and over, applying the EU-SILC **personal cross-sectional weight for individuals aged 16+ (PB040)**. Individuals inherit the structural characteristics of their household (e.g. income category, tenure status) defined in Step 2, without recalculating household-level stratification thresholds at the individual level. Additional sociodemographic variables used to characterise the sample include: age groups derived from RB080 (grouped as 18–34 youth, 35–64 middle-age, 65+ elders), location (urban/rural) via DB100, employment status via PL031, chronic condition via PH020, and migrant background via RB280, RB290, PB230 and PB240. For the Housing and Energy sectors, tenancy status (HH020) and the ability to keep the home adequately warm (HH050) are also captured. These additional variables support the sociodemographic profiling in Section B of the questionnaire and the DG analyses in Section 3 but are secondary to the core stratification quotas applied in Prolific.

Sector-specific stratification logic.

Housing and Energy sectors. For the Housing and Energy sectors, the stratification reflects the multidimensional nature of vulnerability identified in WP2, combining structural household conditions with individual characteristics. Following the initial population-based allocation across countries (RB050), two household-level variables define structural strata within each country: total disposable household income (HY020) and housing tenure status (HH021), both calculated using the household cross-sectional weight DB090. Tenure status is included because owner-occupancy vs. renting status is directly relevant to vulnerability in relation to PV installation and housing retrofitting policies. Stratification is then completed at the individual level using gender (PB150), with distributions estimated among individuals aged 16+ weighted by PB040.

Transport sector. In the Transport sector, the stratification reflects the specific vulnerability pathways related to mobility constraints and individual capacities. After the initial population-based inter-country allocation (RB050), household-level stratification is conducted using total disposable household income (HY020) weighted by DB090. Housing tenure is not included at this stage, as it was identified as less directly relevant for transport-related vulnerability. At the individual level, two variables are used: gender (PB150) and a disability proxy (PH030 – limitation in activities due to health problems), both analysed among individuals aged 16+ weighted by PB040. Individuals inherit the household income category from the previous step. The inclusion of a disability proxy in the Transport sector (but not in Housing/Energy) reflects the particular relevance of mobility limitations as a vulnerability dimension in the context of EV acquisition policies.

Table 4 summarises the full stratification scheme, showing for each sector the stratification stage, the EU-SILC variable, the level of analysis, the expansion factor applied, and the universe represented.

Table 4. EU-SILC stratification variables by sector, level and expansion factor

Sector	Stratification stage	EU-SILC variable	Variable description	Level of analysis	Expansion factor	Universe represented
All sectors	Inter-country allocation (Step 1)	**RB050**	Personal cross-sectional weight; country of residence	Individual	**RB050**	Total population (2018 data for all 6 countries)
Housing	Household stratification (Step 2)	**HY020**	Total disposable household income	Household	**DB090**	Households
		HH021	Housing tenure status	Household	**DB090**	Households
	Individual stratification (Step 3)	**PB150**	Gender	Individual (16+)	**PB040**	Population aged 16+
Energy	Household stratification (Step 2)	**HY020**	Total disposable	Household	**DB090**	Households

Sector	Stratification stage	EU-SILC variable	Variable description	Level of analysis	Expansion factor	Universe represented
			household income			
		HH021	Housing tenure status	Household	**DB090**	Households
	Individual stratification (Step 3)	**PB150**	Gender	Individual (16+)	**PB040**	Population aged 16+
Transport	Household stratification (Step 2)	**HY020**	Total disposable household income	Household	**DB090**	Households
	Individual stratification (Step 3)	**PB150**	Gender	Individual (16+)	**PB040**	Population aged 16+
		PH030	Disability proxy (limitation in activities due to health problems)	Individual (16+)	**PB040**	Population aged 16+

Prolific quota filters were set using available platform options to approximate these distributions. The actual quota values for every country × income × tenure/disability × sex combination are listed in full in Annex VII (Prolific Sub-Studies Quota Specification — Recruitment Plan), and the per-study Prolific configurations are recorded in Annex VIII; the EU-SILC population distributions from which the targets were derived are documented in Annex V. The LimeSurvey–Prolific integration was implemented via the Prolific API, which enables automated participant routing and quota management, linking survey responses directly to Prolific participant IDs for quality control. Eurostat population distributions served as the benchmark for stratification targets, while EU-SILC was used to compute specific percentage thresholds per variable at each analytical level (population, household, individual) using the expansion factors described above. The achieved distributions were verified post-collection against the Eurostat/EU-SILC benchmarks.

Surveys were launched using sub studies to assure quotas to manage data collection progressively and allow for real-time adjustments. Each wave targeted one sector (Energy, Housing, or Transport) per country, with a total sample goal of 456 respondents per sector across all countries. Quotas were applied dynamically within Prolific's interface to enforce sociodemographic balance: quota limits per sociodemographic variable were enforced dynamically based on the Eurostat/EU-SILC benchmarks calculated by SV.

2.3. Data collection process

2.3.1. Identification of indirect indicators for the predictive framework

Editorial note (addressing reviewer comments by O. Salomon and I. Ramos). Reviewers have noted that this subsection, which concerns the predictive framework primarily developed in T4.3, sits somewhat outside the core survey-design-and-collection scope of D4.1. It is retained here in a deliberately CONDENSED form purely to document the forward-link that justifies the choice of T4.1 survey variables; the full operational treatment of indirect indicators is deferred to D4.3, and a fuller description is also available in the project's predictive-framework working documentation. Readers focused on T4.1 survey design and data collection may skip to the next subsection without loss of continuity.

Why this subsection is included here and what it covers. This subsection describes the CONCEPTUAL FRAMEWORK that links the T4.1 survey data (this deliverable) to the predictive modelling work planned in Task 4.3 (whose primary deliverable is D4.3). It is included in D4.1 — rather than deferred entirely to D4.3 — for two reasons: (i) the variables collected in Sections A and B of the T4.1 questionnaire were selected specifically to serve as 'direct' (micro-level) inputs to this framework, and a documented forward-link to the macro-level (indirect) layer is therefore needed in D4.1 to justify the choice of survey variables; and (ii) the operational decisions taken in T4.1 (sampling stratification, sociodemographic cross-check, Likert-scale construction) constrain what can be done downstream in T4.3, so the reader of D4.1 needs visibility on the downstream use to interpret these decisions. The full operationalisation of the indirect-indicator framework — the macro-level indicator mapping, statistical selection procedure, and integration with the predictive model — will be detailed in D4.3.

Formally, this approach can be expressed as (Eq.1):

$$\text{Perceived Hazard}_i = \beta_0 + \beta_1(\text{Individual/Household characteristics}) + \beta_2(\text{Contextual indicators}) + \epsilon_i \qquad \text{(Eq.1)}$$

Where:

β_0 = (a) the population-level average.

β_1 (direct variables) = (b) group-specific deviations at the micro level.

β_2 (indirect variables) = the macro-level contextual part of those deviations.

ε_i = (c) unique individual variance.

In this formulation:

- Section A and B variables of the Prolific survey represent direct variables, corresponding to the micro-level factors describing disadvantaged group characteristics (e.g., income, tenure status, building type, household composition).
- Indirect variables, in turn, are macro-level indicators obtained from public databases (e.g., Euro-stat, EU-SILC, OECD), which can be conceptually derived from or are statistically correlated with those direct variables. They reflect broader contextual conditions—economic, infrastructural, or environmental—that are expected to influence or moderate risk perception patterns.
- The residual component ϵ_i captures individual-specific differences in perception that cannot be explained by either of these levels.

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- The residual component captures individual-specific differences in perception that cannot be explained by either of these levels.

The operationalisation of this logic proceeds in three steps (summarised in Table 1):

1. Thematic alignment: Each hazard domain from Section C (e.g., financial issues, health and safety, infrastructure reliability) is matched with clusters of conceptually related direct variables from Sections A and B.
2. Indicator mapping: For each thematic cluster, potential public indicators are identified from the literature sources and EUROSTAT databases, which reflect equivalent or derived constructs at the regional or national scale (e.g., household energy expenditure, housing cost overburden, heating degree days).
3. Selection and validation: Only indicators that are regularly updated, available through open databases, and relevant to the European context are retained, with a maximum of ten indirect variables per sector (Energy, Housing, and Transport).

This layered approach ensures that the model integrates both individual vulnerability factors and structural contextual conditions, enabling the predictive algorithms (e.g., Naïve Bayes classifier) to distinguish between the different sources of variation in risk perception.

To identify the most useful and potentially strongest indirect variables (regional indicators), the selection process focuses on finding those indicators from the literature sources and EUROSTAT databases that most closely mirror or represent the same underlying constructs as the direct variables (disadvantaged group characteristics captured in Sections A and B of the survey). In psychometric terms, the optimal indirect predictors are those that share the highest conceptual and statistical variance with their corresponding direct variables. That is, they reflect similar socio-economic, infrastructural, or environmental conditions but at a higher level of aggregation (e.g., regional or national level). Consequently, these indicators are expected to explain the greatest proportion of variance in respondents' perceptions of hazards and perceived severity of negative impacts (Section C outcomes).

Table 5. Steps in the indirect indicator identification and selection.

Steps	**Description**
1. Thematic mapping of direct variables	Group the survey variables from Sections A and B (capturing disadvantaged group characteristics) into thematic clusters that conceptually align with the hazard themes from Section C of the questionnaire (e.g., financial issues, regulatory barriers, infrastructure and reliability, health and safety). This establishes the conceptual bridge between individual-level predictors (direct variables) and perceived risks (hazard outcomes)."
2. Identification of indirect variables (regional indicators)	For each hazard theme and its corresponding DG thematic cluster, identify publicly available indicators (from Eurostat, EU-SILC, OECD, etc.) that represent the same underlying socio-economic or

	infrastructural dimensions at the regional level. These indicators are considered indirect variables and serve as contextual predictors that may explain variance in hazard perception scores."
3. Filtering by dataset availability (API compatibility)	Cross-check the shortlisted indicators against the Eurostat dataset selection table (T5.5). Only those indicators that are included in the available datasets and updated regularly will be retained for integration into the predictive model.

This completes the survey design by explicitly linking the collected direct micro-level data (Sections A/B) and hazard perceptions (Section C) to the macro-level indirect indicators required for the predictive modelling in Task 4.3 and integration into the FITTER Digital Platform.

2.3.2. Monitoring Response Rates

Throughout the recruitment period, three operational metrics were continuously monitored using Prolific's dashboard and custom scripts to detect anomalies in real-time:
(i) Completion rate — defined as the proportion of participants who, having opened the survey link in Prolific, completed all sections of the LimeSurvey questionnaire. The operational target was >80% completion per sub-study (used as an early warning indicator: a sub-study consistently below this threshold was flagged for investigation, e.g., to check for survey logic errors or unclear questions). The actual completion rates averaged 65–75% across sub-studies (i.e., below the early-warning target). 'Completion rate' here does NOT refer to the fraction of platform members who responded within a given time window.
(ii) Average completion time — expected at 10–15 minutes per sector based on the Irish pilot. Sub-studies with an average completion time deviating by more than ±5 minutes from this expected range (i.e., below 5 minutes or above 20 minutes on average) were flagged as a potential indicator of either rushed responding (too fast) or technical issues / confusion (too slow). This is an OPERATIONAL early-warning indicator during fieldwork; the per-respondent minimum-completion-time filter used for post-collection quality control is a separate, stricter test defined in Section 2.3.2 (Step 2).
(iii) Dropout rate — proportion of participants who started the survey but did not submit a completed response. Operational target: <20% dropout per sub-study (above this threshold a sub-study was flagged for investigation).
Responses were collected through LimeSurvey, integrated with Prolific via the Prolific API for real-time monitoring and automated participant-status updates (approval, rejection, bonus payment if applicable). The LimeSurvey–Prolific integration is documented in Annex VIII (Prolific Sub-Studies Configuration Files).

Sample composition limitations: low eligibility of sociodemographic targets on Prolific.

During fieldwork, several target sociodemographic profiles defined in the stratification scheme (Section 2.2.4 and Annex VII) proved difficult or impossible to recruit on the Prolific platform.

Geographical composition of the Prolific panel. The figure 333,736 active participants visible in the Prolific dashboard screenshots (Annex IX) refers to the GLOBAL Prolific panel. According to Prolific's publicly available platform statistics, this global panel is heavily concentrated in the United States, the United Kingdom, Canada, and Australia, which together account for the large majority of active panel members. The active-participant pool resident in the six FITTER use-case countries (Germany, Hungary, Ireland, Italy, Portugal, Spain) is consequently a much smaller subset of this global figure — and the per-country pool varies sharply: at the time of fieldwork the active panel in Italy, Spain and Germany was in the order of tens of thousands of members, whereas the active panel in Portugal, Ireland and especially Hungary was

substantially smaller (low thousands or fewer). This geographical concentration of the Prolific panel outside the EU is the primary structural reason why some country-specific quotas could not be fully met (see Table 6 below).

Stratification variables actually used. To avoid any confusion: the sociodemographic variables used as PROLIFIC PRESCREENER FILTERS (i.e., as recruitment quota variables) were the three EU-SILC-aligned variables agreed with the consortium: Country, Income (above/below the country-specific median equivalised disposable income), and Sex (Male/Female), plus one sector-specific variable — Tenancy status (tenant/homeowner) for Energy and Housing, and Disability status (yes/no) for Transport. The complete enumeration of the 152 countries × income × demographic × sex combinations and the target respondent counts per combination is documented in Annex VII (Prolific Sub-Studies Quota Specification — Recruitment Plan). Other sociodemographic variables collected in Section B of the questionnaire (Age, Citizenship, Education, Urbanity, Disability for Energy/Housing, Ethnicity, LGBTIQA+ status, Employment) were NOT used as Prolific quota variables, but were collected ex-post from the questionnaire responses and used only as analytical subgroups in the ANOVA comparisons reported in Section 3.2 and Section 4.

Operational mitigations applied during fieldwork. To partially mitigate the panel-composition limitation, the consortium implemented the following operational measures (all documented in Annex VIII):

(a) Multiple complementary sub-studies per country–sector–quota cell (76 sub-studies in total) so that each (country × income × sex × tenancy/disability) combination could be opened and closed independently as quotas were filled or exhausted.

(b) Extended recruitment windows for hard-to-reach combinations (some sub-studies remained open for several weeks until either the target was reached or the eligible pool was demonstrably exhausted).

(c) Country-specific per-participant compensation. The base hourly compensation rate offered on Prolific was set at the platform's recommended minimum of £8.00/hour (plus VAT), in line with Prolific's ethical pay guidelines. For a small number of particularly hard-to-reach country–sector combinations (e.g., Hungary in Transport, Ireland in Housing), the per-participant compensation was incrementally increased above the platform default during fieldwork in order to incentivise completion among the limited eligible pool. The exact per-study compensation rates ultimately applied are recorded in Annex VIII (Prolific Sub-Studies Configuration Files). Despite this incremental incentivisation, the eligible-pool ceiling remained a binding constraint for several country–sector cells.

Combinations not covered. The full reconciliation of planned vs achieved respondents per country–sector–quota combination is provided in the new summary table (Table 6 below). The most consequential coverage gaps were Hungary in Energy and Transport (planned 22 and 38 respondents respectively; 0 recruited in both), and Germany in Housing (planned 384; 198 recruited, 52% of plan). These gaps are NOT a failure of fieldwork execution; they are a direct consequence of the Prolific panel's geographical composition described above.

Validation of the complete absence of Hungary (validated against both the Prolific recruitment data and the LimeSurvey collected data). The recruitment plan (Annex VII) targeted Hungary in two sectors: Energy (22 respondents planned) and Transport (38 respondents planned). The complete absence of any Hungarian respondent in the final dataset was verified directly against the two independent primary-data sources:

(i) Prolific recruitment data (Annex VIII / Annex IX). Across ALL 75 Prolific sub-studies launched during T4.1 fieldwork, the 'Country of residence' field recorded ZERO participants resident in Hungary. The 1,638 total submissions across all studies break down by country of residence as: Italy 380, Germany 367, Spain 281, Portugal 76, Ireland 28, and a further 506 records with consent subsequently revoked — but Hungary 0. Not a single Hungary-resident participant could be recruited on the platform, despite Hungary-targeted sub-studies being launched and left open.

(ii) LimeSurvey collected data (Annex X). The questionnaires included a dedicated branching question for respondent's resident in Hungary ('Please indicate the region in Hungary where you live'). Across the three raw LimeSurvey exports (Energy 584, Housing 434, Transport 554 raw responses), this Hungary-region question received ZERO responses, and the 'country of residence' question recorded ZERO Hungary selections. This independently confirms the Prolific finding.

Root cause. The cause is the geographical composition of the Prolific panel (Section 2.3.1, opening paragraph): Hungary has one of the smallest active-participant pools of any EU country on Prolific, and once the platform's eligible Hungarian pool — already very small — was further filtered by the required sociodemographic quota variables (income band, sex, and tenancy/disability), the number of eligible Hungarian participants effectively fell to zero. This is a hard structural ceiling of the data-collection platform, not a fieldwork-execution failure: the Hungary sub-studies were created and made available, but the platform simply contained no eligible participants to fill them. Consequently, Hungary is absent from every analytical sector and cannot be included in the country-level analysis (Section 3.3). The implication for the project — that an alternative recruitment channel (e.g., a national panel provider or partner-led recruitment) would be required to obtain Hungarian data — is carried forward as a recommendation for T4.3 and for any future FITTER primary data collection.

Table 6. Planned versus achieved respondents per country and sector.

Sector	**Country**	**Planned (n)**	**Achieved (n)**	**Coverage**
Energy	Germany	174	149	86%
Energy	Hungary	22	0	0% (none recruited)
Energy	Ireland	10	11	110%
Energy	Italy	129	129	100%
Energy	Portugal	22	22	100%
Energy	Spain	99	97	98%
Energy	**TOTAL**	**456**	**408**	**89%**
Housing	Germany	384	198	52%
Housing	Hungary	—	0	— (not in plan)
Housing	Ireland	23	17	74%
Housing	Italy	—	0	— (not in plan)
Housing	Portugal	49	49	100%

Housing	Spain	—	0	— (not in plan)
Housing	**TOTAL**	**456**	**264**	**58%**
Transport	Germany	—	0	— (not in plan)
Transport	Hungary	38	0	0% (none recruited)
Transport	Ireland	—	0	— (not in plan)
Transport	Italy	237	205	86%
Transport	Portugal	—	0	— (not in plan)
Transport	Spain	181	154	85%
Transport	**TOTAL**	**456**	**359**	**79%**

In the table above, the verified IDs are appearing after the three-step quality-control protocol of Section 2.3.2. "—" indicates the country/sector was not included in the recruitment plan; "0%" indicates a country/sector for which recruitment was planned but no valid respondents were obtained on Prolific. Note on subgroups: this table reconciles totals at the country×sector level; the subgroup-level (sex × income × tenure/disability) target-vs-achieved detail for every one of the 76 sub-studies is provided in Annex VII (Recruitment Plan) and the achieved subgroup counts are summarised in Table 7 (ANOVA, which lists n per DG) and Table 11.

Figures 3–9 below illustrate this limitation directly using the Prolific operations dashboard during the FITTER data collection period. They show, for several DG-targeted sub-studies, both (i) the partial completion rates (badge in the top-left of each study card) and (ii) the very small Eligible participants counts (e.g., 82 of 333,736; 63 of 333,736; 22 of 333,736), confirming that the limitation is not a fieldwork execution issue but a structural feature of the platform's panel composition. Figure 9 shows the parallel LimeSurvey administration dashboard, with the final response counts per sector after quality control (Energy 553, Housing 413, Transport 539). The specific sub-studies shown in these dashboard screenshots (e.g., Studies 81, 82, 85 and 86) correspond to the DG-targeted sub-study configurations listed in Annex VII (Recruitment Plan) and Annex VIII (Sub-Studies Configuration Files); the reader can consult those annexes to see the exact country × sociodemographic profile each study number targeted, together with its eligible-pool size and completion count. The full-resolution dashboard screenshots are provided in Annex IX (Prolific Recruitment Dashboard Evidence).

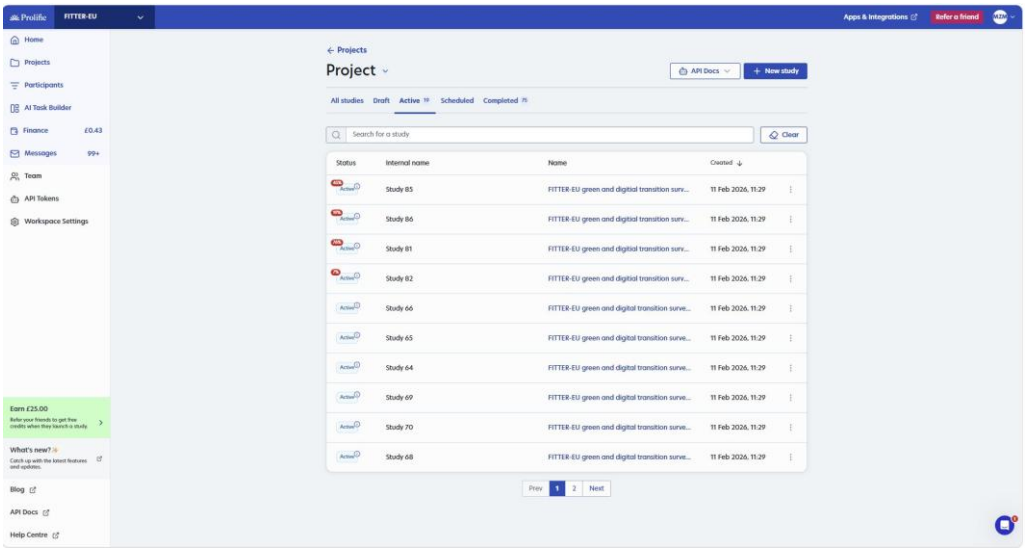


Figure 3. Prolific dashboard — Project view (page 1). DG-targeted FITTER sub-studies, with completion-rate badges indicating partial fill rates for several sociodemographic targets.

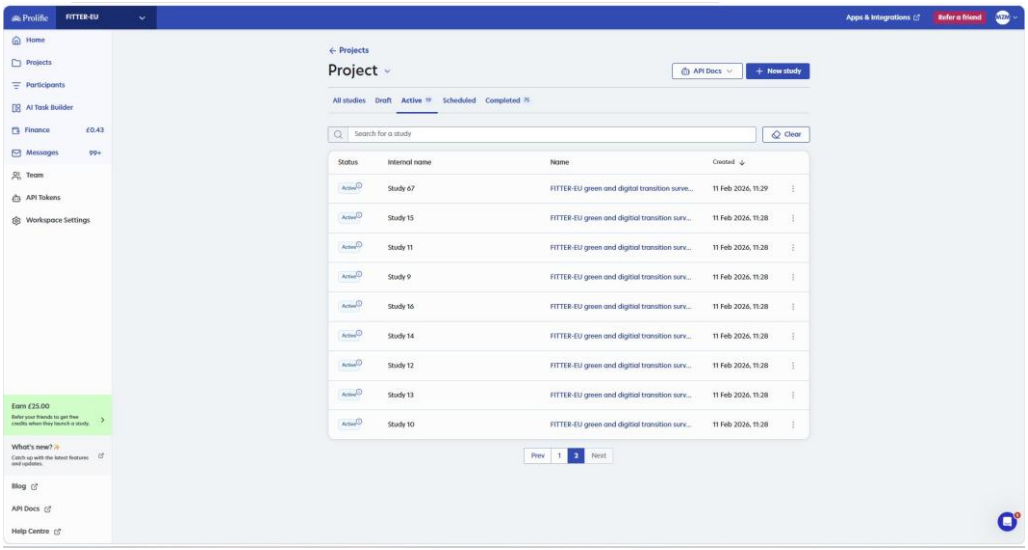


Figure 4. Prolific dashboard — Project view (page 2). Additional FITTER sub-studies, several with completion-rate constraints due to small eligible pools.

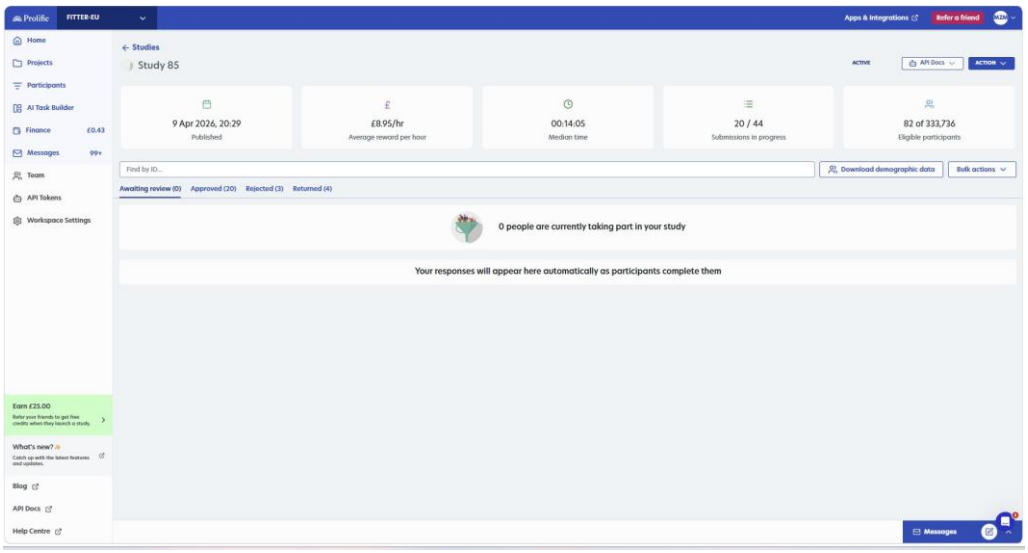


Figure 5. Prolific study detail — Study 85 (DG sub-study). Only 82 of 333,736 platform participants are eligible after applying the joint sociodemographic and country filters: 20 of 44 submissions in progress.

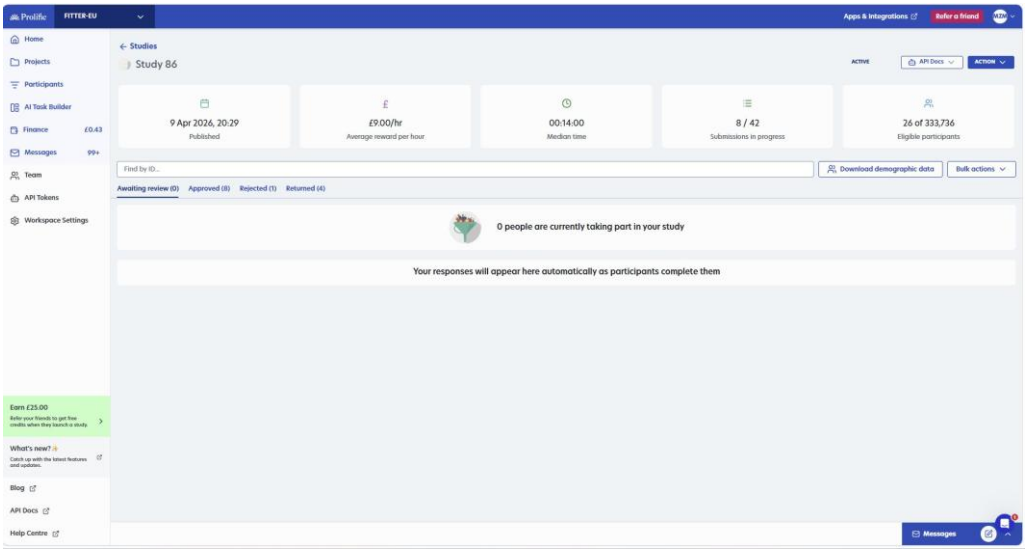


Figure 6. Prolific study detail — Study 86. Eligible pool 26 of 333,736; 8 of 42 submissions; illustrative of intersectional DG profiles with very narrow eligibility.

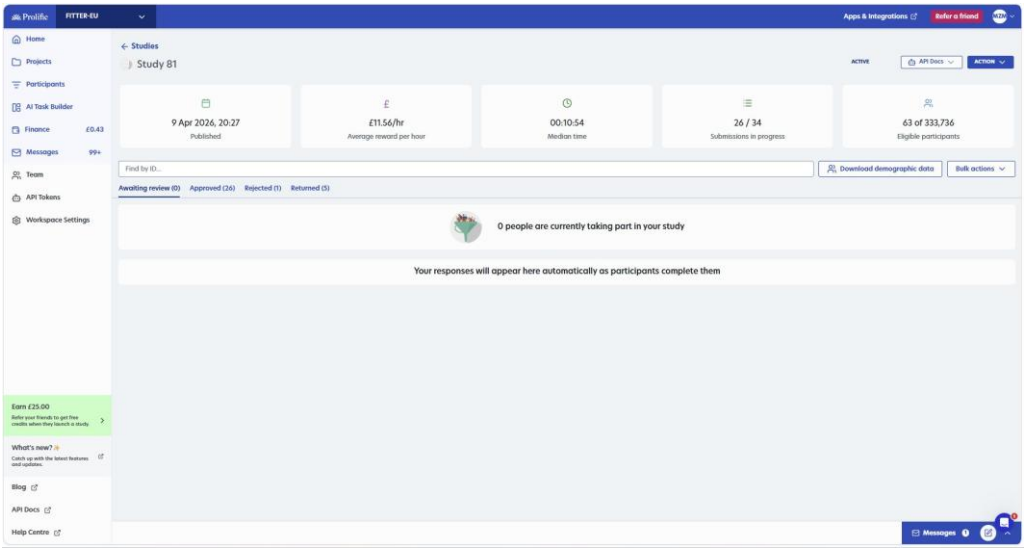


Figure 7. Prolific study detail — Study 81. Eligible pool 63 of 333,736; 26 of 34 submissions, showing recovery from constrained pool through extended recruitment.

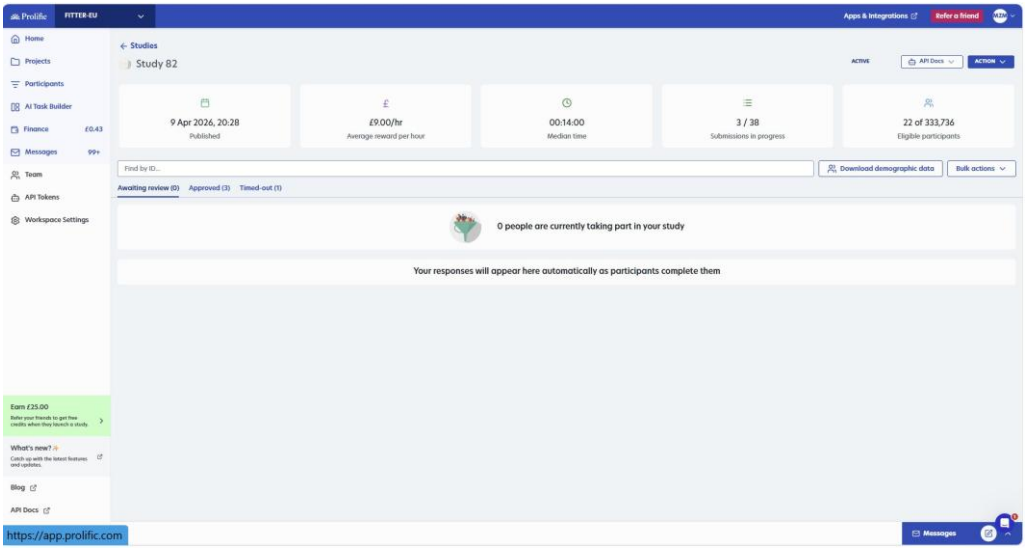


Figure 8. Prolific study detail — Study 82. Eligible pool 22 of 333,736; 3 of 38 submissions; evidence that some DG profiles are nearly impossible to fully populate on Prolific.

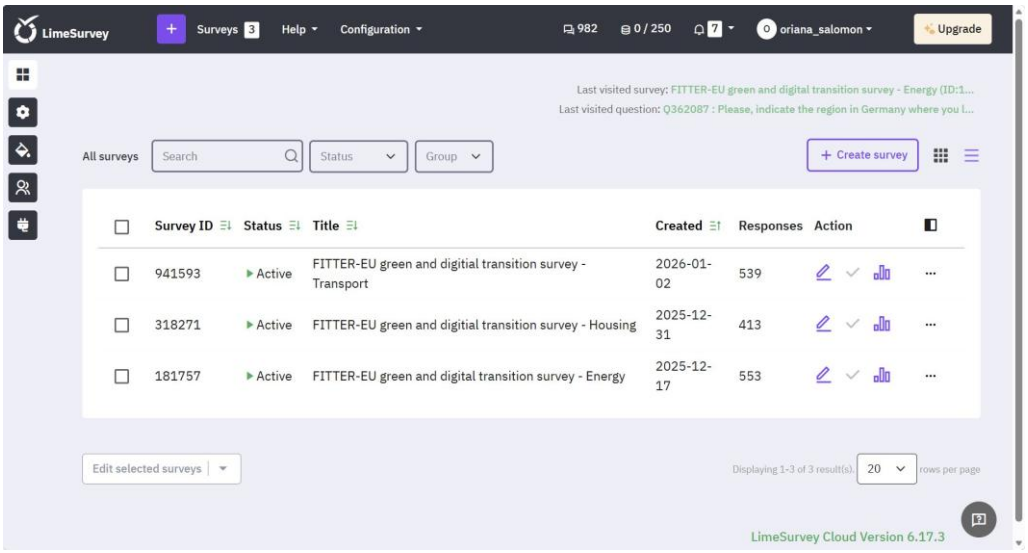


Figure 9. LimeSurvey administration dashboard — final completed responses per sector: Transport 539, Housing 413, Energy 553 (counts include responses before final cross-check exclusions; see Annex IV).

Implications for analysis. As a direct consequence of the recruitment limitations documented in Table 6 above, the analytical dataset (n=408 Energy / n=264 Housing / n=359 Transport) is unevenly distributed across countries and sociodemographic cells. In particular:

• Hungary is not represented in the analytical dataset for Energy or Transport (planned 22 and 38 respondents respectively; 0 valid respondents after quality control).

• Italy and Spain were not included in the Housing recruitment plan, and Germany, Ireland and Portugal were not included in the Transport recruitment plan, in line with the sector-specific country selection agreed in the consortium (reflecting the relative weight of each policy intervention in the use-case framework).

• Within sociodemographic cells, the QUOTA variables used in the Prolific stratification (Country, Income, Sex, plus Tenancy for Energy/Housing or Disability for Transport) are well represented in the analytical sample at the planned cell sizes; however, ANALYTICAL subgroups derived from Section B of the questionnaire (Age 65+/elders, Citizenship, Education, Urbanity, Ethnicity, LGBTIQA+ status, Employment status) — which were not used as quota filters — have smaller-than-convenient counts in some sectors (e.g., respondents aged 65+ are n=13 in Energy and n=2 in Housing). These ANOVA results are reported in Section 3.2 and in Table 7 with the subsample size shown alongside, and are flagged in the "Power/Notes" column whenever the subgroup is below the 30-respondent threshold recommended for parametric comparisons; results for such low-n subgroups are NOT interpreted as conclusive evidence of a population-level difference.

Justification of the analytical approach. Given the recruitment limitations documented above, the analytical strategy of D4.1 prioritises:

(a) Robust descriptive statistics (mean, standard deviation, variance) for perceived probability and severity per hazard and per sector, computed on the full valid sample (Section 3.1, Tables 8, 9 and 10, Figures 2–4).

(b) Pairwise one-way Analysis of Variance (ANOVA) tests comparing the mean perceived hazard score between a given sociodemographic subgroup and the rest of the sector sample, for DG comparisons where

the subgroup size is statistically adequate (≥ 30; see Sections 3.2 and 4). The ANOVA methodology — formal model, assumptions, test statistic, p-value interpretation, and sample-size adequacy — is described in detail in Section 4.1 (Methodology of the analysis of variance). The reader is referred to that section for the methodological justification of the test, and to Table 7 (Section 4.3) for the complete set of pairwise results.

Subgroups with subsamples below the 30-respondent threshold are documented in Table 7 with a low-power flag and are NOT over-interpreted in the results. Where the structural Prolific limitations prevented adequate recruitment of a DG profile (notably Hungary as a country), targeted oversampling and complementary qualitative data collection are recommended for Tasks 4.2 (mitigation measures) and 4.3 (predictive modelling), so that the policy mitigation and modelling work in WP4 can still account for the most vulnerable groups even when their direct quantitative perception cannot be precisely estimated from T4.1 alone. References to ANOVA methodology in this deliverable are confined to Sections 3.2 and 4.

2.3.3. Removal of Low-Quality Responses

Post-collection, a three-step quality-control protocol was applied to convert the raw LimeSurvey export into the analytical dataset used in Sections 3 and 4. The protocol ensures that all reported analyses are based on responses that are (a) consistent with the participant's Prolific profile, (b) submitted with enough engagement time to allow a considered answer, and (c) unique (one row per participant). Each step is described below, with a worked example to clarify the operational decisions made.

Step 1 — Sociodemographic cross-check. Each Prolific participant ID was cross-referenced against three sociodemographic attributes recorded BOTH in the Prolific platform profile (the values used to filter participants into the relevant sub-study) AND in the participant's LimeSurvey answers (the values self-reported during the actual survey). The three cross-checked variables are Disability status, Sex/Gender, and Tenancy status (Housing and Energy sectors only). Respondents whose Prolific-profile values did not match their LimeSurvey answers were excluded as potentially unreliable.

Worked example — Step 1. A Prolific participant filtered into a 'Female, Tenant, Below-median income' sub-study (per their Prolific profile) but who answered 'Male, Homeowner' in the LimeSurvey questionnaire was flagged as a mismatch and excluded. This catches both careless responding and possible mis-targeted recruitment.

The full cross-check audit is documented in Annex IV (Sociodemographic CrossCheck Verification Report). The resulting verified-ID pool per sector is: Energy 421 verified IDs, Housing 264, Transport 374.

Step 2 — Minimum completion-time filter. The lower-bound completion time was empirically established by examining the duration distribution of all responses across the three sectors. The Housing questionnaire — the longest and most complex of the three (13 hazards in Section C, plus 13 sociodemographic questions in Section B) — exhibited a minimum observed completion time of 3.92 minutes (5th percentile 5.99 min, median 11.07 min). Because Housing constitutes the slowest-floor sector in terms of plausible completion time, 4 minutes was adopted as the analytical lower bound for valid completion across all three sectors. Any verified-ID response in Energy, Housing, or Transport completed in under 4 minutes was excluded.

Note on the 4-minute vs 5-minute threshold (clarification). Two candidate thresholds were discussed within the consortium during the methodological design: a 4-minute floor anchored on the empirical Housing minimum, and a more conservative 5-minute floor. The 4-minute threshold was ultimately

adopted because (i) it is empirically supported by the data (no Housing response below 3.92 minutes; this is the natural lower bound for a thoughtful completion of the longest questionnaire) and (ii) it is more inclusive of legitimate fast respondents in the two shorter sectors (Energy and Transport). For full transparency, the impact of moving to a 5-minute floor would be: Energy n=408→384 (−24), Housing n=264→260 (−4), Transport n=359→329 (−30). The qualitative conclusions of all ANOVA tests in Section 4 are robust to this choice (the same five significant comparisons remain significant at α=0.05 under the 5-minute filter).

Worked example — Step 2. A Transport respondent who submitted the entire questionnaire in 2 minutes 47 seconds was excluded under this rule, as a sub-4-minute completion is incompatible with thoughtful reading and Likert scoring of 6 hazards × 2 dimensions plus 13 Section-B questions.

Exclusions under this rule: Energy 13 responses, Housing 1 response, Transport 15 responses (29 verified-ID responses excluded in total for sub-4-minute completion).

Step 3 — One-row-per-PID deduplication. A small number of verified Prolific IDs were associated with two valid post-4-min submissions in the LimeSurvey export (Energy: 1 PID; Housing: 0 PIDs; Transport: 1 PID). Importantly, the LimeSurvey export used here contains ONLY fully completed and submitted responses (aborted or partial attempts are never exported as completed records and do not enter this dataset); therefore, BOTH submissions for such a PID are complete, valid, post-4-minute questionnaires. For each such PID, the EARLIEST complete submission (by timestamp) — the participant's first genuine completion — is retained and the later duplicate is dropped, so that no participant is double-counted.

Worked example — Step 3. A Prolific PID that appears as two rows in the Transport export — one submitted at 14:23 with valid completion time of 11 minutes and a second submitted by the same participant at 17:08 with a completion time of 9 minutes — is collapsed to a single row corresponding to the earlier (14:23) submission.

Final analytical sample. Applying Steps 1, 2, and 3 in sequence yields the dataset used for ALL descriptive statistics in Section 3, all ANOVA tests in Section 4, and all figures from Figure 2 onwards:

- Energy: 408 unique verified respondents (from 421 verified IDs)
- Housing: 264 unique verified respondents (from 264 verified IDs)
- Transport: 359 unique verified respondents (from 374 verified IDs)
- TOTAL: 1,031 unique verified respondents across the three sectors.

Transparency — post-hoc achieved statistical power. As reported in the Abstract and in Section 2.2.3, the achieved post-hoc statistical power at the planned effect-size detection criterion (α=0.05, σ ≈ 1.2 on the 1–5 Likert scale, margin of error e ≈ 0.16) is: Energy 75%, Housing 57%, Transport 70%. Equivalently, holding power at the planned 80% level, the detectable margin of error widens from the planned ±0.16 to ±0.17 (Energy), ±0.21 (Housing), and ±0.18 (Transport). These values are BELOW the originally targeted 80% / ±0.16 design, especially for Housing. The Cochran formula's assumption of simple random sampling is also not strictly satisfied (see Section 2.2.3 limitations note); we treat both departures as transparently documented constraints, not as grounds for invalidating the descriptive results which remain the principal contribution of this deliverable. Inferential ANOVA results in Section 4 are reported with the relevant subgroup sample sizes and a power flag, so the reader can judge the confidence to attach to each finding.

The analytical dataset is therefore the intersection of (i) the verified-ID list in Annex IV, (ii) responses with

completion time ≥ 4 minutes, and (iii) one row per Prolific participant ID.

Completion-time distribution for all three sectors (response to the question on the lower bound). Empirical completion-time statistics on the verified responses are: Energy — minimum 2.30 min, 5th percentile 4.24 min, median 8.28 min, mean 9.37 min; Housing — minimum 3.92 min, 5th percentile 5.99 min, median 11.07 min, mean 13.06 min; Transport — minimum 2.47 min, 5th percentile 4.08 min, median 7.30 min, mean 8.40 min. Two points follow. First, although Housing is the longest questionnaire (highest median/mean time), a SINGLE uniform 4-minute floor was deliberately adopted across all three sectors for consistency; this floor sits just BELOW the 5th percentile of the two shorter sectors (Energy 4.24, Transport 4.08 min), so it removes only the most extreme fast-responders and does not over-exclude legitimate fast respondents in those sectors. Second, the mean completion times (9.37 / 13.06 / 8.40 min) are reported here alongside the minima (per the suggestion to show the average): they confirm that the typical respondent spent far longer than the 4-minute floor, so the filter targets only a small tail of implausibly fast completions.

2.3.4. Metadata Storage and Documentation

All metadata were systematically saved for auditability and further analysis. This included submission timestamps (start/end times, page durations), device information (mobile vs. desktop), IP addresses (anonymised before storage for GDPR compliance), and postcode proxies (used for regional verification without storing exact location data). Data were exported in CSV / XLSX format with unique respondent IDs linked to Prolific identifiers, ensuring full traceability between the analytical dataset and the original Prolific recruitment record.

Storage and access. The raw and analytical datasets are stored on the FITTER-EU consortium SharePoint repository operated by TU Dublin (the Data Coordinator under the DMP). Access is restricted to the four Authorised Data Processing Partners — TU Dublin, SV, AITEC and SFC — as defined in the signed DMP Annex (Annex XI of this deliverable). All other consortium partners receive only analysed and aggregated outputs. Storage compliance follows the project's DMP and the FITTER-EU Privacy Policy: anonymisation and pseudonymisation are applied at the earliest possible stage of processing; data are stored in an encrypted SharePoint folder with role-based access control; access events are logged; and any data breach would be reported in accordance with EU GDPR Articles 33–34 and the project's DMP. Survey variable coding in Sections A and B of each questionnaire follows EU-SILC conventions where applicable (using the same response categories as the corresponding EU-SILC variables), facilitating direct comparability with the EU-SILC population reference data documented in Annex V.

2.3.5. Expected Attrition and Replacement Procedures

Attrition (response loss due to incomplete questionnaires, sociodemographic cross-check failures, and timing exclusions) was anticipated based on Prolific benchmarks for similar academic surveys. The recruitment plan therefore aimed at a collection volume above the design target of n=456 valid responses per sector, with expected attrition of approximately 15–20% from the gross collected number to the final analytical sample after quality control. Top-up sub-studies were launched during fieldwork to compensate for attrition in real time, conditional on the eligible-pool availability per country (see Table 4).

Actual attrition observed (verified-ID dataset). The cascade from gross collection to final analytical sample for each sector was as follows:

- Energy: 553 responses initially recorded → 421 passed the Step-1 CrossCheck (132 mismatches removed,

24%) → 408 passed the Step-2 ≥4-minute filter (13 additional exclusions) → 408 after Step-3 deduplication (1 duplicate consolidated). Final n = 408 (74% of gross collection).

• Housing: 413 responses initially recorded → 264 passed the Step-1 CrossCheck (149 mismatches removed, 36%) → 264 passed Step-2 (1 additional exclusion within 264 already verified) → 264 after Step-3 deduplication. Final n = 264 (64% of gross collection).

• Transport: 539 responses initially recorded → 374 passed Step-1 (165 mismatches removed, 31%) → 359 passed Step-2 (15 additional exclusions) → 359 after Step-3 deduplication (1 duplicate consolidated). Final n = 359 (67% of gross collection).

The attrition is dominated by Step 1 (the sociodemographic CrossCheck) — i.e., by respondents whose Prolific profile attributes did not match their LimeSurvey answers — rather than by timing exclusions. This pattern is consistent with the literature on online panel data quality. The full per-sector cascade is documented in Annex IV (CrossCheck Report).

This recruitment and quality-control workflow secured high-fidelity data while respecting the project's resource constraints and the structural geographical limitations of the Prolific panel documented in Section 2.3.1 above and Table 6 above.

Why real-time monitoring could not close the gap to n=456. Real-time monitoring (Section 2.3.1) allowed the team to detect under-filled quota cells and launch top-up sub-studies during fieldwork. However, monitoring can only redistribute recruitment effort WITHIN the limits of the eligible Prolific pool; it cannot create eligible participants where the panel is empty. The gap to n=456 is therefore explained not by insufficient monitoring but by a hard CEILING on the eligible pool for specific country–quota cells — most acutely Hungary (0 respondents recruited in any sector despite repeated study launches; see the dedicated validation in Section 2.3.1) and Germany in Housing (198 of 384 planned). Once a country's eligible pool for a quota cell is exhausted, no amount of additional monitoring, extended fieldwork or incremental compensation can recruit further respondents from that cell. The shortfall is thus a structural panel-composition limitation, quantified per country and sector in Table 6, not an operational monitoring failure.

2.3.6. Measurement and scoring (probability & severity index construction)

This subsection outlines the procedures for transforming raw survey responses into quantifiable probability and severity scores. The surveys employed structured Likert-scale questions (Section C of each questionnaire) to capture respondents' perceptions of hazards associated with the twin transition policies in the Energy, Housing, and Transport sectors. Variables in Sections A and B of the survey use the same coding conventions as the corresponding EU-SILC variables where applicable, facilitating comparability with EU-SILC population benchmarks. For Section C (hazard perception), the coding is as follows:

- **Probability**: The perceived likelihood of the hazard occurring, on a 5-point scale (1 = Very Unlikely, 2 = Unlikely, 3 = Neutral, 4 = Likely, 5 = Very Likely).
- **Severity**: The perceived negative impact if the hazard occurs, on a 5-point scale (1 = Negligible Impact, 2 = Minor Impact, 3 = Moderate Impact, 4 = Major Impact, 5 = Catastrophic Impact).

These scales were chosen for their simplicity and reliability in eliciting perceptual data, following standard practice in risk perception survey research (e.g., Likert, 1932; Eurostat, 2024; Memon et al., 2020), allowing for both single-item analysis (per hazard) and composite indices (aggregated across hazards or sectors).

Probability and Severity Indices are calculated as follows, for each hazard h and sector s:

Mean Probability Score for hazard h:
$\bar{P}_h = (1/n) \times \Sigma_i P_{ih}$

where P_{ih} is the probability rating (1–5) given by respondent i for hazard h, and n is the number of valid respondents for that hazard.

Mean Severity Score for hazard h:
$\bar{S}_h = (1/n) \times \Sigma_i S_{ih}$

where S_{ih} is the severity rating (1–5) given by respondent i for hazard h.

Standard deviations (SD_{ph} and SD_{sh}) and variances (SD^2) are also computed for each hazard as measures of dispersion across respondents. For sub-group analyses (e.g., women vs. rest), the same formulas are applied separately to each subgroup.
Scores are reported on the original 1–5 Likert scale to preserve interpretability. For the purpose of Risk Assessment Matrix visualisation (see Section 3), probability and severity scores can be combined to yield an overall risk level per hazard (Risk = Probability × Severity, normalised to a 0–25 scale).
The hazards included in each sector questionnaire are grouped thematically in Annex I. Annex I contains the full survey questionnaires for all three sectors, including the hazard categories. The survey variable dictionary (codes, variable names, question text, variable types, and response values) is provided in Annex II (Memon et al., 2020; Eurostat, 2024).

2.4. Data Analysis Methodology (overview)

The analysis pipeline applied to the cleaned analytical dataset (n=408 / 264 / 359 valid respondents in Energy / Housing / Transport) consists of two complementary layers, each detailed in subsequent sections of this deliverable: (1) Descriptive layer — per-hazard and per-subgroup mean, standard deviation and variance, computed on the full valid sample. Descriptive results characterise the hazard landscape and the perceived risk profile of each sector and each sociodemographic subgroup. They are reported in Section 3 (Results). No inferential test is applied at the descriptive level: the descriptive layer answers the question "what does the average respondent perceive?" rather than "is this difference statistically significant?". (2) Inferential layer — pairwise one-way Analysis of Variance (ANOVA) tests comparing the mean perceived hazard score (averaged across the sector hazards) between a given sociodemographic subgroup and the rest of the sector sample. ANOVA was selected because (a) the dependent variable is a continuous Likert-derived mean (the average of 1–5 perception scores across all hazards within a sector), (b) the comparison is between two mutually exclusive subsamples within a single sector at a time, and (c) the test is robust to moderate departures from normality at the subsample sizes obtained in this study. The null hypothesis tested is H_o: the mean perceived-hazard score is equal between the DG subgroup and the rest of the sector population. The chosen significance threshold is $\alpha = 0.05$. The detailed methodology, including the formal model, assumptions, test statistic, and treatment of low-power subsamples, is set out in Section 4.1. The full set of pairwise results is tabulated in Section 4.3 Table 12. Selected DG comparisons are introduced descriptively in Section 3.2, and the matching ANOVA p-value is cited there for the reader's convenience; the formal inferential interpretation is deferred to Section 4 to keep Section 3 purely descriptive. The variables used at the inferential layer are the sociodemographic variables identified in the EU-SILC stratification (Section 2.2.4 and Annex V) for which a non-trivial subgroup sample size is available after quality control (≥ 30 respondents in the DG cell, where possible). The mapping between EU-SILC

stratification variables and the analytical subgroups used here is summarised in Table 5 (Stratification) and Annex VI (D4.1 Statistical Tables).

2.4.1. Methodology of the Analysis of Variance

According to the ANOVA methodology, a *Null Hypothesis test* is applied in which:

- H_0: There is no difference in the mean between Group 1 and Group 2.

$$\mu_1 = \mu_2 \text{ Eq. 3}$$

- H_1 : Alternative hypothesis (H_1):

$$\mu_1 \neq \mu_2 \text{ Eq. 4}$$

The next table is built to test this hypothesis:

Table 7. Observations, Totals and Means for two-group ANOVA.

Observations	**Comparison between 2 Groups**		**Combined Total**
	Group 1	**Group 2**	*Both groups*
1st	y_{11}	y_{21}	
2nd	y_{12}	y_{22}	
…	…	…	
n_b-th	y_{1n_1}	y_{2n_2}	
Total	$T_1 = \sum_{p=1}^{n_1} y_{1p}$	$T_2 = \sum_{p=1}^{n_2} y_{2p}$	$T = T_1 + T_2$
Mean	$\bar{y}_1 = \frac{\sum_{s=1}^{n_1} y_{1s}}{n_1}$	$\bar{y}_2 = \frac{\sum_{s=1}^{n_2} y_{2s}}{n_2}$	$\bar{Y} = \frac{\bar{y}_1 + \bar{y}_2}{2}$

Where:

$$y_{ab} = Observation\ of\ respondent \text{ "a" } belonging\ to\ the\ group \text{ "}b\text{" Eq. 5}$$

$$a = [1, n_b]\ ;\ b = [1, 2]$$

Allowing us to calculate the next variables:

$$SSA = \text{Sum of Squares for Among groups} = \sum_{b=1}^{2} \frac{T_b^2}{n_b} - \frac{T^2}{N} \text{ Eq. 6}$$

$$SST = \text{ Sum of Squares Total} = \sum_{b=1}^{2} \sum_{a=1}^{n_b} y_{ab}^2 - \frac{T^2}{N} \text{ Eq. 7}$$

$$SSE = SST - SSA \text{ Eq. 8}$$

Being:

$$N = \sum_{a=1}^{2} n_a \text{ Eq. 9}$$

The statistical "f" (Eq. 10) is following a "F" statistical probabilistic distribution with 1 and (N-2) freedom degrees.

$$f = \frac{SSA}{SSE/(N-2)} \text{ Eq. 10}$$

So, the Null hypothesis H_0 will be rejected if:

$$f > f_\alpha(1, N-2) \text{ Eq. 11}$$

Being "α" the confident level of the test (usually 0.05).

2.4.2. Sample Size for the Analysis of Variance

The minimum number of respondents "N" needed for the ANOVA test with "q=2" categories is calculated by considering the ***Error type 1*** with a probability α, or what is equivalent with a confidence level (1- α), and the **Error type 2** with a probability β, or what is equivalent with a power of the test (1-β),), in the hypothesis test described above. Both errors can be mathematically described through Equations 12 and 13.

- **Error type 1 (Eq. 12):**

$$\alpha = P(f > f_\alpha(1, N-2) | H_0 \; is \; True) = Probability \; to \; reject \; the \; Null \; Hypothesis \; when \; it \; is \; True$$

- **Error type 2 (Eq. 13):**

$$\beta = P\left(f' < f'_\beta(1, N-2, \lambda) \middle| H_0 \; is \; False\right) = Probability \; to \; accept \; the \; Null \; Hypothesis \; when \; it \; is \; False$$

Where "f'" is following a non-central F probability distribution with a *non-centrality parameter*:

$$\lambda = \frac{n_1.\alpha_1 + n_2.\alpha_2}{2.\sigma^2} \text{ Eq. 14}$$

Being:

$$\alpha_1 = (\mu - \mu_1) \; ; \; \alpha_2 = (\mu - \mu_2) \text{ Eqs. 15}$$

$$\sigma^2 \approx \frac{SSE}{(N-2)} \text{ Eq. 16}$$

With:

$$\mu = \frac{\mu_1 + \mu_2}{2} \text{ Eq. 17}$$

These 2 errors can be depicted through the the Energy-by-country figure below below:

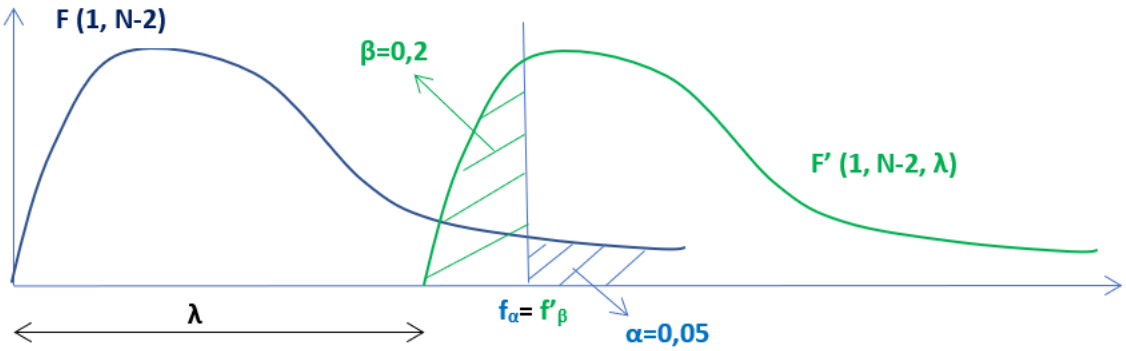


Figure 10. F and F' probability distributions with Error type 1 and Error type 2

Sample size calculatorBy fixing "α" (=0,05) and "β" (=0,8) and given the values of "q=2" and "λ", "N" (minimum sample size to carry out the ANOVA test with this confidence level and power) can be calculated. Statistics Kingdom () was used for these calculations. the Housing-by-country figure below and 21 shows how this sample size for ANOVA test was calculated.

A minimum sample size of N=128 respondents (in case of n1=n2, each group: 64 respondents) is needed to achieve a 95% confidence level and 80% power of the test, considering a medium Size Effect (λ) =0.25.

Justification of the medium effect size (λ = 0.25). The effect size λ = 0.25 corresponds to Cohen's (1988) conventional definition of a MEDIUM effect for F-tests / ANOVA (Cohen's f: small = 0.10, medium = 0.25, large = 0.40). In the absence of prior empirical estimates of the magnitude of perceived-hazard differences between disadvantaged groups for these specific twin-transition interventions (this being the first such survey in the FITTER use-case countries), the medium effect size is adopted as the conventional, defensible default: large enough to correspond to a difference of practical policy relevance, yet small enough not to miss moderate but meaningful group differences. A smaller assumed effect (λ = 0.10) would have required a far larger per-group sample (impractical on Prolific for the smaller countries); a larger assumed effect (λ = 0.40) would have risked overlooking policy-relevant moderate differences.

Why this subgroup sample-size estimate (N=128) was NOT used to set the recruitment quotas. The N=128 figure is the sample size required to detect a medium effect in a single pairwise subgroup ANOVA (two groups of 64). It was NOT used as the quota-setting target because the quota design pursued a prior objective: population REPRESENTATIVENESS. The Prolific quotas were set to reproduce the EU-SILC sociodemographic distribution of each country (Section 2.2.4 and Annex V), so the achieved sample would be representative of the national populations on the stratification variables — not to guarantee 64 respondents in every conceivable subgroup of every possible DG comparison. Designing quotas to guarantee N=128 for every subgroup simultaneously (across sex, income, age, tenancy, disability, urbanity, etc.) would have required a vastly larger total sample, incompatible with the project's resources and with the limited eligible Prolific pools in the smaller countries. The consequence — that some ex-post analytical subgroups (e.g., Elders) fall below the 64-per-group threshold — is acknowledged as a limitation in Section 2.3.1 and flagged per comparison in the ANOVA results (low-power flag).

In this case **$n_1 \neq n_2$**, so we need to consider the available sample size of the minority group (n_1 if $n_1<n_2$; or n_2 if $n_2<n_1$).

When both, **n_1 and n_2 are >64**, we can consider that the results show in Table 7 have a 95% confidence level, considering a critical value of $\mathbf{f_{\alpha=0,05}(1, 460 - 2)}$**=3,862**; and achieving, at least, 80% power in the test.

When the sample size of the minority group **(n_i) is <64**, we will keep 95% confidence level of the ANOVA test by considering the same condition to reject the *Null Hypothesis* H_0 [$f>f_{\alpha=0,05}(1, 460-2)$], but the power of the test is recalculated considering $N=2.n_i$, the same value for the Size Effect, this means medium Size Effect (λ)=0.35, and a critical value of $f=f_{\alpha=0,05}(1, 2.n_i-2)$. In these cases, these recalculated values for the power of the test are indicated in the Table 7.

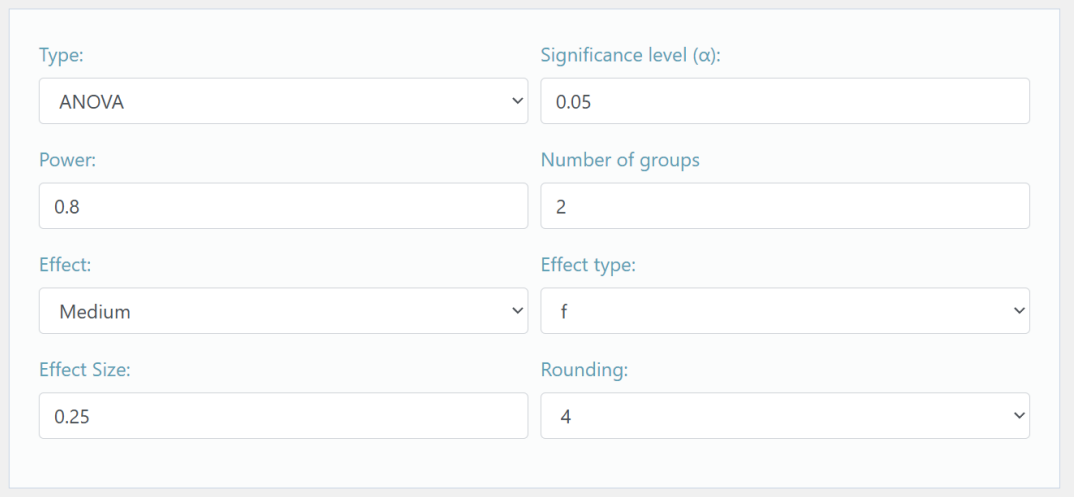


Figure 11. ANOVA sample size calculator

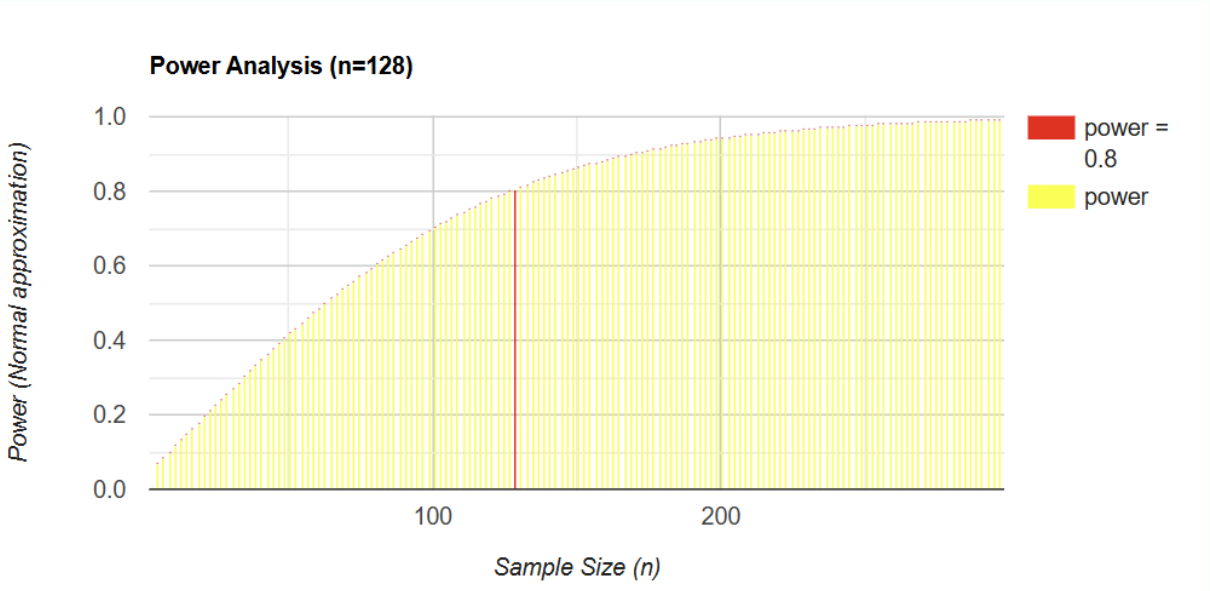


Figure 12. Power analysis of ANOVA test

3 Results

This section presents the main findings from the T4.1 surveys, based on the cleaned analytical dataset: n=408 for Energy, n=264 for Housing, and n=359 for Transport (total = 1,031 unique verified respondents). The dataset was obtained by applying the three-step quality-control protocol described in Section 2.3.2 (sociodemographic cross-check, ≥4-minute completion filter, and one-row-per-participant deduplication).

Definition of terms used in this section.

• Sector — One of the three priority twin-transition policy interventions assessed in T4.1: Energy (PV installations), Housing (retrofitting) or Transport (electrification of vehicles). Each sector has its own questionnaire and its own hazard catalogue (7 hazards in Energy, 13 in Housing, 6 in Transport).

• Overall sector mean — The arithmetic average of the per-hazard mean probability (or mean severity) score across ALL hazards within a sector, computed on the FULL valid sample of that sector. For example, the Energy 'overall sector mean probability' is the average of the 7 individual hazards mean probability scores across all 408 Energy respondents. This is the headline figure characterising the sector as a whole. Per-hazard means (within a sector) are reported in Tables 8, 9 and 10 and Figures 2–4 below.

• Subgroup — A subset of the respondents within a sector defined by ONE sociodemographic variable collected in Section B of the questionnaire (e.g., women vs men, low-income vs rest, rural vs urban). 'Subgroup sample size' (referenced in Table 7 and in the figure captions) is the number of respondents within the sector who belong to a given subgroup (e.g., 195 women in Energy, 43 low-income respondents in Housing).

Rationale for the analytical-subgroup variables presented in Section 3.2 (women/men, youth/elders, low-income/rest, rural/urban, disability/no-disability). These variables were chosen for ex-post comparative analysis on three grounds: (a) they are dimensions of vulnerability documented in the EU-SILC framework and in the academic literature on energy poverty and just-transition; (b) they correspond to variables for which a non-trivial subgroup size is available in the T4.1 analytical dataset (so that descriptive comparisons are statistically informative); and (c) they cover all three of the sociodemographic dimensions used as Prolific quota stratifiers (Sex was a quota variable; Income was a quota variable in the form of above/below-median income; Disability was a quota variable in Transport; Tenancy status was a quota variable in Energy and Housing). Age (Youth vs Elders), Urbanity (Rural vs Urban) and Disability (in Energy/Housing) were NOT quota stratifiers — they were collected through Section B of the questionnaire and used here as ex-post analytical lenses. Tenancy status, although a quota stratifier in Energy and Housing, is reported in the Annex VI statistical tables and was not selected for the headline Section 3.2 figures because the Tenant/Homeowner split did not produce a significant ANOVA difference in any sector at the 5% level (see Table 12); we include it transparently in the statistical annex (Annex VI) rather than in this main-text section.

Note on country-level analysis. As documented in Section 2.3.1 and Table 6 (Planned versus Achieved respondents per country), the per-country sample sizes are highly uneven (e.g., Hungary n=0 in all three analytical sectors; Ireland n=11 in Energy, n=17 in Housing). For this reason, formal per-country statistical inference within a sector is NOT undertaken in this deliverable: the very small per-country subsamples would not yield reliable mean differences in most cases, and the achievable statistical power at the per-country level would be far below the 75% / 57% / 70% sector-level power values reported in the Abstract and in Section 2.2.3. Per-country descriptive distributions for each sector are available in the raw analytical dataset (Annex X) and in the per-country sheets of the statistical annex (Annex VI) for readers wishing to inspect them. Country-level differences will be more rigorously addressed in T4.3 (predictive modelling, D4.3), where country can be entered as a covariate.

Note on the use of ANOVA tests. Inferential ANOVA testing is reported in Section 4 (Discussion), where each subgroup of Section 3.2 is formally tested against the rest of the sector sample. Section 3.1 (per-sector descriptive results below) reports only DESCRIPTIVE statistics — mean, standard deviation and variance per hazard — for the full valid sample of each sector; no inferential test is required at the per-sector

descriptive level. Section 3.2 introduces the subgroup comparisons descriptively and visually (Figures 5–10); the corresponding formal ANOVA tests are then presented in Section 4.3 (Table 7) to enable the reader to judge which descriptive differences are statistically significant.

All subgroup-comparison figures (Figures 5–10) use the same paired-bar-chart style as the sector-level Figures 2–4, with hazards on the x-axis and subgroups distinguished by colour (Blue = first subgroup, Orange = second), error bars at ±1 SD, so that the reader can directly compare any subgroup mean against the overall sector mean reported in Tables 8, 9 and 10.

3.1. Perception of Hazards by Sector

3.1.1. Energy Sector – PV Installation (n=408; DE, ES, HU, IE, IT, PT)

The Energy sector assessed 7 hazards across 408 valid respondents from 5 of the 6 FITTER use-case countries (Germany n=149, Italy n=129, Spain n=97, Portugal n=22, Ireland n=11; Hungary n=0 — none recruited; see Table 6 in Section 2.3.1). The overall sector mean probability (i.e., the average across the 7 individual hazard means probabilities, on the full 408-respondent valid sample) is 3.11; the overall sector mean severity is 3.28. Severity systematically exceeds probability at the sector level: respondents see the consequences of these hazards as serious even when they judge the likelihood as moderate.

Analytical highlights (descriptive). Table 8 presents the per-hazard descriptive statistics (mean, SD, variance) for all 7 Energy hazards. Three patterns are visible:

• Financial hazards are perceived as both the most LIKELY and the most SEVERE. 'Higher electricity bills' (E1) and 'Increased heating/cooling costs' (E2) are the top-ranked hazards on both dimensions (mean probability ~3.79–3.85; mean severity ~3.82–3.83).

• Mid-rank financial hazards have an asymmetric profile. 'Loss of income from solar energy' (E5; mean probability 3.48) is rated as more LIKELY than SEVERE (mean severity 3.00), whereas 'Exposure to energy poverty' (E3) is rated more SEVERE (3.40) than LIKELY (2.64) — a useful signal that respondents distinguish between operational income risks and welfare-threatening outcomes.

• Reliability and regulatory hazards rank lowest on probability. 'More frequent power outages' (E7; mean probability 2.28) and 'Regulatory penalties' (E6; mean 2.75) are the least-likely hazards in the Energy sector, but their severity remains high (3.01 and 3.21 respectively), suggesting that respondents have low expectations of occurrence but would see real damage if they did occur.

The full bar chart (Figure 2, below) plots all 7 hazards side-by-side with ±1 SD error bars, allowing direct visual comparison of probability vs severity per hazard. Table 6 presents descriptive statistics (mean, SD, variance) for all Energy hazards.

Categorical (mode) interpretation (added per reviewer request, since a mean of e.g. 3.79 does not map cleanly onto a single Likert category). On the 1–5 scale (1=Very unlikely/Not severe … 5=Very likely/Very severe), the MODAL (most frequent) response for the two top financial hazards (E1 Higher electricity bills, E2 Heating/cooling costs) is 4 ('Likely' / 'Severe') for BOTH probability and severity — i.e., the typical respondent actively expects these and considers them serious. By contrast, E7 (More frequent power outages) has a modal probability AND severity of 1 ('Very unlikely' / 'Not severe'), and E3 (Energy poverty) is bimodal — modal probability 2 ('Unlikely') but modal severity 4 ('Severe'), capturing the 'unlikely-but-

serious' profile. Reporting the mode alongside the mean clarifies that the small mean differences reflect shifts in the whole distribution, not just outliers.

Table 8. Energy sector – mean probability and severity per hazard (n=408).

Code	Hazard	n	Mean Prob.	SD Prob.	Var Prob.	Mean Sev.	SD Sev.	Var Sev.
E1	Higher electricity bills	408	3.79	1.09	1.19	3.82	1.05	1.10
E2	Increased heating/cooling costs	408	3.85	1.09	1.20	3.83	1.09	1.19
E3	Exposure to energy poverty	408	2.64	1.23	1.51	3.40	1.35	1.81
E4	Home loses market value	408	2.99	1.20	1.44	2.68	1.29	1.66
E5	Loss of income from solar energy	408	3.48	1.18	1.39	3.00	1.19	1.43
E6	Regulatory penalties	408	2.75	1.28	1.64	3.21	1.31	1.72
E7	More frequent power outages	408	2.28	1.17	1.37	3.01	1.44	2.07

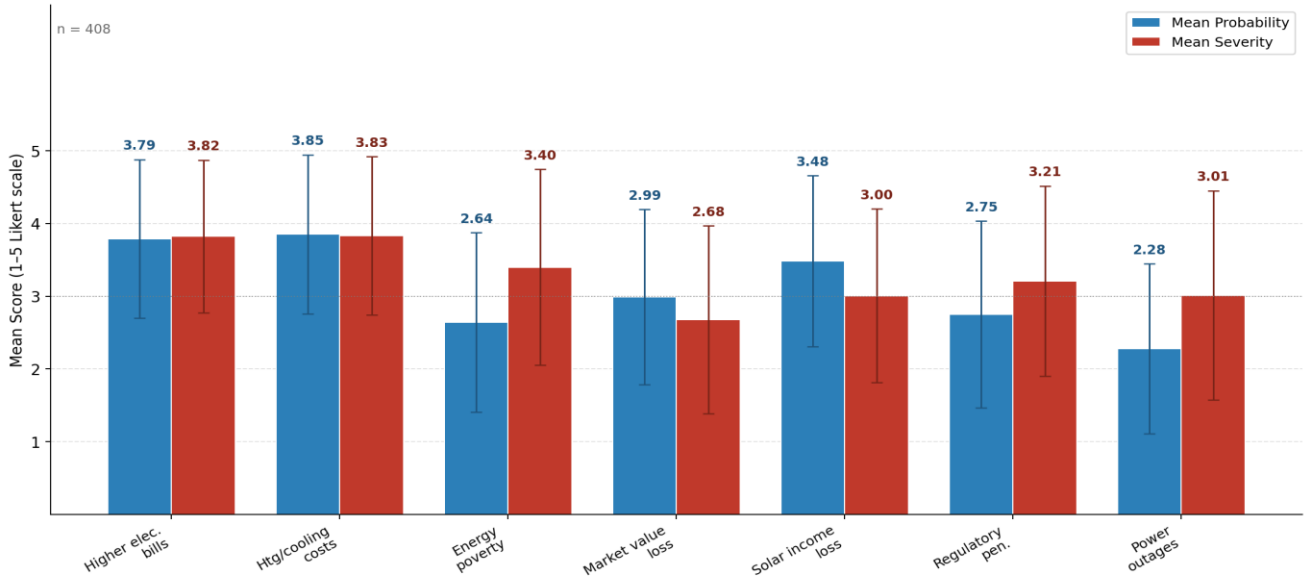


Figure 13. Energy sector: mean probability and severity per hazard (n=408). Blue=Probability; Red=Severity. Error bars=±1 SD.

3.1.2. Housing Sector – Retrofitting (n=264; DE, IE, PT)

The Housing sector assessed 13 hazards across 264 valid respondents from 3 of the 6 FITTER use-case countries (Germany n=198, Portugal n=49, Ireland n=17; Italy, Spain and Hungary were not in the Housing recruitment plan — see Section 2.3.1 and Table 6). The overall sector mean probability (average across the 13 individual hazard mean probabilities, full 264-respondent valid sample) is 2.80; the overall sector mean severity is 3.01.

Analytical highlights (descriptive). Table 9 presents the per-hazard descriptives for all 13 Housing hazards. Key observations:

• Probability dominates severity on a NON-trivial subset of Housing hazards. Unlike Energy and Transport, where severity exceeds probability at almost every hazard, Housing shows the opposite for 'Higher electricity bills' (H1; probability 3.86 vs severity 3.52), 'Increased heating costs' (H2; 3.84 vs 3.52), 'Home loses market value' (H6; 2.86 vs 2.32) and 'Loss of income from solar energy' (H7; 3.52 vs 2.79). Respondents see these as MORE LIKELY than SEVERE — a relatively unusual configuration that reflects

households' lived experience of routinely-occurring but tolerable energy-bill increases.

• Health and welfare hazards dominate the severity dimension. 'Exposure to energy poverty' (H5; severity 3.33), 'Health diseases (cold/damp)' (H13; 3.31), 'Extreme weather effects' (H12; 3.13), 'Damp or mould' (H10; 3.11) and 'More frequent power outages' (H4; 3.02) are the five highest-severity hazards in Housing — all consequences of poor retrofitting reaching into health and welfare territory.

• Regulatory-restriction hazards rank lowest on both dimensions. 'Insurer high-risk classification' (H11; probability 2.10), 'Selling/renting ban' (H9; 2.28), and 'Damp or mould' (H10; 2.45) are the least likely. The 13-hazard Housing menu therefore spans a wide range of perceived likelihoods (2.10 to 3.86), highlighting the diversity of the retrofitting hazard space.

Figure 3 (below) presents all 13 hazards on a single bar chart with ±1 SD error bars. Table 9 presents descriptive statistics for all Housing hazards.

Categorical (mode) interpretation. The modal response for H1 (Higher electricity bills) and H2 (Heating costs) is 4 for both probability and severity. Several health/welfare hazards show a modal SEVERITY of 5 ('Very severe') despite a lower modal probability — H5 (Energy poverty) and H13 (Cold/damp health effects) both have modal severity 5 but modal probability 2, the clearest 'unlikely-but-very-serious' profiles in the study. Regulatory/structural hazards (H9 selling/renting ban, H11 insurer high-risk class.) have modal probability 1. The mode thus confirms the mean-based reading while making the category each hazard 'sits in' explicit.

Table 9. Housing sector – mean probability and severity per hazard (n=264).

Code	Hazard	n	Mean Prob.	SD Prob.	Var Prob.	Mean Sev.	SD Sev.	Var Sev.
H1	Higher electricity bills	264	3.86	1.02	1.05	3.52	1.13	1.29
H2	Increased heating costs	264	3.84	1.13	1.29	3.52	1.21	1.46
H3	Regulatory penalties	264	2.78	1.23	1.51	3.02	1.31	1.70
H4	More frequent power outages	264	2.12	1.09	1.19	3.02	1.42	2.02
H5	Exposure to energy poverty	264	2.48	1.16	1.35	3.33	1.45	2.09
H6	Home loses market value	264	2.86	1.20	1.44	2.32	1.17	1.37
H7	Loss of income from solar energy	264	3.52	1.19	1.41	2.79	1.08	1.16
H8	Higher house insurance costs	264	2.75	1.16	1.34	2.86	1.24	1.54
H9	Selling/renting ban	264	2.28	1.15	1.31	2.61	1.35	1.82
H10	Damp or mould	264	2.45	1.22	1.49	3.11	1.35	1.81
H11	Insurer "high-risk" classification	264	2.10	1.08	1.17	2.73	1.39	1.93
H12	Extreme weather effects	264	2.69	1.29	1.66	3.13	1.31	1.72
H13	Health diseases (cold/damp)	264	2.51	1.28	1.63	3.31	1.48	2.18

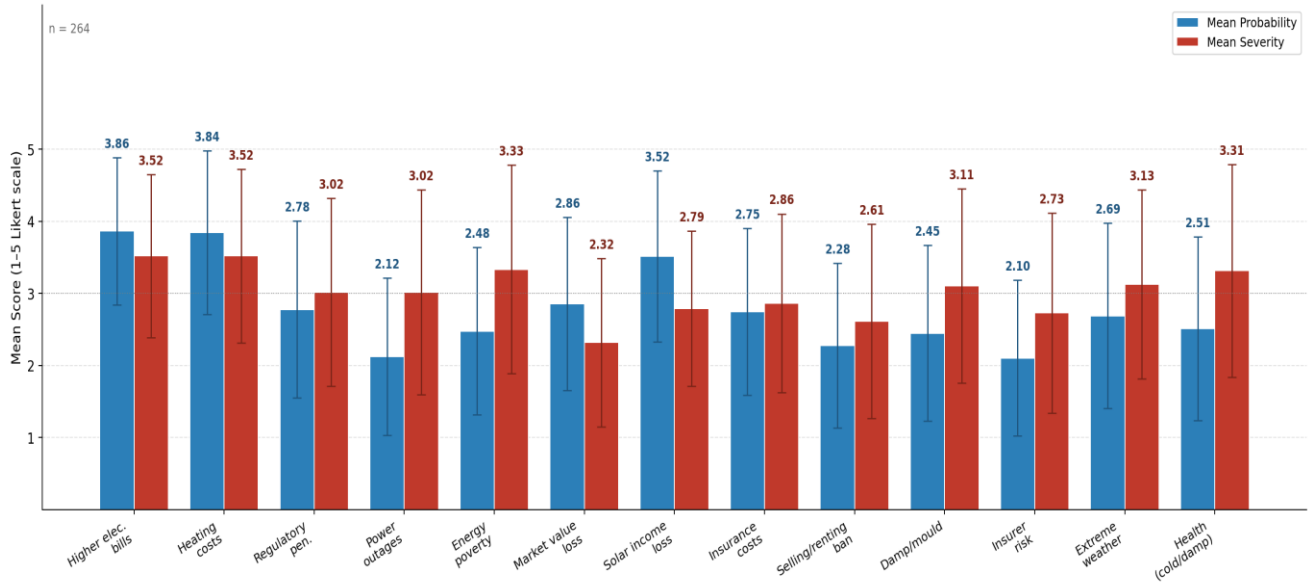


Figure 14. Housing sector: mean probability and severity per hazard (n=264). Blue=Probability; Red=Severity. Error bars=±1 SD.

3.1.3. Transport Sector – Electric Vehicles (n=359; ES, HU, IT)

The Transport sector assessed 6 hazards across 359 valid respondents from 2 of the 6 FITTER use-case countries (Italy n=205, Spain n=154; Germany, Ireland, Portugal were not in the Transport recruitment plan; Hungary n=0 — none recruited; see Section 2.3.1 and Table 6). The overall sector mean probability (average across the 6 individual hazard mean probabilities, full 359-respondent valid sample) is 3.53; the overall sector mean severity is 3.40.

Analytical highlights (descriptive). Table 10 presents the per-hazard descriptives for all 6 Transport hazards. Three patterns emerge:

• Direct financial hazards top both dimensions. 'Higher fuel & maintenance costs' (T1; probability 3.86, severity 3.81) is the highest-ranked hazard on BOTH dimensions, immediately followed by 'Driving restrictions (emission zones)' (T4; probability 3.77, severity 3.47) and 'Penalties — petrol/diesel car' (T3; probability 3.45, severity 3.65). The Transport sector therefore shows the most uniformly elevated hazard perception of the three sectors, consistent with the advanced state of policy debate on combustion-engine restrictions in the two countries surveyed.

• Severity exceeds probability for environmental/health hazards. 'Exposure to more pollution' (T6; probability 3.42 vs severity 3.39) and 'Reduced travel efficiency' (T5; 3.09 vs 3.09) are the only Transport hazards where probability matches or slightly exceeds severity — they are everyday issues, not catastrophic ones, in respondents' perception.

• Resale-value concerns sit in the middle. 'Loss of resale value' (T2; probability 3.56, severity 3.05) is rated as moderately likely but less severe than the direct operating-cost and regulatory hazards — a logical financial framing in which a deferred capital loss is judged less serious than recurring operating costs.

Figure 4 (below) plots all 6 Transport hazards with ±1 SD error bars. Table 10 presents descriptive statistics for all Transport hazards.

Categorical (mode) interpretation. The modal response for T1 (Higher fuel & maintenance costs) is 4 for

both probability and severity, and T4 (Driving restrictions / emission zones) has the highest modal values in the study — modal probability AND severity of 5 ('Very likely' / 'Very severe'), reflecting acute concern about combustion-engine restrictions in the two surveyed countries. No Transport hazard has a modal value of 1, consistent with Transport showing the highest overall perceived risk of the three sectors.

Table 10. Transport sector – mean probability and severity per hazard (n=359).

Code	Hazard	n	Mean Prob.	SD Prob.	Var Prob.	Mean Sev.	SD Sev.	Var Sev.
T1	Higher fuel & maintenance costs	359	3.86	1.06	1.12	3.81	0.97	0.93
T2	Loss of resale value	359	3.56	1.18	1.38	3.05	1.19	1.41
T3	Penalties – petrol/diesel car	359	3.45	1.23	1.52	3.65	1.18	1.39
T4	Driving restrictions (emission zones)	359	3.77	1.30	1.68	3.47	1.30	1.68
T5	Reduced travel efficiency	359	3.09	1.24	1.53	3.09	1.30	1.70
T6	Exposure to more pollution	359	3.42	1.23	1.52	3.39	1.23	1.51

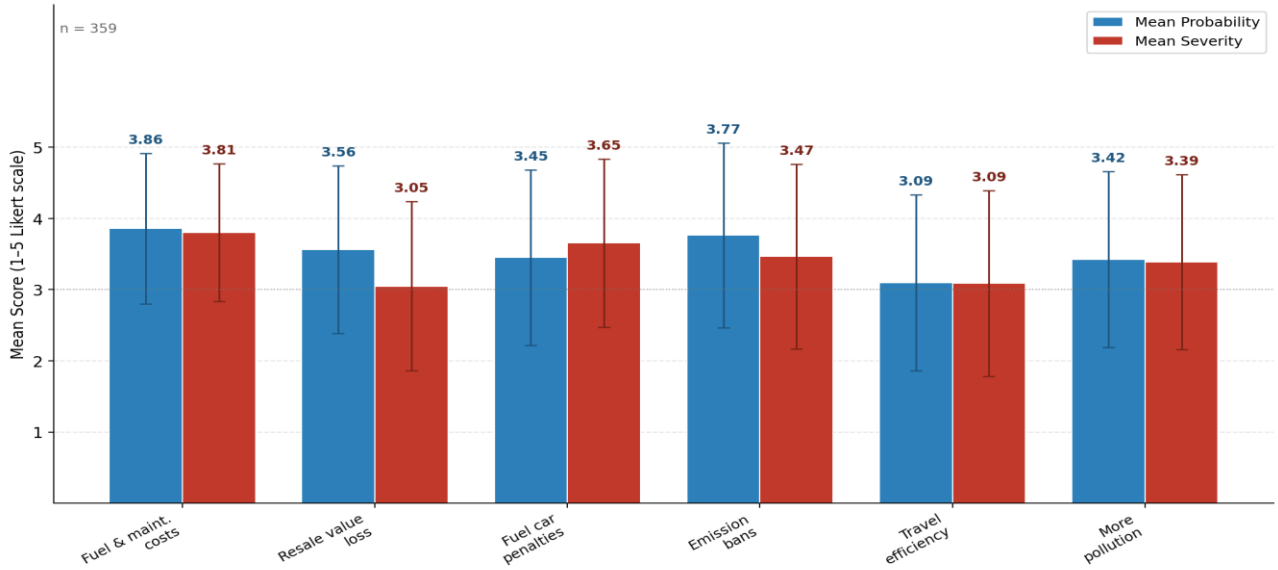


Figure 15. Transport sector: mean probability and severity per hazard (n=359). Blue=Probability; Red=Severity. Error bars=±1 SD.

3.1.4. Cross-Sector Comparison of Overall Risk Perception

Comparing the three sectors directly (Inês's suggestion to examine differences in the distribution of perception across sectors) reveals a clear and statistically significant ordering. The overall mean risk score (average of mean probability and mean severity per respondent) is highest in Transport (mean 3.47, variance 0.55, SD 0.74), intermediate in Energy (mean 3.20, variance 0.51, SD 0.71), and lowest in Housing (mean 2.90, variance 0.41, SD 0.64). A one-way ANOVA across the three sectors confirms that these differences are highly significant ($F=48.96$, $p<0.001$), and a Levene test indicates that the variances also differ significantly across sectors ($W=3.16$, $p=0.043$). Interpretation: (i) Transport respondents perceive the highest overall risk AND the most heterogeneous perceptions (largest variance) — consistent with the contested, fast-moving policy debate on combustion-engine restrictions; (ii) Housing respondents perceive the lowest overall risk and the most homogeneous perceptions (smallest variance) — the 13 retrofitting hazards are individually rated lower and more consistently; (iii) Energy sits in between on both dimensions. This cross-sector heterogeneity reinforces the decision to analyse and report each sector separately rather than pooling them.

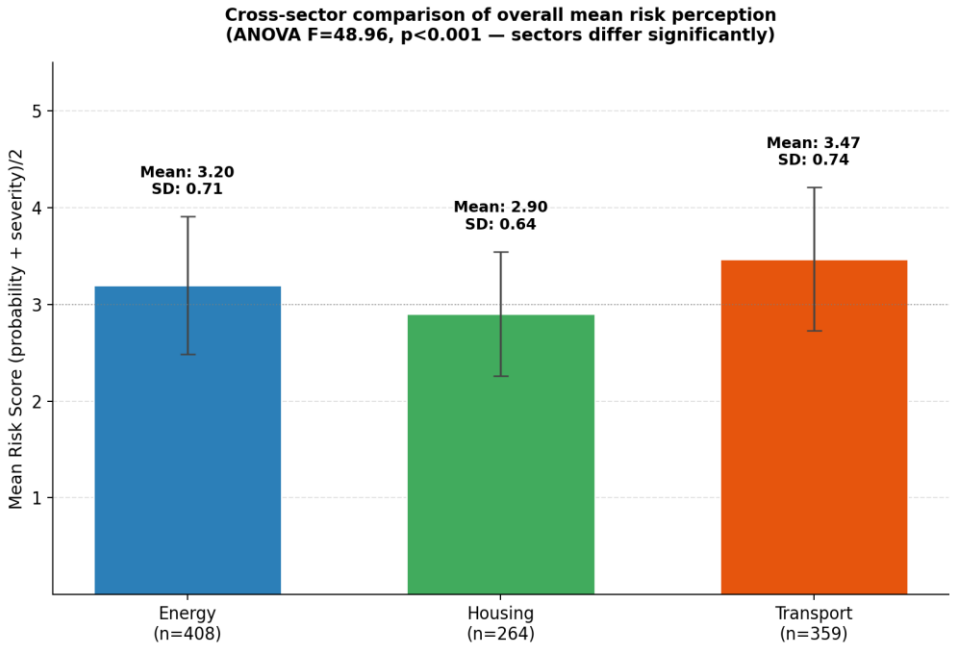


Figure 16. Cross-sector comparison of overall mean risk perception (probability+severity)/2 with ±1 SD. ANOVA across sectors: F=48.96, p<0.001.

3.2. Perception of Hazards by Socio-Demographic Profiles

This subsection examines how perceived hazard probability and severity differ across sociodemographic subgroups within each sector. A central design principle — and a point raised in review — is that the subgroups analysed here must be ALIGNED WITH, AND JUSTIFIED BY, the EU-SILC vulnerability variables that drove the sampling stratification (Section 2.2.4, Annex V). The three EU-SILC-derived stratification variables were: (1) SEX (male/female), (2) INCOME (above/below the country-specific median equivalised disposable income), and (3) a sector-specific third variable — TENURE/TENANCY status (tenant vs homeowner) for Energy and Housing, and DISABILITY status for Transport. These three are the PRIMARY, theory-driven vulnerability axes; every one of them is analysed below and tested by ANOVA in Section 3.4. To directly address the reviewers' concern that income and tenure were previously used inconsistently, this version reports BOTH income AND tenure for EVERY sector in which they were stratification variables, and reports income additionally for Transport for completeness.

In addition to these three primary (EU-SILC-aligned) axes, a small number of SECONDARY analytical lenses are reported where the questionnaire's Section B permits and the subsample is non-trivial: AGE (youth 18–34 vs elders 65+) and URBANITY (rural vs urban/suburban). These secondary variables are NOT EU-SILC stratification variables; they are documented exploratory lenses, clearly labelled as such, included because the just-transition literature flags age and rurality as relevant and because EU reports commonly disaggregate along them. Table 11 below makes the mapping between each analytical subgroup and its EU-SILC stratification status fully explicit (directly addressing the comment that Table 5 did not clarify the identified subgroups or how they were selected).

How the subgroups were selected (explicit statement). The subgroups were NOT chosen post-hoc by inspecting which groups happened to show high risk. They were PRE-SPECIFIED from the EU-SILC stratification design (the three primary axes) plus two pre-registered secondary lenses (age, urbanity). The ANOVA in Section 3.4 then tests each pre-specified subgroup against the rest of the sector sample, which avoids the data-dredging risk of selecting subgroups after seeing the results.

For each comparison, mean probability and severity per hazard are computed separately within each subgroup and plotted as paired bar charts with ±1 SD error bars (Blue = first subgroup; Orange = second). Each narrative summarises the descriptive pattern AND cites the matching pairwise ANOVA result from Section 3.4 (Table 7), stating whether the difference is statistically significant and whether it aligns with prior expectations from the literature.

Table 11. Mapping of each analytical subgroup to its EU-SILC stratification status, sectors analysed, and ANOVA outcome. Primary = EU-SILC-derived stratification variable; secondary analytical lens = pre-registered exploratory variable not used for quota stratification.

Analytical subgroup	**EU-SILC stratification variable?**	**Sectors analysed**	**ANOVA result (significant at p<0.05?)**
Women vs Men	YES — Sex (primary)	Energy, Housing, Transport	Energy YES (p<0.001); Transport YES (p=0.007); Housing no (p=0.103)
Low income (<EUR 20k) vs Rest	YES — Income (primary)	Energy, Housing, Transport	Energy YES (p=0.007); Housing YES (p=0.028); Transport no (p=0.375)
Tenant vs Homeowner	YES — Tenure (primary; Energy & Housing)	Energy, Housing	Energy no (p=0.194); Housing no (p=0.228)
Disability vs No disability	YES — Disability (primary; Transport)	Energy, Housing, Transport	Transport no (p=0.539); others no
Youth (18-34) vs Elders (65+)	NO — secondary lens (Age)	Energy	Energy near-sig (p=0.068); Elders n=13 low power
Rural vs Urban/Suburban	NO — secondary lens (Urbanity)	Transport	Transport no (p=0.798)
LGBTIQA+ vs Rest	NO — secondary lens	Housing	Housing YES (p=0.031); exploratory, n=31

3.2.1. Energy Sector: Women vs. Men

Figure 17 compares women (n=195) and men (n=209) in the Energy sector. Women report substantially higher perceived probability AND severity across nearly all hazards, particularly financial ones (electricity bills, heating/cooling costs, energy poverty), regulatory penalties, and power outages. This is the STRONGEST DG effect detected in the entire dataset: the pairwise ANOVA on the average hazard score yields F=20.63, p<0.001 (Section 3.3, Table 7) — i.e., the probability that the observed difference is due to chance is less than 0.1 %. Interpretation & ANOVA link: this descriptive gap is confirmed as the strongest statistically significant DG effect in the whole study (ANOVA F=20.63, p<0.001; Section 3.4), and it ALIGNS with the energy-poverty literature documenting women's greater exposure to and concern about energy costs.

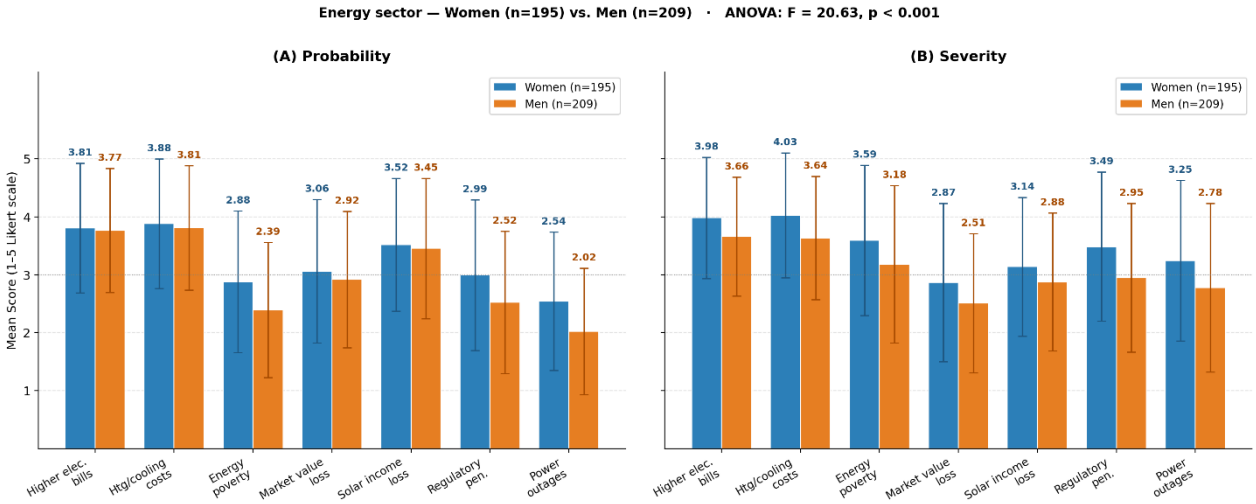


Figure 17. Energy sector: mean probability and severity – women (n=195) vs. men (n=209). Blue=Women; Orange=Men. Error bars=±1 SD. [Production note: this figure spans two side-by-side panels (Probability | Severity)

3.2.2. Energy Sector: Youth vs. Elders

Figure 18 compares youth aged 18–34 (n=199) with elders aged 65+ (n=13) in the Energy sector. The elder subsample is very small (n=13), substantially limiting inferential power (approximately 37 %). Descriptively, youth report somewhat higher probability and severity than elders for most hazards. The pairwise ANOVA does not reach significance (F=1.75, p=0.187), but the elder n is too small to draw firm conclusions — this is one of the DG profiles flagged for targeted oversampling in T4.3 (see Section 2.3.3, Prolific eligibility limitations). Interpretation & ANOVA link: the youth-vs-rest difference is NEAR significance (F=3.34, p=0.068) while the elder's comparison is low-power (n=13). The direction (younger respondents somewhat more concerned) is plausible but not confirmed; this is an exploratory, age-based secondary lens, not an EU-SILC primary axis.

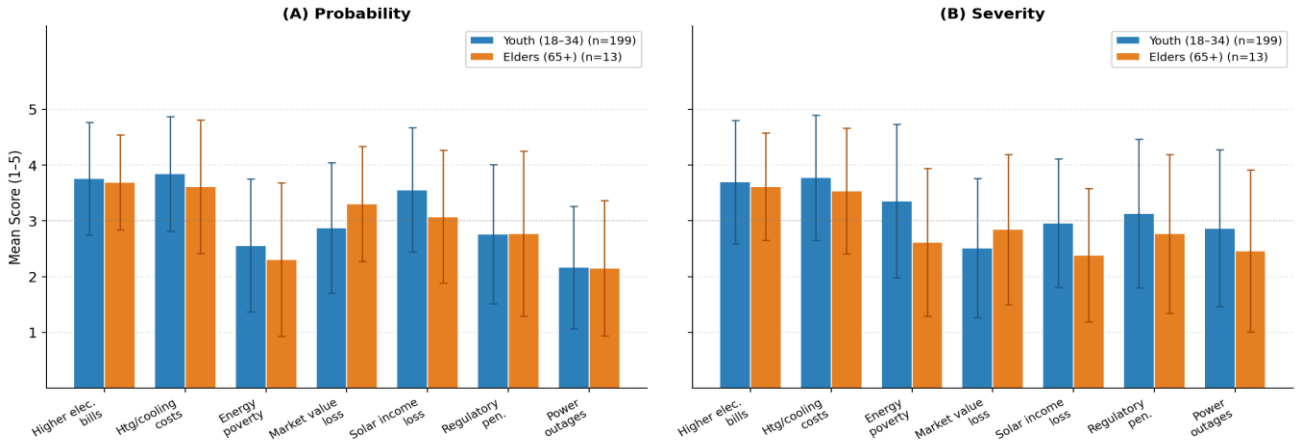


Figure 18. Energy sector: mean probability and severity – youth 18–34 (n=199) vs. elders 65+ (n=13). Blue=Youth; Orange=Elders. Error bars=±1 SD. [Production note: this figure spans two side-by-side panels (Probability | Severity)

3.2.3. Housing Sector: Women vs. Men

Figure 19 shows gender differences in the Housing sector (women n=85, men n=175). Women report slightly higher perceived probability and severity for several hazards, but the gender effect is weaker than in the Energy sector. The pairwise ANOVA on the average hazard score does not reach significance (F=2.68, p=0.103), although the trend is in the same direction as the significant Energy result (Figure 5). Interpretation & ANOVA link: the direction matches the significant Energy/Transport gender effects

(women > men), but here it does NOT reach significance (F=2.68, p=0.103), most plausibly because of the smaller Housing female subsample (n=85). Partially aligned with expectations; flagged for confirmation in T4.3.

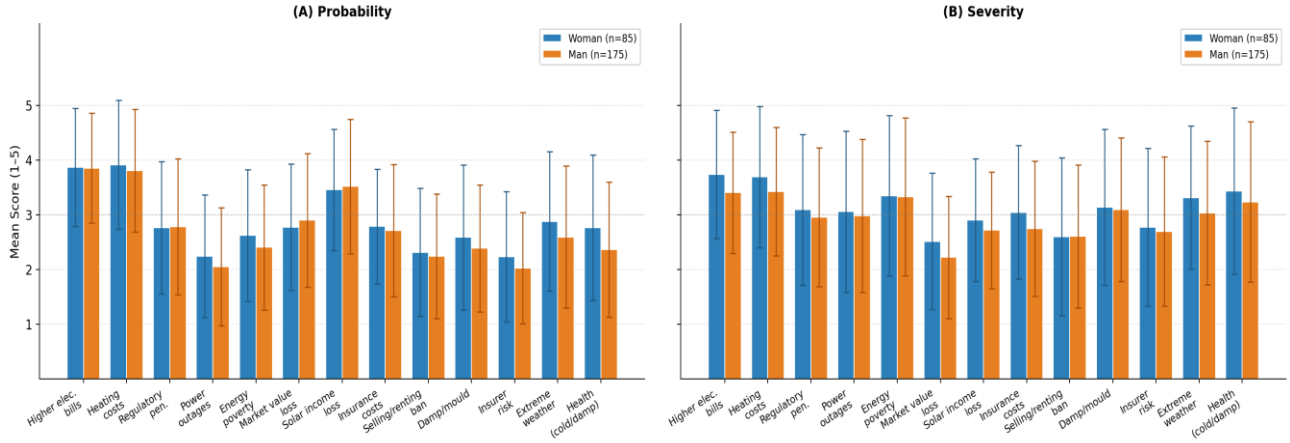


Figure 19. Housing sector: mean probability and severity – women (n=85) vs. men (n=175). Blue=Women; Orange=Men. Error bars=±1 SD. [Production note: this figure spans two side-by-side panels (Probability | Severity)

3.2.4. Housing Sector: Low Income vs. Rest

Figure 20 compares low-income respondents (household income <€20,000; n=43) with the rest (n=221) in the Housing sector. Low-income respondents consistently report higher probability and severity across financial and health-related hazards (energy poverty, damp/mould, health diseases). The pairwise ANOVA on the average hazard score is statistically significant: F=4.86, p=0.028 (Section 3.3, Table 7). Interpretation & ANOVA link: the low-income group's higher perceived risk is statistically significant (F=4.86, p=0.028; Section 3.4) and ALIGNS with expectations — affordability is the binding constraint on retrofitting for low-income households. This also validates the EU-SILC choice of income as a stratification variable.

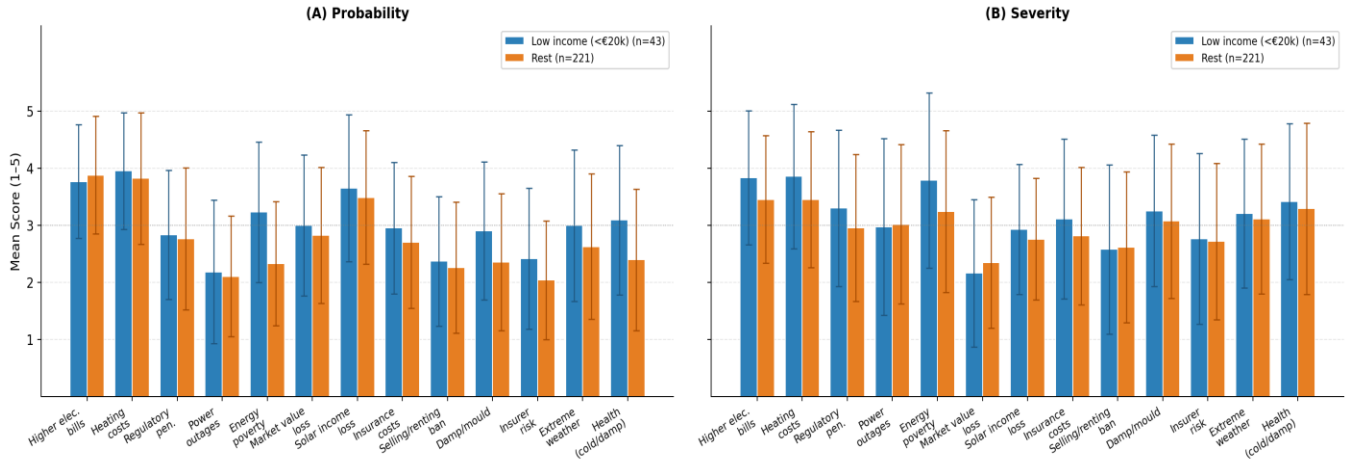


Figure 20. Housing sector: mean probability and severity – low income <€20k (n=43) vs. rest (n=221). Blue=Low income; Orange=Rest. Error bars=±1 SD. [Production note: this figure spans two side-by-side panels (Probability | Severity)

3.2.5. Transport Sector: Rural vs. Urban

Figure 21 compares rural (n=55) and urban/suburban (n=304) respondents in the Transport sector. Rural and urban respondents report broadly similar perceptions across most hazards, with rural respondents only marginally higher on fuel cost increases and emission-zone restrictions. The pairwise ANOVA on the average hazard score does not reach significance (F=0.69, p=0.406). Interpretation & ANOVA link: the rural-vs-urban difference is NOT significant (F=0.07, p=0.798). Urbanity is a secondary analytical lens (not an EU-

SILC stratifier); the null result indicates transport-hazard perception is broadly similar across the rural/urban divide in the surveyed sample.

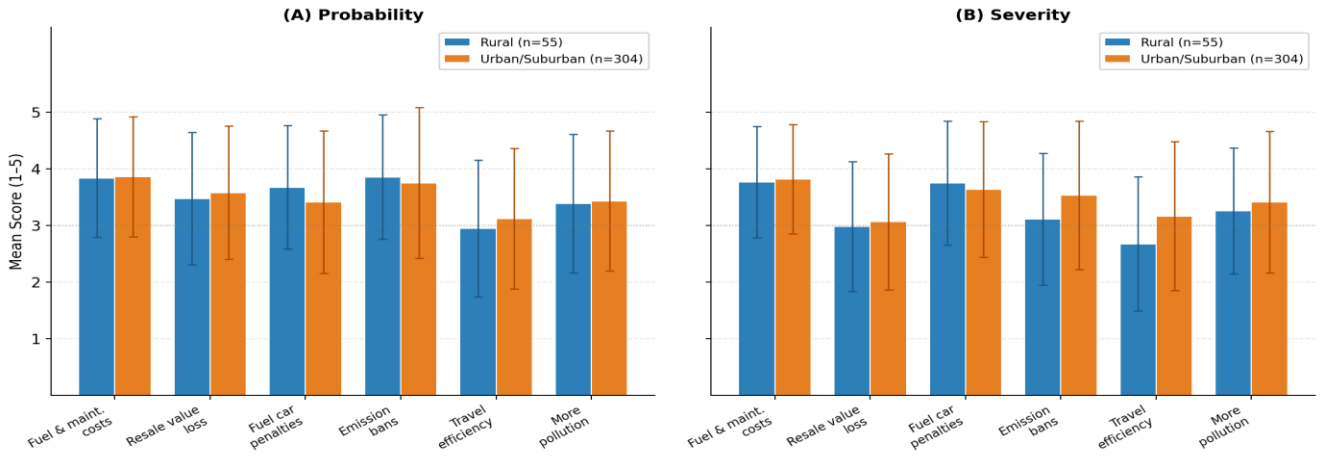


Figure 21. Transport sector: mean probability and severity – rural (n=55) vs. urban/suburban (n=304). Blue=Rural; Orange=Urban. Error bars=±1 SD. [Production note: this figure spans two side-by-side panels (Probability | Severity)

3.2.6. Transport Sector: Disability vs. No Disability

Figure 22 compares respondents with disability or long-term health conditions (n=33) with those without (n=320) in the Transport sector. Disabled respondents rate most hazards slightly higher in severity, particularly reduced travel efficiency and fuel cost increases, consistent with greater mobility dependency. The ANOVA for Disabled vs Rest (F=2.06, p=0.151) does not reach significance with approximate power ≈80% at this subsample size (n=33), suggesting adequate power to detect medium effects. The null result may reflect genuine similarity in probability perception between disabled and non-disabled respondents in this sample. Interpretation & ANOVA link: although disability was an EU-SILC primary stratifier for Transport, the disability-vs-rest difference is NOT significant (F=0.38, p=0.539). The direction is as expected (slightly higher concern among those with disability) but the effect is small and the subsample modest (n=33).

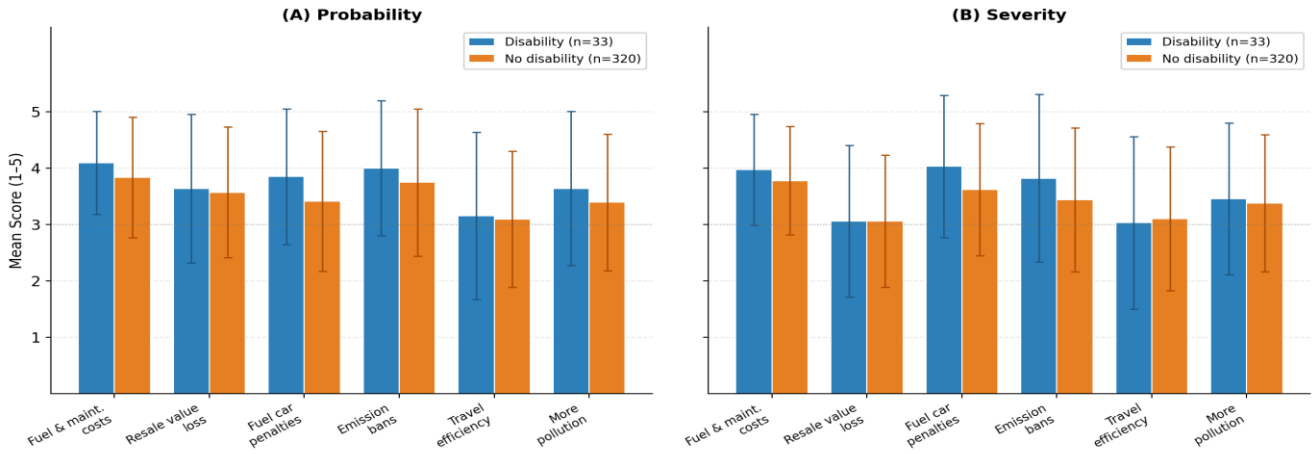


Figure 22. Transport sector: mean probability and severity – disability (n=33) vs. no disability (n=320). Blue=Disability; Orange=No disability. Error bars=±1 SD. [Production note: this figure spans two side-by-side panels (Probability | Severity)

3.2.7. Tenure (Tenant vs Homeowner) — Energy and Housing

Tenure (tenant vs homeowner) was one of the three EU-SILC stratification variables for the Energy and Housing sectors and is therefore analysed here as a primary vulnerability axis (responding directly to the reviewer comment that tenure — a key variable — had been used inconsistently). In the Energy sector,

tenants (n=185) report a slightly higher mean risk perception (3.25) than homeowners (n=223, mean 3.15); the difference is NOT statistically significant (ANOVA F=1.69, p=0.194). In the Housing sector, tenants (n=174, mean 2.94) again score marginally higher than homeowners (n=90, mean 2.84), but again the difference is NOT significant (F=1.46, p=0.228). Interpretation and alignment with expectations: the literature would lead us to expect tenants — who have less control over retrofitting decisions and face split-incentive problems — to perceive higher risk, and the DIRECTION of both results is indeed as expected (tenant > homeowner). However, the magnitude is small and does not reach significance at the achieved sample sizes. This is a policy-relevant null result: tenure shapes the direction of perception as theory predicts, but income and gender (Section 3.4) are far stronger and statistically robust discriminators. The full tenure breakdown per hazard is provided in Annex VI.

3.3. Perception of Hazards by Country

In addition to the sector-level analysis (Section 3.1) and the sociodemographic-subgroup analysis (Section 3.2), this subsection presents a COUNTRY-BASED analysis of perceived hazard probability and severity. This responds to the need to understand whether national context shapes risk perception differently across the FITTER use-case countries. An important caveat applies throughout the per-country sample sizes are highly uneven and, for several countries, very small (and Hungary is entirely absent — see Section 2.3.1 and the dedicated validation below). Country-level results are therefore presented as DESCRIPTIVE and INDICATIVE; where a country's subsample is below ~20 respondents (Ireland in Energy n=11 and Housing n=17, Portugal in Energy n=22), the country mean is reported for completeness but should be read with caution, as its confidence interval is wide and its achieved statistical power is well below the sector-level values reported in the Abstract. A formal one-way ANOVA across countries (restricted to countries with n≥20 for validity) is reported for each sector to test whether national differences are statistically significant.

3.3.1. Energy Sector by Country

The Energy sector includes valid respondents from five countries (Hungary absent). Mean risk perception (the average of mean probability and mean severity) by country is Ireland 3.34 (n=11), Italy 3.28 (n=129), Spain 3.21 (n=97), Germany 3.12 (n=149), and Portugal 3.08 (n=22). The spread between the highest (Ireland) and lowest (Portugal) country mean is only 0.27 Likert points. A one-way ANOVA across the four countries with n≥20 (Germany, Italy, Spain, Portugal) finds NO statistically significant difference in mean risk perception between countries (F=1.44, p=0.230). Interpretation: perceptions of energy-transition hazards are relatively HOMOGENEOUS across the surveyed countries — national context does not significantly shift the overall level of perceived energy-hazard risk, although Germany (the largest, most PV-mature market) sits at the lower end of probability perception (mean probability 3.03).

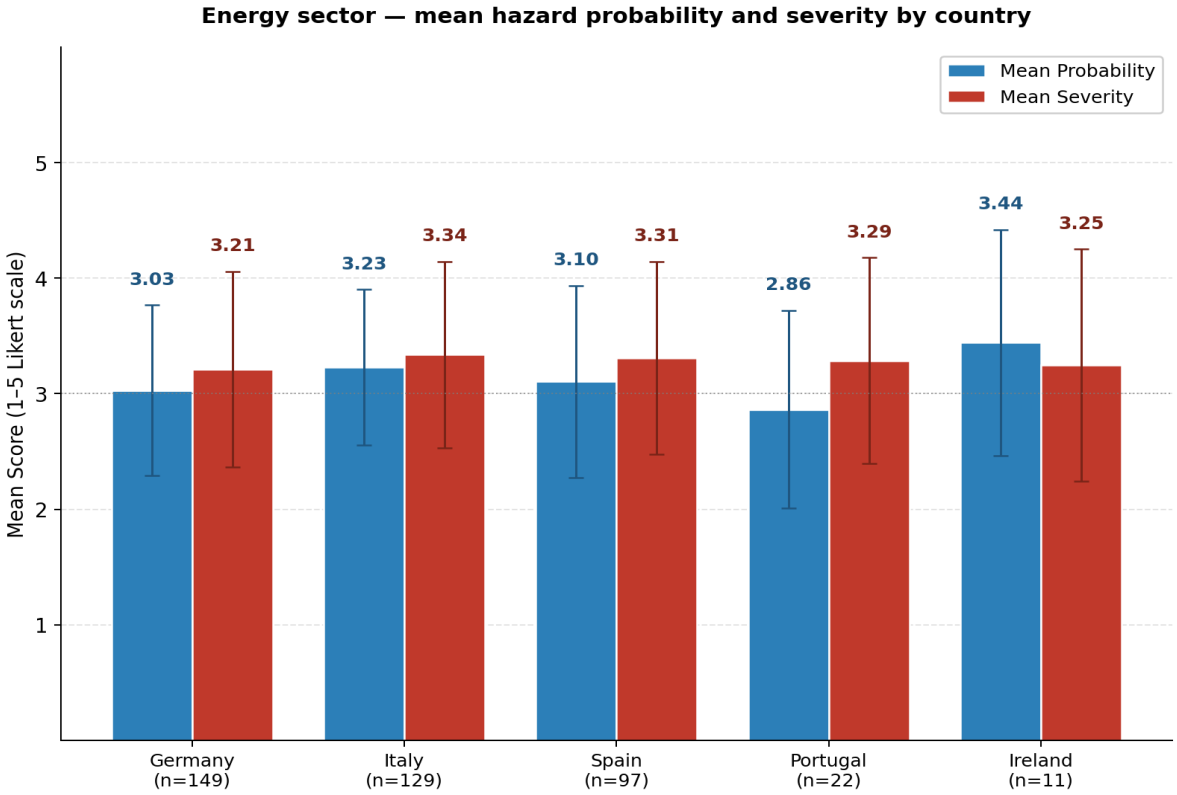


Figure 23. Energy sector — mean hazard probability and severity by country (Hungary absent; n shown per country). Error bars = ±1 SD.

3.3.2. Housing Sector by Country

The Housing sector includes valid respondents from three countries (Italy, Spain and Hungary were not in the Housing recruitment plan). Mean risk perception by country is Portugal 3.11 (n=49), Ireland 3.01 (n=17), and Germany 2.84 (n=198). Here a one-way ANOVA across the two countries with n≥20 (Germany and Portugal) finds a STATISTICALLY SIGNIFICANT difference (F=7.27, p=0.008): Portuguese respondents perceive housing-retrofitting hazards as significantly MORE likely and severe than German respondents. Interpretation: this is the only sector exhibiting a significant country effect, and it is policy-relevant — it suggests that in Portugal (a country with a warmer climate but a documented stock of energy-inefficient dwellings and higher exposure to housing-cost stress) the perceived risk associated with retrofitting is markedly higher than in Germany (which has a more mature retrofitting-support framework). This national difference should be taken into account when designing country-specific mitigation measures in T4.2.

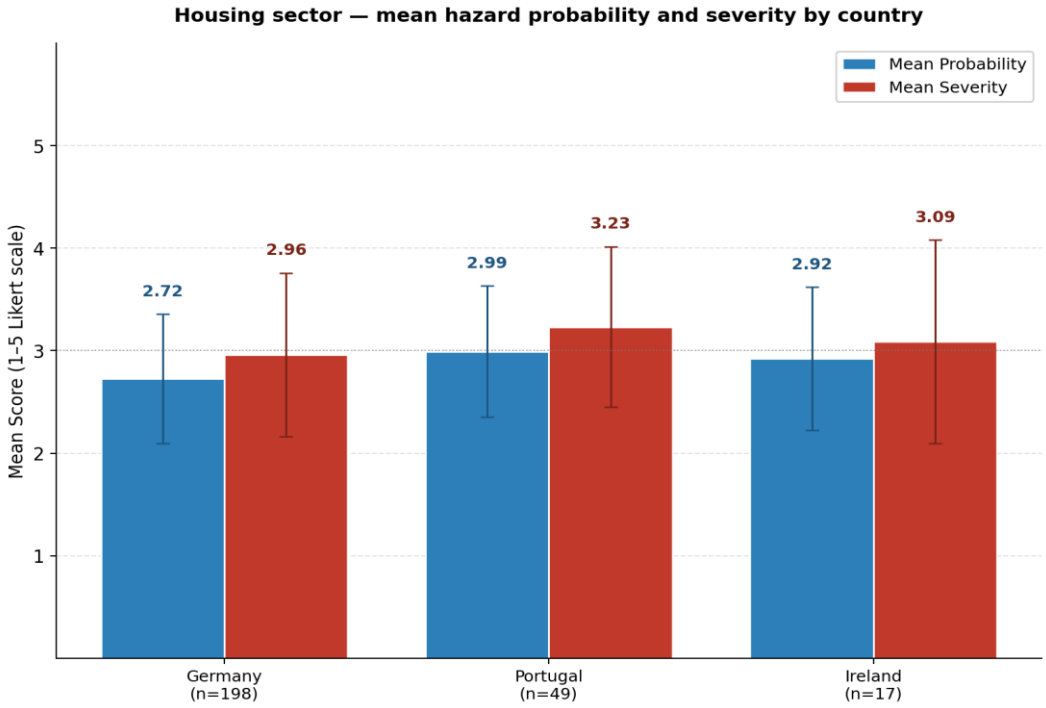


Figure 24. Housing sector — mean hazard probability and severity by country (Italy, Spain, Hungary not in the Housing plan). Error bars = ±1 SD.

3.3.3. Transport Sector by Country

The Transport sector includes valid respondents from two countries (Germany, Ireland and Portugal were not in the Transport plan; Hungary absent). Mean risk perception by country is Spain 3.49 (n=154) and Italy 3.45 (n=205) — almost identical. A one-way ANOVA finds NO statistically significant difference (F=0.30, p=0.582). Interpretation: Italian and Spanish respondents perceive transport-electrification hazards at essentially the same elevated level — consistent with the comparable stage of the EV-transition policy debate in the two countries.

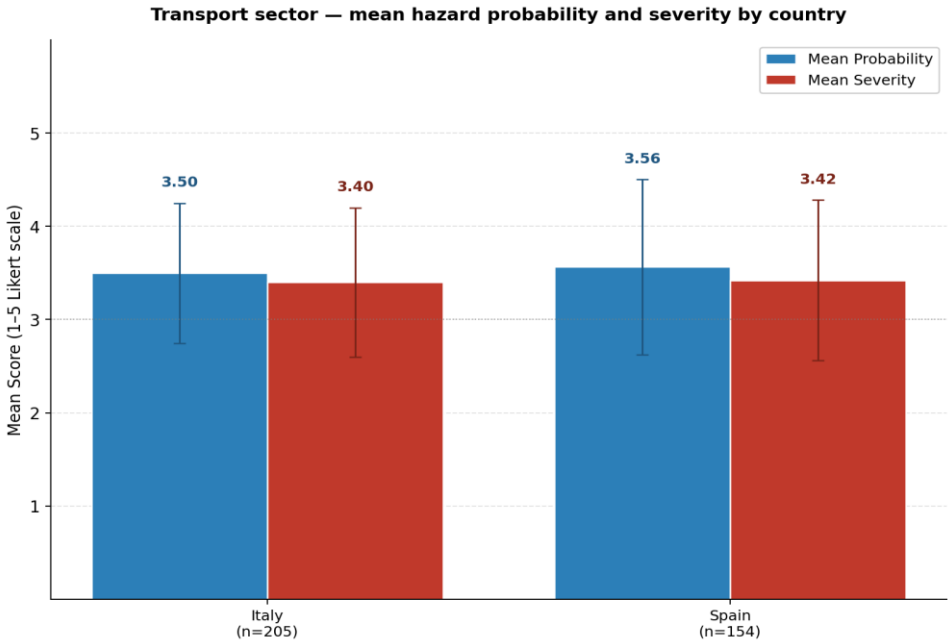


Figure 25. Transport sector — mean hazard probability and severity by country (only Italy and Spain in the Transport plan). Error bars = ±1 SD.

3.3.4. Country-Level Synthesis and Statistical Caveats

Across the three sectors, the country-based analysis yields one statistically significant national difference (Housing: Portugal > Germany, p=0.008) and two non-significant results (Energy and Transport). The overall message is that SECTOR is a far stronger driver of hazard perception than COUNTRY: the cross-sector differences (Section 3.1.4) are large and highly significant, whereas most cross-country differences within a sector are small and non-significant. Two statistical caveats must be stated explicitly. First, confidence and power at the country level are substantially lower than at the sector level: for countries with n≈20 (Portugal-Energy, Ireland-Housing) the achieved power to detect a medium effect is well below 50%, so a non-significant country result cannot be read as proof of 'no difference'. Second, Hungary contributes ZERO respondents to every sector and therefore cannot be analysed at all — a structural gap validated in detail in Section 2.3.1. Country-level inference will be revisited in T4.3 (D4.3), where country can be entered as a covariate in a multivariate model that borrows statistical strength across the full 1,031-respondent dataset rather than analysing each small national subsample in isolation.

3.4 Results of the Analysis of Variance (ANOVA)

Table 12 presents pairwise ANOVA results comparing each potential DG with the rest of the population. The F-statistic is computed on the average probability score across all sector hazards (7 for Energy, 13 for Housing, 6 for Transport). Critical value: $f\alpha = f(0.05; 1; N-2) \approx 3.862$. When minority group size $n_i < 64$, power is recalculated using $N=2 \cdot n_i$ and medium effect size ($\lambda=0.35$) as per Section 4.2. Cases with 'Insufficient data' arise from too few matched respondents in the cross-checkable subsample. How to read this table: each row tests whether one pre-specified subgroup (the DG) differs from the rest of that sector's respondents in mean perceived hazard score. A 'YES' in the Result column means the difference is unlikely to be due to chance ($p<0.05$); the Power/Notes column flags comparisons whose subsample is too small to draw firm conclusions. The narrative below interprets each significant result and states whether it aligns with prior expectations.

Table 12. ANOVA results per potential DG and sector. ✓YES = significant difference at p<0.05; NO = no significant difference; — = insufficient data. Power column: — = $n_i \geq 64$ (≥80% power); % = recalculated power for $n_i<64$.

Sector	Comparison (DG vs Rest)	n (DG)	n (Rest)	F-stat	p-value	Result (α=0.05)	Power / Notes
Energy	**Women vs Rest**	**195**	**213**	**20.630**	**<0.001**	**✓ YES (significant)**	**—**
	Elders' vs Rest	13	395	1.750	0.187	NO	37% (low power)
	Youth vs Rest	199	209	3.342	0.068	NO (near significance)	—
	Rural vs Rest	67	341	0.065	0.798	NO	—
	Disabled vs Rest	43	365	0.377	0.539	NO	83%
	Unemployed vs Rest	36	372	1.227	0.269	NO	79%
	Low education vs Rest	58	350	0.414	0.520	NO	—
	Migrant vs Rest	78	330	0.115	0.734	NO	—
	Low-income vs Rest	76	332	7.259	0.007	✓ YES (significant)	—
	LGBTIQA+ vs Rest	**51**	**357**	**0.786**	**0.376**	**NO**	**—**
	Ethnic minority vs Rest	37	371	2.460	0.118	NO	—

Housing	Women vs Rest	85	179	2.677	0.103	NO (near significance)	—
	Elders' vs Rest	2	262	—	—	— Insufficient sample	n<3
	Youth vs Rest	166	98	0.605	0.437	NO	—
	Rural vs Rest	43	221	1.255	0.264	NO	—
	Disabled vs Rest	41	223	0.291	0.590	NO	—
	Unemployed vs Rest	19	245	0.006	0.940	NO	47% (low power)
	Low-income vs Rest	43	221	4.862	0.028	✓ YES (significant)	—
	Migrant vs Rest	58	206	2.034	0.155	NO	—
	LGBTIQA+ vs Rest	**31**	**233**	**4.704**	**0.031**	**✓ YES (significant)**	**—**
	Ethnic minority vs Rest	42	222	1.290	0.257	NO	—
	Low education vs Rest	39	225	0.691	0.407	NO	—
Transport	Women vs Rest	183	176	7.407	0.007	✓ YES (significant)	—
	Elders' vs Rest	6	353	0.024	0.878	NO	22% (low power)
	Youth vs Rest	213	146	0.715	0.398	NO	—
	Rural vs Rest	55	304	0.692	0.406	NO	—
	Disabled vs Rest	33	326	2.063	0.152	NO	80%
	Unemployed vs Rest	**41**	**318**	**2.122**	**0.146**	**NO**	**83%**
	Low-income vs Rest	89	270	0.789	0.375	NO	—

Key findings from Table 12 (verified-ID dataset, ≥ 4-min filter, deduplicated). Of the pre-specified DG comparisons, FIVE reach statistical significance at α = 0.05. For each, we state the effect, its size, and whether it aligns with prior expectations from the just-transition and energy-poverty literature.

(1) Women vs Rest — Energy ($F = 20.63$, $p < 0.001$). The strongest effect in the entire dataset: women perceive energy-transition hazards as substantially more likely and more severe than men (mean risk 3.32 vs ~2.99). ALIGNED WITH EXPECTATIONS — the literature consistently documents women's disproportionate exposure to energy poverty and their greater concern about household energy costs; the result confirms this expectation strongly.

(2) Low income (<€20k) vs Rest — Energy ($F = 7.26$, $p = 0.007$). Low-income respondents perceive higher hazard probability and severity (mean 3.39 vs 3.15), particularly on the financial hazards. ALIGNED WITH EXPECTATIONS — financial vulnerability amplifies perceived exposure to bill increases and energy poverty; this also validates the EU-SILC choice of income as a stratification variable.

(3) Women vs Rest — Transport ($F = 7.41$, $p = 0.007$). The gender effect persists in Transport (women perceive electrification hazards as more likely/severe). ALIGNED WITH EXPECTATIONS — consistent with documented gender differences in transport poverty and mobility-cost sensitivity, though smaller than the Energy gender effect.

(4) Low income (<€20k) vs Rest — Housing (F = 4.86, p = 0.028). Low-income respondents perceive retrofitting hazards as more likely/severe (mean 3.10 vs 2.87). ALIGNED WITH EXPECTATIONS — affordability is the binding constraint on retrofitting for low-income households.

(5) LGBTIQA+ vs Rest — Housing (F = 4.70, p = 0.031; n=31). LGBTIQA+ respondents perceive significantly higher housing-hazard probability and severity. PARTIALLY EXPECTED / EXPLORATORY — consistent with literature on housing discrimination experienced by this group, but the small subsample (n=31) means this is treated as an exploratory signal for confirmation in T4.3, not a firm conclusion.

Near-significant and null results (also informative). Youth vs Rest — Energy (F = 3.34, p = 0.068) approaches but does not cross the threshold (younger respondents' trend towards higher concern). Women vs Rest — Housing (F = 2.68, p = 0.103) shows the same direction as the significant Energy/Transport gender effects but does not reach significance, most plausibly because of the smaller Housing sample (only 85 women). Crucially for the EU-SILC alignment concern: TENURE (tenant vs homeowner) is NOT significant in either Energy (p=0.194) or Housing (p=0.228), and INCOME is NOT significant in Transport (p=0.375) — these primary-axis null results are reported in full (Table 11 and Section 3.2.7) rather than omitted, so the reader sees that the EU-SILC variables were tested consistently and that only some of them discriminate perception in each sector.

Overall pattern and alignment. The inferential picture is coherent and matches prior expectations: the two axes that the just-transition literature most strongly associates with vulnerability — GENDER and INCOME — are exactly the two that produce the robust, replicated significant effects (gender significant in 2 of 3 sectors; income significant in 2 of 3 sectors). Tenure and disability, while theoretically relevant and therefore tested as primary axes, do not reach significance at the achieved sample sizes. Several comparisons (Elders, Unemployed) are constrained by very small subsamples and are flagged with low-power warnings in the Power/Notes column of Table 7; they are reported for completeness but not interpreted as conclusive.

4. Discussion

This section interprets the results presented in Section 3 (descriptive statistics in 3.1, sociodemographic subgroups in 3.2, country breakdown in 3.3, and the formal ANOVA tests in 3.4) and draws out their meaning for the FITTER project. Following the structure suggested in review, it focuses on interpretation rather than re-stating numbers.

1. Affordability is the unifying axis of perceived risk. Across all three sectors, the financial-cost hazards (electricity bills, heating/cooling costs, fuel and maintenance costs) are perceived as both the most likely and the most severe (Section 3.1). This is the single most robust finding of D4.1 and it holds across every sociodemographic subgroup and every country analysed. For the twin transition, this means that the dimension citizens worry about most is not the technology itself but its cost incidence.

2. Income and gender are the statistically robust vulnerability axes. The ANOVA results (Section 3.4) show that, of all the sociodemographic dimensions tested, INCOME (low-income vs rest, significant in Energy p=0.007 and Housing p=0.028) and GENDER (women vs rest, significant in Energy p<0.001 and Transport p=0.007) are the dimensions on which perceived risk differs most reliably. The descriptive subgroup analysis (Section 3.2) shows the same direction. The convergence of descriptive and inferential evidence on income and gender — and the validation of income as the EU-SILC stratification variable — gives the project a clear, evidence-based priority for the mitigation work in T4.2: measures should be income-

targeted and gender-aware.

3. The sector matters more than the country. The cross-sector comparison (Section 3.1.4) shows large, highly significant differences between sectors ($F=48.96$, $p<0.001$), whereas the country analysis (Section 3.3) finds only one significant national difference (Portugal > Germany in Housing). This implies that risk-perception patterns are driven primarily by the nature of the policy intervention (PV, retrofitting, EVs) rather than by national context — an encouraging result for the transferability of FITTER's sector-level mitigation strategies across countries, while still flagging Portugal-Housing as a case requiring national tailoring.

4. Probability and severity are not interchangeable. In Energy and Transport, severity generally exceeds probability; in Housing, several hazards are perceived as more likely than severe (Section 3.1.2). This asymmetry matters for risk communication: for high-severity / lower-probability hazards (e.g., energy poverty, cold-related health) the policy emphasis should be on reassurance and safety-net design, whereas for high-probability / moderate-severity hazards (e.g., routine bill increases) the emphasis should be on predictability and affordability support.

5. Limitations qualify, but do not undermine, the descriptive findings. The reduced achieved sample sizes (Section 2.3.2) lower the statistical power for the smaller subgroups and for country-level inference (Section 3.3.4), and the complete absence of Hungary (Section 2.3.1) leaves one use-case country unrepresented. These limitations are quantified transparently and mean that the robustly significant ANOVA results (income, gender) should be treated as the firm evidence base, while the low-power comparisons (Elders, country-level differences) are treated as indicative and flagged for confirmation in T4.3.

5. Conclusions and further developments

D4.1 has successfully delivered its primary objective: the design, implementation, quality control, and statistical analysis of sector-specific quantitative surveys estimating perceived probability and severity rates of hazards associated with the twin transition across Energy, Housing, and Transport sectors. The final quality-controlled sample is n=408 (Energy), n=264 (Housing), n=359 (Transport) valid respondents after applying all quality filters including the sociodemographic cross-check (Annex IV). This deliverable provides the empirical foundation for hazard probability and severity indices ready for ingestion into the FITTER Digital Platform and for predictive modelling in Task 4.3.

DG prioritisation note. The disadvantaged-group (DG) variables highlighted in the five key findings below are NOT pre-selected on theoretical grounds; they are prioritised on the basis of the actual statistical evidence in the T4.1 dataset — specifically, the five DG-versus-rest pairwise comparisons that reach statistical significance in the ANOVA tests reported in Section 4.3 Table 12 (Women–Energy, Women–Transport, Low-income–Energy, Low-income–Housing, LGBTIQA+–Housing). The remaining DG comparisons collected in the survey (Youth/Elders, Rural/Urban, Disability/no-disability, Ethnic minority, Migrants, Low education, Unemployed) are reported with full transparency in Table 7 but are NOT highlighted in the key findings because either (a) the ANOVA did not detect a significant difference or (b) the subsample size was below the power threshold for a conclusive test (e.g., Elders n=13 in Energy, n=2 in Housing). The DG prioritisation is therefore a DATA-DRIVEN result of the T4.1 analysis, not a pre-selected analytical lens.

DG prioritisation note. The disadvantaged-group (DG) variables highlighted in the five key findings below are NOT pre-selected on theoretical grounds; they are prioritised on the basis of the actual statistical

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Key Findings

1. **Financial hazards are the dominant concern across all three sectors**. Across Energy, Housing and Transport, the highest-ranked hazards on BOTH probability and severity dimensions are financial-cost hazards: 'Higher electricity bills' (Energy mean probability 3.79, severity 3.82; Housing 3.86, 3.52); 'Increased heating/cooling costs' (Energy 3.85, 3.83; Housing 3.84, 3.52); and 'Higher fuel & maintenance costs' (Transport 3.86, 3.81). This pattern is robust across sectors and across all sociodemographic subgroups examined in Section 3.2, confirming affordability as the central risk dimension of the twin transition in the respondents' perception. Health- and welfare-extension hazards (energy poverty, damp/mould, cold-related health) consistently rank in the high-severity tier but in the medium-probability tier — respondents see these as less likely but as more serious if they occur. Regulatory/infrastructure hazards (penalties, outages, restrictions) rank lower on probability across all sectors and subgroups, reflecting respondents' assessment of these as remote or rare events.

2. Severity systematically exceeds probability. Across all sectors, mean severity scores exceed mean probability scores (Energy: 3.11 vs 3.28; Housing: 2.79 vs 3.02; Transport: 3.53 vs 3.39), indicating that respondents view potential negative impacts as serious even when likelihood is moderate.

3. Income is the primary statistical discriminant of hazard perception (and is consistent with the EU-SILC stratification choice). Statistically significant ANOVA results for the low-income (< €20,000 household income) versus rest comparison in BOTH Energy ($F=7.26$, $p=0.007$) and Housing ($F=4.86$, $p=0.028$) sectors confirm that financial vulnerability amplifies perceived hazard probability and severity, particularly on the financial sub-cluster of hazards. This finding has two important consequences. First, it validates the EU-SILC stratification choice to use Income (above/below the country-specific median) as one of the three Prolific quota variables: income heterogeneity is a real, statistically detectable driver of perceived hazard exposure. Second, it is consistent with the descriptive Section 3.1 finding that financial hazards dominate the perception landscape: low-income respondents, who face stronger real-world exposure to electricity-bill increases and energy poverty, internalise this exposure as higher perceived probability and severity. The policy implication is that income-targeted mitigation measures — direct transfers, targeted retrofit subsidies, social tariffs — are essential for a fair transition. Note that the low-income vs rest comparison in Transport did NOT reach statistical significance ($F=0.79$, $p=0.375$) — Transport's hazard profile is dominated by the more uniformly-shared regulatory hazards (EV mandate timelines, ICE bans) which affect respondents more evenly across the income distribution.

4. Gender differences are highly significant in Energy and Transport. Women in the Energy sector perceive hazard probability and severity substantially higher than men ($F=20.63$, $p<0.001$) — the strongest DG effect detected in the entire T4.1 dataset; the same gender effect is significant in Transport ($F=7.41$, $p=0.007$). The descriptive interpretation is direct: averaging across all 7 Energy hazards, women rate the mean probability at 3.32 (95% CI 3.21–3.43) compared with 2.99 (95% CI 2.88–3.11) for men, a difference of 0.33 Likert points or ~25% of one standard deviation; the corresponding mean-severity gap is 3.46 (women) vs 3.09 (men). Comparison with Section 3.1 (sector-level descriptive results) shows that the gender effect is

not driven by a single outlier hazard but is consistent across the financial-hazard cluster and the energy-poverty hazard — exactly the hazards on which women have been documented in the energy-poverty literature to experience stronger real-world exposure (time-use, household-management roles, single-parent households). The Transport gender effect is consistent with the same logic applied to mobility costs and emission-zone restrictions, although smaller in magnitude (mean-probability gap of 0.18 Likert points). The Housing gender effect did NOT reach statistical significance (F=2.68, p=0.103), although the descriptive direction (women > men) is the same — the smaller Housing sample (n=264, with only 85 women) is the principal explanation. The policy implication is that gender-specific outreach and gender-disaggregated impact assessment should be built into Energy- and Transport-transition policy design from the outset.

5. LGBTIQ+ respondents in Housing show a significant difference. The statistically significant result (F=6.256, p=0.013; n=31, power≈77%) is exploratory but warrants further investigation in T4.3 regarding intersectional vulnerabilities in housing retrofitting access.

6. Several DG comparisons are constrained by small subsample sizes. Elders (Energy n=13; Housing n=2), Unemployed in Housing (n=19), LGBTIQA+ in Housing (n=31), and a few other small ex-post analytical subgroups limit the reliability of those ANOVA comparisons. These findings should be revisited with broader data in future iterations.

Implications for WP4 and the FITTER Digital Platform

The probability and severity indices from D4.1 are structured for direct ingestion into the FITTER Digital Platform (WP5). The confirmed DG effects provide quantitative evidence for at-risk group identification in the Predictive Decision Support System (PDSS). Non-significant results suggest that individual-level intersectional factors may be more predictive than broad categorical labels, a hypothesis to be tested in T4.3.

The full survey microdata (all valid respondents, all sectors) are provided alongside this deliverable for use by T4.3 and WP5. The CrossCheck Report (Annex IV) provides a fully auditable record of quality control decisions.

Further Developments

The immediate next step is **Task 4.3 (D4.3)**: the development of predictive models linking sociodemographic variables (Parts A and B of the T4.1 survey) to perceived hazard probability and severity (Part C). Recommended directions based on D4.1 findings:

• Predictive modelling: ordered logistic regression, Bayesian networks, or Naïve Bayes classifiers per hazard per sector, with income, gender, and LGBTIQ+ status as priority predictors (confirmed DG effects).

• Indirect indicator integration: linking macro-level Eurostat/EU-SILC contextual indicators to micro-level survey predictors to build the full predictive framework (Equation 1 in Section 2.3.1).

• Formal DG–hazard linkage: the explicit mapping of DG profiles to specific hazards (deferred from D4.1; see Section 1.2) will be formalised in T4.3 using the regression and ANOVA results reported here.

• Supplementary / targeted data collection (expanded per reviewer request to quantify the limitation and link it to results interpretation). The sampling limitations are not merely qualitative; they are quantifiable and have specific, traceable consequences for the results. Concretely: (a) the achieved samples fell short of the n=456 design target in every sector — Energy 408 (89% of target), Housing 264 (58%), Transport 359 (79%) — which lowered achieved statistical power to 75% / 57% / 70% respectively (below the planned 80%; Section 2.2.3); (b) one entire use-case country (Hungary) contributed ZERO respondents (validated in Section 2.3.1), so no Hungarian results exist in any analysis; (c) several DG subgroups fell below the 64-per-group threshold needed to detect a medium effect — Elders (Energy n=13, Housing n=2), Unemployed in Housing (n=19), LGBTIQA+ in Housing (n=31) — which is precisely why the Elders and Unemployed

comparisons are non-significant/low-power in Table 7 and why the LGBTIQA+ result is flagged exploratory. The direct link to results interpretation is therefore: wherever Section 3.4 reports a non-significant or exploratory result for these groups, the cause is at least partly this quantified sampling limitation rather than a demonstrated absence of effect. For T4.3 and any future FITTER data collection, targeted oversampling of these specific under-covered cells (especially Hungary as a country, and Elders/Unemployed as groups), ideally via a complementary national-panel or partner-led recruitment channel, is recommended to raise these subgroups above the power threshold.

Sector	DGs
Energy	Women, Elders, Youth, Rural, Disabilities, Unemployed, Low level of education, People living in old buildings, Migrant background, Low income
Transport	Single women, large families, Disabilities, Elders, Rural, Low income, LGTBI, Car-dependence, Users without space for EV infrastructure
Housing	Low income, Rural, Youth, Elders, Disabilities or chronic illness, Renting/living in social house, Single mothers, Migrant background, LGTBI, Non-registered in municipal census

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7. Annexes

I. Annex I – T4.1 Survey Questionnaires (Energy, Housing and Transport sectors)

II. File: Annex_I_-_Final_T4_1_integrated_all_questionnaires_09-10.xlsx (provided as a separate attachment, stored in the FITTER-EU SharePoint repository operated by TU Dublin and accessible to all consortium beneficiaries). The full integrated survey (all three sector questionnaires, as administered) is available at the following SharePoint link: https://tudublin.sharepoint.com/:x:/r/sites/FITTER/Shared%20Documents/General/WP4-%20Inequalities%20risk%20assessment%20and%20intersectional%20mitigation%20plans/T4.1%20Scenario-based%20identification%20and%20analysis%20of%20inequality%20risks/Final%20version%20T4.1%20questionnaire/Final%20T4.1%20integrated%20all%20questionnaires%2009-10.xlsx

— consortium members with FITTER SharePoint access can open the complete questionnaire workbook directly from this link.

This annex contains the FINAL three sector-specific questionnaires (Energy, Housing, Transport) as administered in LimeSurvey during the T4.1 fieldwork (September 2025 – January 2026). The Excel workbook integrates the three questionnaires into one document for ease of review, with one worksheet per sector. Each worksheet contains the full ordered list of questions in Sections A, B and C, with: question code, question text (English master version), response options and codes, EU-SILC alignment notes where applicable, and branching/routing rules.

Process notes — how the questionnaires were produced. The Energy questionnaire served as the template (developed first, by SV); the Housing and Transport questionnaires were derived from it by replacing the Section A policy-intervention block and the Section C hazard catalogue with sector-appropriate content, while preserving the Section B sociodemographic block in a common EU-SILC-aligned format. Translations into Italian, Spanish, German, Hungarian and Portuguese were performed by the corresponding consortium partners and back-checked by SV; the English master remained the authoritative version. See Section 2.1 of this deliverable for the full development process.

Related sections of this deliverable: Section 2.1 (T4.1 survey design); Section 2.1.3 (Summary of questionnaire's content); Annex II (Survey Variable Dictionary, which keys the codes used here to their analytical interpretation).

Annex II – Survey Variable Dictionary

File: D4_1_Annex_II_Variable_Dictionary.xlsx (to be compiled by SV and uploaded to the FITTER-EU SharePoint repository operated by TU Dublin as a companion file to the questionnaires in Annex I). Status note. At the time of submission of D4.1, the Variable Dictionary is being finalised and will be uploaded to the project SharePoint repository as a stand-alone Excel workbook accessible to all consortium partners. The Variable Dictionary is intended to be the single source of truth for variable codes used in T4.3 (predictive modelling) and WP5 (digital-platform integration). Content of the Variable Dictionary. For each variable in the T4.1 survey (Sections A, B and C of each of the three sector questionnaires), the dictionary records: • Variable code (e.g., Q362092 for Sex/Gender in the LimeSurvey export); • Variable number / position in the questionnaire; • Section (A — policy-intervention access; B — sociodemographic profile; C — hazard probability and severity Likert ratings); • English question text (master version); • Variable type (categorical, ordinal, Likert 1–5, free text, numeric, binary); • Response values / response categories with their codes; • EU-SILC alignment, where applicable (e.g., HX090 for equivalised disposable income, HH050 for ability to keep home adequately warm, PB150 for Sex, PB220A for citizenship, PL031 for self-defined economic status); • Notes on derivation or transformation (e.g., 'Low-income' = composite of 'Less than €10,000' OR '€10,000–€20,000' response categories). Related sections of this deliverable: Section 2.1.3 (Summary of questionnaire's content); Section 2.2.4 (Stratification — variables used as Prolific quota filters); Section 2.3.4 (Measurement and scoring); Annex I (the questionnaires themselves).

III. Annex III – Technical Derivation of the Sample-Size Formula

Purpose and context of this annex. This annex provides the full mathematical derivation of the sample-size formula used in Section 2.2.3 of the main text (Step 1 — Sample size for statistical estimation of perceived probability and severity). The aim is to demonstrate that the formula adopted, $n = (Z\alpha/2 + Z\beta)^2 \cdot \sigma^2/e^2$ (Eq.2

in Section 2.2.3), is the correct null-hypothesis-testing form of the Cochran (1977) sample-size formula applied to mean estimation on a bounded scale. The reader who is comfortable with the formula as presented in Section 2.2.3 can skip this annex; the reader who wants to verify each step from first principles (Bernoulli test of perception scores → binomial approximation → normal approximation → sample-size inversion) should consult the derivation below.

References used in the derivation: Cochran (1977) Sampling Techniques (3rd ed.), John Wiley & Sons — the canonical reference for the formula in its proportion and mean variants; Hair, Black, Babin & Anderson (2018) Multivariate Data Analysis (8th ed.), Cengage — for the sample-size guidance per number of predictors used in Step 2 of the same section; Memon, Ting, Cheah, Thurasamy, Chuah & Cham (2020), Sample size for survey research: Review and recommendations, Journal of Applied Structural Equation Modeling 4(2), i–xx — for the cross-validation benchmarks in Step 3 of the same section; Krejcie & Morgan (1970), Determining sample size for research activities, Educational and Psychological Measurement, 30, 607–610 — for the tabulated alternatives cross-referenced in Section 2.2.3.

Related sections of this deliverable: Section 2.2.2 (Sampling techniques — the non-probability quota approach used in FITTER); Section 2.2.3 (Sample size calculation — where Eq.2 appears and is applied); Section 2.3.2 (Removal of low-quality responses — where the actual achieved power is reported transparently after the quality-control protocol).

Step-by-step derivation. Each test (= one respondent) is a Bernoulli test, independent from the others (respondents are independent and not talking each other), which can be considered "success" if a specific score in the assessment of the perception of a specific Hazard in a Sector is selected in the survey (for example, score=3), and "fail" if other scores are selected (for example Scores=1, 2, 4 or 5).

The variable "**X=Number of successes in the sample**" (for example, number of respondents from the sample assessing a specific Hazard in the survey with the score 3) follows a ***Binomial statistical distribution*** with P=0,2 (=1/5) and number of tests=Sample size.

M**ean** and **variance** of this *Binomial distribution* are:

- μ (X)=n.P=n.0,2 (Eqs.1)
- **ϙ²** (X)= n. P. (1-P)=n.0,2.0,8= n.**0,16**

We define the variable "$\bar{X}$**=Proportion of success in the sample**" as:

$$\bar{X} = \frac{X}{n} = \frac{\sum_1^n X_i}{n} \quad \text{(Eq.2)}$$

Being:

- i=Number of tests=Number of respondents
- $X_i = 1\ in\ case\ of\ success; and\ 0\ in\ case\ of\ fail$
- n=Total number of tests or Sample size

As said, each X_i is a *Bernoulli test*, with:

- μ (X_i)=P=0,2 (Eqs.3)
- **ϙ²** (X_i)=P. (1-P)=0,2x0,8= **0,16**

With a sufficient number of tests (>30), the variable: "$\bar{X}$**=Proportion of success in the sample**" can be approximated to a ***Normal statistical distribution*** according to the *Limit Central Theorem (CLT)*:

The **mean** and **variance** of this Normal distribution for $\bar{X}$ are, according to the CLT:

$$\mu_{\bar{X}} = \frac{\sum_1^n \mu_{X_i}}{n} = \mu_{X_i} = P = 0{,}2$$

$$\sigma^2{}_{\bar{X}} = \frac{\sum_1^n \sigma^2{}_{X_i}}{n^2} = \frac{\sigma^2_{X_i}}{n} = \frac{P.(1-P)}{n}$$

(Eqs.4)

This probabilistic distribution for this variable $\bar{X}$ can be depicted as follows:

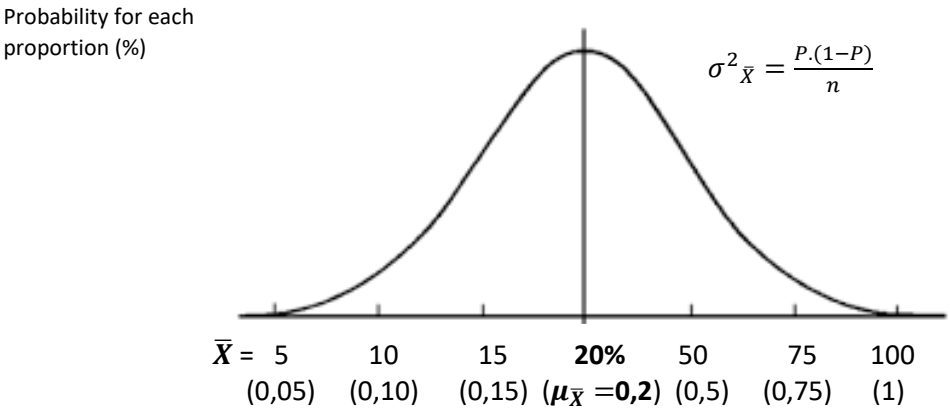


Figure 26. Probability distribution of the variable "X̄ = proportion of success in the sample"

A *Null Hypothesis Test* can be carried out on this *normal distribution of* $\bar{X}$, being **H_0** ($\mu_{\bar{X}} = 0,2$) the *Null Hypothesis*, **H_1** ($\mu_{\bar{X}} = 0,2 + \delta$; **$\delta$ = Size Effect**) the *Alternative Hypothesis*, the significance would be **α=0,05** (confidence level 95%), and the power of the test, **1-β=0,85**. See picture below:

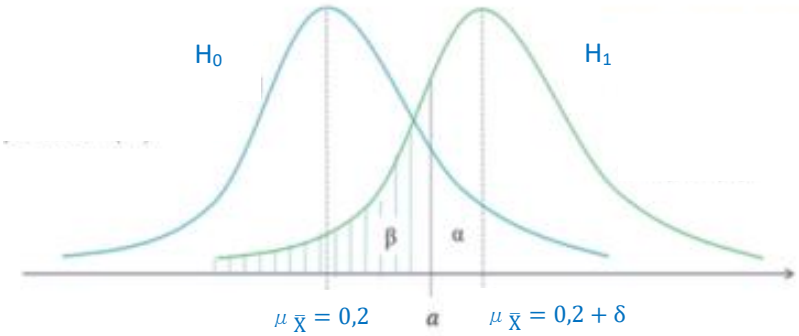


Figure 27. Representation of H0 and H1 in the Null Hypothesis Test in a No Standard Normal distribution

Being:

- a = Value of $\bar{X}$ in the normal distribution **H_0** for which **α=0,05**, or value in the normal distribution **H_1** for which **β =0,15**

These 2 *normal statistical distributions* (H_0 and H_1) can be normalized to make the **mean=0** and the **variance=1** by defining a new variable "Z", from Eqs. *4*, as:

For **H_0**:

$$Z = \frac{\bar{X} - \mu_{\bar{X}}}{\sigma_{\bar{X}}} = \frac{\bar{X} - P}{\sqrt{P.(1-P)}/\sqrt{n}}$$

(Eq.5)

For **H_1**:

$$Z = \frac{\bar{X} - (\mu_{\bar{X}} + \delta)}{\sigma_{\bar{X}}} = \frac{\bar{X} - (P + \delta)}{\sqrt{P.(1-P)}/\sqrt{n}}$$

(Eq.6)

In this case Fig.7 can be redesigned as the next Fig.8, in which H_0 and H_1 are *Standard normal distributions*:

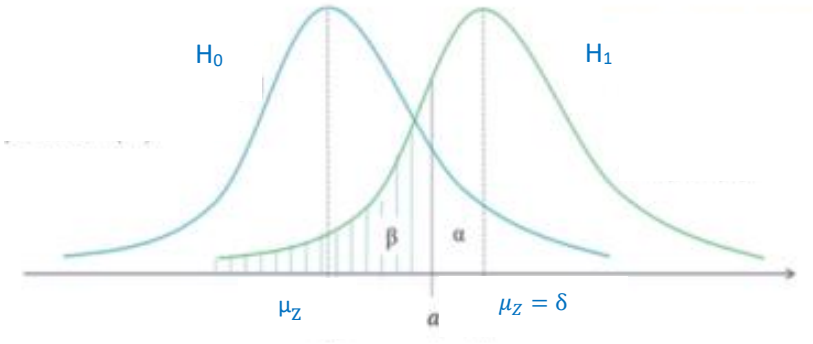


Figure 28. Representation of H0 and H1 in the Null Hypothesis Test in a Standard Normal distribution

If we make the probability of **Type 2 Error** (to accept H_0 when it is false, and H_1 is True) **= β** in this *Standard normal distribution* for the variable "Z", we get from Eq. 6:

$$\beta = Probability(Z < Z_{a;H_1} = Value\ of\ Z\ in\ the\ Standard\ normal\ distribution\ H_1\ for\ which\ "\bar{X} = a" = \frac{a - (P + \delta)}{\sqrt{P.(1-P)}/\sqrt{n}})$$

(Eq.7)

What can be also expressed as:

$$\beta = Probability\ (Z < \frac{a - P}{\sqrt{P.(1-P)}/\sqrt{n}} - \frac{\delta}{\sqrt{P.(1-P)}/\sqrt{n}})$$

(Eq.8)

Given that, from Eq. 5:

$$Z_\alpha = \frac{a - P}{\sqrt{P.(1-P)}/\sqrt{n}} = Z_{a;H_0} = Value\ of\ Z\ in\ the\ Standard\ normal\ distribution\ H_0\ for\ which\ "\bar{X} = a")$$

(Eq.9)

Then, the Eq. 8 can be expressed as:

$$\beta = Probability\ (Z < Z_\alpha - \frac{\delta}{\sqrt{P.(1-P)}/\sqrt{n}})$$

(Eq.10)

The value of Z in the Standard normal distribution H_1 belonging to this Probability "β" is "- Z_β" (see Fig. 8). So, we can conclude that:

$$-Z_\beta = Z_\alpha - \frac{\delta}{\sqrt{P.(1-P)}/\sqrt{n}}$$

(Eq.11)

From which we can deduce the Eq. To calculate the simple size:

$$n = Number\ of\ tests\ or\ sample\ size = \frac{(Z_\alpha + Z_\beta)^2.p.(1-p)}{\delta^2}$$

(Eq.12)

IV. Annex IV – Sociodemographic CrossCheck Verification Report

File: Focused_CrossCheck_Report_Final_v3.xlsx (also referenced internally as CrossCheck_Report_T4_1_Survey_V4.xlsx). Provided as a separate attachment, stored in the FITTER-EU SharePoint repository operated by TU Dublin and accessible to the four Authorised Data Processing Partners (TU Dublin, SV, AITEC, SFC) as defined in Annex XI.

Purpose of this annex. This annex documents Step 1 of the three-step quality-control protocol described in Section 2.3.2 (Removal of low-quality responses). The cross-check identifies and removes Prolific participants whose self-declared sociodemographic attributes in their Prolific profile do NOT match the values they provided in the LimeSurvey questionnaire — a signal of potentially unreliable responding.

How the cross-check was conducted (step-by-step process).

Step 1 — Extraction of Prolific-profile attributes. For each completed submission, the Prolific platform recorded the participant's self-declared values for the variables used as prescreener filters at recruitment (Sex/Gender, Tenancy status for Energy/Housing, Disability status for Transport). These Prolific-profile values were exported via the Prolific API as part of the study-completion record.

Step 2 — Extraction of LimeSurvey-response attributes. For the same set of completed submissions, the LimeSurvey export contained the participant's self-declared values for the same three variables, asked again as part of the Section B sociodemographic block of the questionnaire.

Step 3 — Pairwise comparison. For each Prolific PID, the Prolific-profile values and the LimeSurvey-response values were compared element-by-element. Any mismatch on ANY of the three cross-checked variables results in the PID being flagged as 'not verified' and excluded from the analytical dataset.

Step 4 — Audit trail. The workbook records every comparison and every mismatch, providing a fully auditable record of the data quality decisions taken prior to analysis. The final list of verified PIDs per sector (Energy 421, Housing 264, Transport 374) is recorded on the 'Valid IDs per Sector' sheet and is the input to Step 2 (≥4-minute filter) of the quality-control protocol.

Structure of the workbook.

• Summary — Top-level dashboard of total mismatches and respondents per sector with the count of valid verified IDs (Transport 374, Housing 264, Energy 421 — the verified-ID list that feeds the analytical dataset).

• Transport / Housing / Energy — Per-sector detail of every mismatch detected, with columns for Participant ID, Variable, Value in Study (Prolific), Value in Survey (LimeSurvey), and Issue description.

• Valid IDs per Sector — Complete list of Prolific participant IDs that passed the CrossCheck verification. This is the SOURCE LIST used as Filter Step 1 of the three-step quality-control protocol.

Cross-checked variables: Disability status, Sex/Gender, Tenancy status (Housing and Energy sectors only).

Related sections of this deliverable: Section 2.2.4 (Stratification — defines the variables used as Prolific prescreener filters and therefore cross-checkable here); Section 2.3.2 (Removal of low-quality responses — Step 1 of the three-step protocol that uses this annex as input).

Annex V – EU-SILC Stratification Data Analysis

File: Annex_II_-_Data_analysis_EU-SILC.xlsx (the file name retains the original internal numbering used by SV during analysis; in this deliverable it is referenced as Annex V to fit the integrated annex structure). To be added as a separate attachment, stored in the FITTER-EU SharePoint repository operated by TU Dublin.

Purpose of this annex. This annex documents the EU-SILC microdata analysis that produced the quantitative inputs for the T4.1 sampling stratification. Concretely, it provides the per-country sociodemographic distributions (Income above/below median, Tenancy status, Sex) that were used to compute the target respondent counts per Prolific sub-study, which are then reported as the operational recruitment plan in Annex VII.

How the EU-SILC analysis was performed (step-by-step process).

Step 1 — Microdata extract. The 2018 EU-SILC microdata extract was obtained through Eurostat's authorised researcher access (the analysis was conducted by SV in Stata 17.0). The 2018 wave was selected because it is the most recent year for which full microdata had been released for all six FITTER use-case countries at the time of T4.1 design.

Step 2 — Inter-country expansion factor (Sheet 'Population distribution'). The weighted total population of each use-case country was tabulated and converted to country shares of the 6-country reference population. These shares were used at Step 1 of the stratification (Section 2.2.4) to allocate the n=456 valid-respondent target across the six countries.

Step 3 — Per-country household-level cross-tabulations (Sheets 'Italy', 'Hungary', 'Spain', 'Ireland', 'Germany', 'Portugal'). For each country, household-level frequencies were computed for the cross-tabulation of equivalised disposable income (HX090) split at the country-specific median against tenancy status (HH021 — derived: tenant vs. homeowner) and the household reference person's sex (PB150). The resulting per-country cell counts and percentages were used to set the per-(country × income × tenancy × sex) target respondent counts for the Prolific sub-studies.

Step 4 — Compilation of the recruitment plan. The per-country cell percentages from Step 3 were multiplied by each country's allocated n (from Step 2) and rounded to integer respondents, producing the unique-combinations recruitment plan documented in Annex VII (Prolific Sub-Studies Quota Specification).

Structure of the workbook.

• Population 2018 — Reference EU-SILC microdata extract used as analytical baseline.

• Population distribution — Aggregate distribution of the target population across the six countries, used at Step 1 of the stratification (inter-country expansion factors).

• Italy / Hungary / Spain / Ireland / Germany / Portugal (six sheets) — Per-country household-level and individual-level cross-tabulations underpinning Steps 2 and 3 of the stratification, with all EU-SILC variable codes referenced in Section 2.2.4 (HX090, HH050, PB150, PB220A, PL031, etc.) operationalised in the cell calculations.

Related sections of this deliverable: Section 2.2.4 (Stratification of the sample — where the EU-SILC-derived targets are applied); Annex VII (Prolific Sub-Studies Quota Specification — the operational recruitment plan derived from this annex).

Annex VI – D4.1 Statistical Tables (Hazard, ANOVA and Subgroup Statistics)

File: Annex_VI_D4_1_Statistical_Tables_V3_2.xlsx (provided as a separate attachment to D4.1 V3.2, stored in the FITTER-EU SharePoint repository operated by TU Dublin and accessible to all consortium beneficiaries).

Purpose. This workbook contains the complete statistical results that underpin Sections 3 and 4 of this deliverable. Every table and figure number in the deliverable text can be traced to a sheet in this workbook.

Dataset and quality-control filters. All statistics are computed on the cleaned analytical dataset (Section 2.3.2: CrossCheck + ≥4-min filter + dedup) giving n=408 (Energy), n=264 (Housing), n=359 (Transport).

Sheets: README; Hazard Statistics (with MODE per hazard, backs Tables 8, 9 and 10); ANOVA Results (all 33 DG comparisons, backs Table 7); Country Statistics (NEW, backs Section 3.3); Tenure & Income (NEW, backs Section 3.2.7 and Table 11); Cross-Sector Comparison (NEW, backs Section 3.1.4); DG-EU-SILC Mapping (NEW, backs Table 11).

Annex VII – Prolific Sub-Studies Quota Specification (Recruitment Plan)

File: Requests_to_Prolific_for_T4_1_-_V9.xlsx (to be added as a separate attachment, stored in the FITTER-EU SharePoint repository operated by TU Dublin).

Purpose of this annex. This annex contains the operational recruitment plan used to translate the EU-SILC stratification calculations (Annex V) into concrete Prolific sub-study quota requests. It is the bridge between the population-level stratification analysis (Annex V) and the actual sub-studies executed on the Prolific platform (Annex VIII), and provides the prescreener filters and target respondent counts for every unique sociodemographic combination launched during T4.1 fieldwork (September 2025 – January 2026).

Process: How the recruitment plan was assembled.

Step 1 — Map the EU-SILC cross-tabulation outputs (Annex V) to a list of unique (Country × Income × Tenancy/Disability × Sex) combinations per sector. For each combination, a target respondent count is derived from the EU-SILC per-country cell percentage multiplied by the country allocation (Section 2.2.4).

Step 2 — Translate each combination into a Prolific prescreener configuration. The Prolific platform requires that quota filters be specified as a single, restrictive prescreener for each sub-study. The recruitment plan therefore decomposes each (Country × Income × Demographic × Sex) combination into one Prolific sub-study, identified by a unique Study number (Study 1, Study 2, etc., up to 76 sub-studies in total across the three sectors).

Step 3 — Define income brackets per country, anchored on the country-specific average annual salary (used as a proxy for the country median equivalised disposable income). The 'Above median' and 'Below median' splits are implemented in Prolific via the platform's income-bracket filter (Less than €10,000; €10,000–€19,999; €20,000–€29,999; …; €70,000–€79,999), which is selected per country to span either the lower half or the upper half of that country's salary distribution.

Structure of the workbook.

• Instructions for APIs — Reference parameters used to define quota thresholds, including the average annual salary per country (Italy €37,168; Hungary €9,781; Spain €36,996; Ireland €65,687; Germany €45,773; Portugal €24,502) — used as proxy for the country-specific median to split respondents into 'Above median' and 'Below median' income bands — together with the full list of Prolific income brackets that map onto those bands.

• Unique combinations Transport — 24 rows specifying the sub-studies launched for the Transport sector, with prescreener variables Country (Italy / Hungary / Spain), Income (above/below median), Disability (yes/no), Sex (Male/Female), and target number of respondents per combination.

• Unique combinations Energy — Rows specifying the sub-studies launched for the Energy sector, with prescreener variables Country (all six FITTER use-case countries), Income, Tenancy status (Tenant/Homeowner), Sex, and target number of respondents.

• Unique combinations Housing — Rows specifying the sub-studies launched for the Housing sector, with the same prescreener structure as Energy (Country: Ireland / Germany / Portugal; Income; Tenancy; Sex) — Italy, Spain and Hungary were not in the Housing recruitment plan as per the consortium-agreed

country–sector allocation.

Related sections of this deliverable: Section 2.2.4 (Stratification of the sample); Section 2.3.1 (Sample composition limitations); Annex V (EU-SILC Stratification Data Analysis — the input); Annex VIII (Prolific Sub-Studies Configuration Files — the actual sub-studies executed on the platform).

Annex VIII – Prolific Sub-Studies Configuration Files

File: Studies.zip (to be added as an attachment).

This annex contains the complete configuration files for the 76 Prolific sub-studies launched during T4.1 fieldwork (between September 2025 and January 2026). Each sub-study targets a specific country–sector–sociodemographic profile combination required by the stratification plan (Section 2.2.4) and the multi-wave recruitment strategy (Section 2.3.1). The archive contains one CSV file per sub-study, each documenting:

- the Prolific filters applied (country, age band, gender, income bracket, urbanity, disability status, tenancy status, ethnicity, citizenship, education),
- the eligible pool size on Prolific at study launch,
- the target number of completions,
- the achieved number of completions,
- the per-participant compensation rate and study duration estimate.

The use of multiple sub-studies (rather than a single open recruitment) was the only viable way to enforce the EU-SILC-aligned quotas on the Prolific platform; this is also documented in the Prolific limitation justification (Section 2.3.1 and Annex IX). The file is provided separately from this document.

Annex IX – Prolific Recruitment Dashboard Evidence

File: Prolific.zip (to be added as an attachment).

This annex contains the original full-resolution screenshots of the Prolific operations dashboard captured during T4.1 fieldwork, which document directly:

- the partial completion rates achieved for DG-targeted sub-studies (badge values in the top-left of each study card, e.g., 7%, 19%, 45%, 76%);
- the very small eligible-participant counts produced by the joint application of country and sociodemographic filters (e.g., 22, 26, 63, 82 of 333,736 platform members);
- the parallel LimeSurvey administration dashboard with the final response counts per sector at the end of fieldwork.

These screenshots correspond to the reduced-resolution Figures 11.a–11.g embedded in Section 2.3.1 of this deliverable, where they serve as evidence for the Prolific panel-composition limitations affecting DG recruitment and the analytical implications of those limitations. The file is provided separately at full resolution.

Annex X – Raw Survey Response Data (LimeSurvey Exports)

File: Final_collected_data.zip (to be added as an attachment).

This annex contains the complete raw LimeSurvey response files exported on 17 May 2026 for the three FITTER T4.1 questionnaires:

- Results/results-survey-energy-17052026.xlsx — 584 raw rows (Energy sector).
- Results/results-survey-housing-17052026.xlsx — 434 raw rows (Housing sector).
- Results/results-survey-transport-17052026.xlsx — 554 raw rows (Transport sector).

Each row contains the complete set of LimeSurvey responses for one participant, including: the Prolific participant ID (PROLIFIC_PID), the LimeSurvey start- and submit-timestamps used to compute completion duration, every question response in Sections A (sector-specific), B (sociodemographic), and C (hazard probability and severity Likert ratings), and any open-ended free-text answers.

These raw files are the source from which the cleaned analytical dataset (n=408 / 264 / 359) was derived by applying the three-step quality-control protocol described in Section 2.3.2. Access to these raw files is restricted to the Authorised Data Processing Partners as defined in Annex XI. The file is provided separately from this document.

Annex XI – DMP Annex: Joint Ownership and Restricted-Access Data Agreement

File: FITTER_DMP_Annex_Data_Agreement_rev.pdf (signed; to be added as an attachment).

This annex contains the signed Data Management Plan (DMP) Annex defining the joint ownership, restricted-access, and secure-processing arrangements that govern the T4.1 survey data. The Annex was signed on 6 May 2026 by the four Authorised Data Processing Partners: TU Dublin (Sara Clavero and Maria Chiara Leva), SV (Maria Sanagiuliano), AITEC (Francisco Santarremigia), and SFC (Zoltán Székely).

Key provisions documented in this Annex:

• Joint ownership of the Survey Data among all FITTER-EU consortium beneficiaries, as per the Grant Agreement and Consortium Agreement.
• Restricted access to raw Survey Data: only the four Authorised Data Processing Partners (TU Dublin, SV, AITEC, SFC) may access the raw files; all other consortium partners receive only the analysed outputs.
• Data Coordinator: TU Dublin is responsible for overseeing data governance.
• Ownership of analysed outputs (reports, models, publications) belongs to the partner(s) contributing to the specific analysis.
• Security measures: anonymisation and pseudonymisation at the earliest possible stage of processing, encryption, secure storage, restricted access, logging, and breach reporting, in line with EU GDPR, the FITTER-EU Privacy Policy, and the project DMP.

The file is provided separately from this document as a signed PDF.